

Diffusion Models for Lattice Field Theory

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On behalf of [DM-QFT collaboration](#)

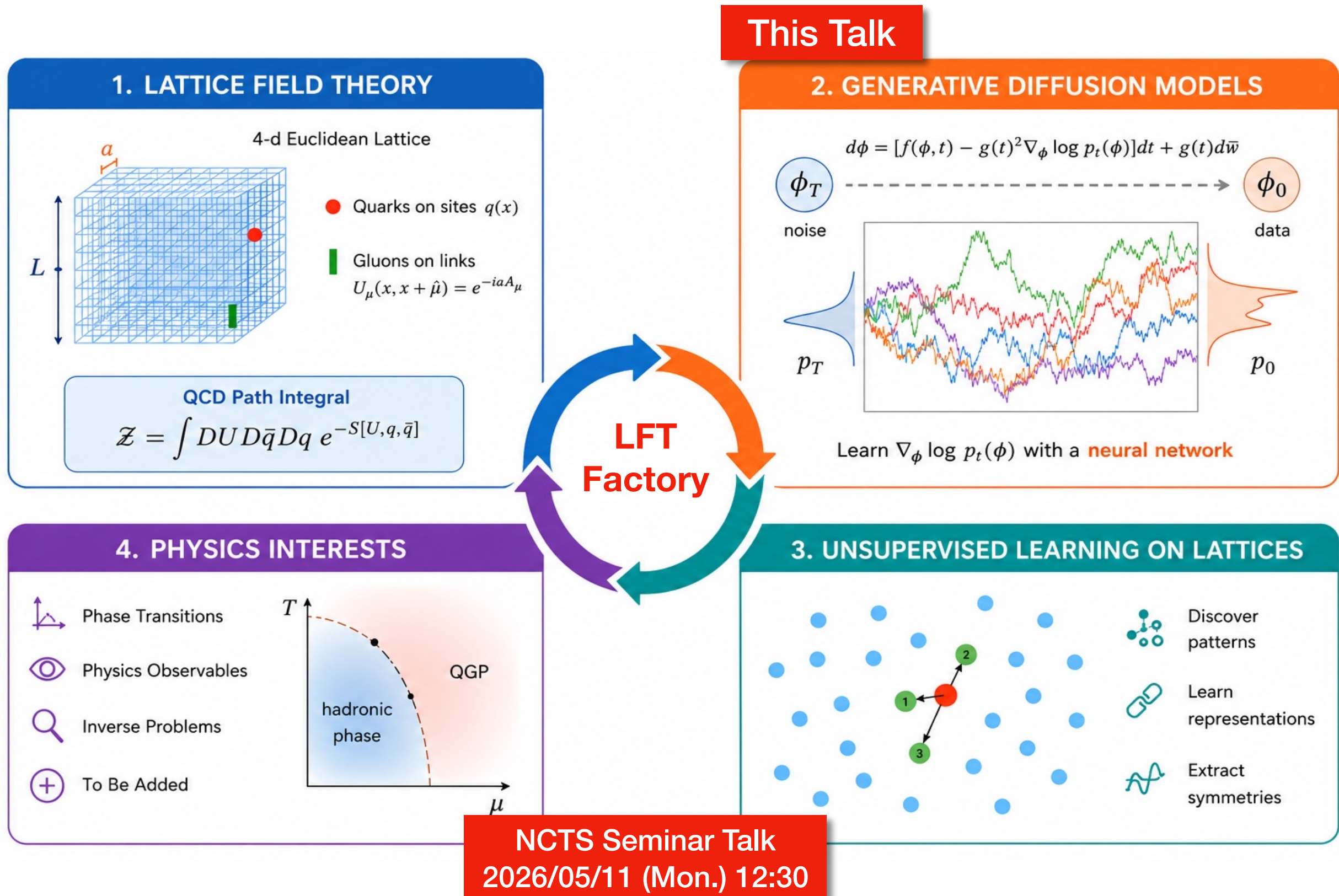
Data Analysis and Machine Learning for Physics Workshop, NTU, Taipei, May 6-8 2026

Background and References



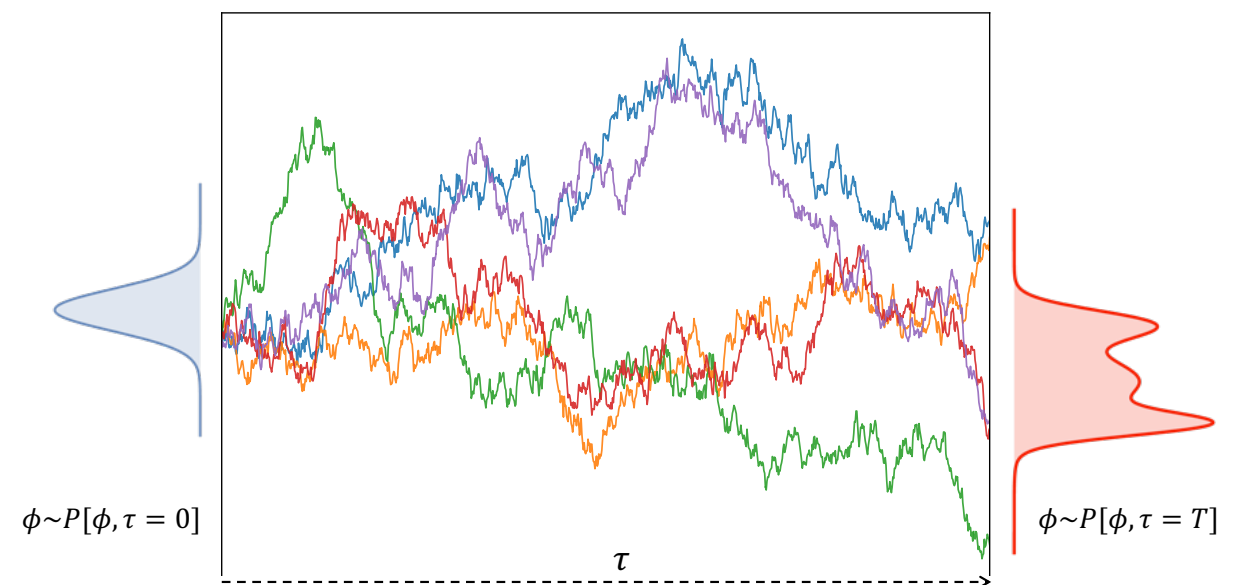
- ▶ **LW**, G Aarts, K Zhou, JHEP 05 (2024) 060 [[2309.17082](#)] [hep-lat]
- ▶ G Aarts, D Habibi, **LW**, K Zhou, Mach.Learn.Sci.Tech. 6 (2025) 2, 025004 and NeurIPS 2024 ML4PS [[2410.21212](#)] [hep-lat]
- ▶ G Aarts, K Fukushima, T Hatsuda, A Ipp, S Shi, **LW**, K Zhou, Nature Rev.Phys. 7 (2025) 3, 154-163, [[2501.05580](#)] [hep-lat]
- ▶ Q Zhu, G Aarts, W Wang, K Zhou, **LW**, JHEP03(2026)111 [[2502.05504](#)] [hep-lat] and NeurIPS 2024 ML4PS [[2410.19602](#)] [hep-lat]
- ▶ G Aarts, D Habibi, A Ipp, D, Müller, T Ranner, **LW**, W Wang, Q Zhu [[2601.19552](#)] [hep-lat]

Background and References



Outline

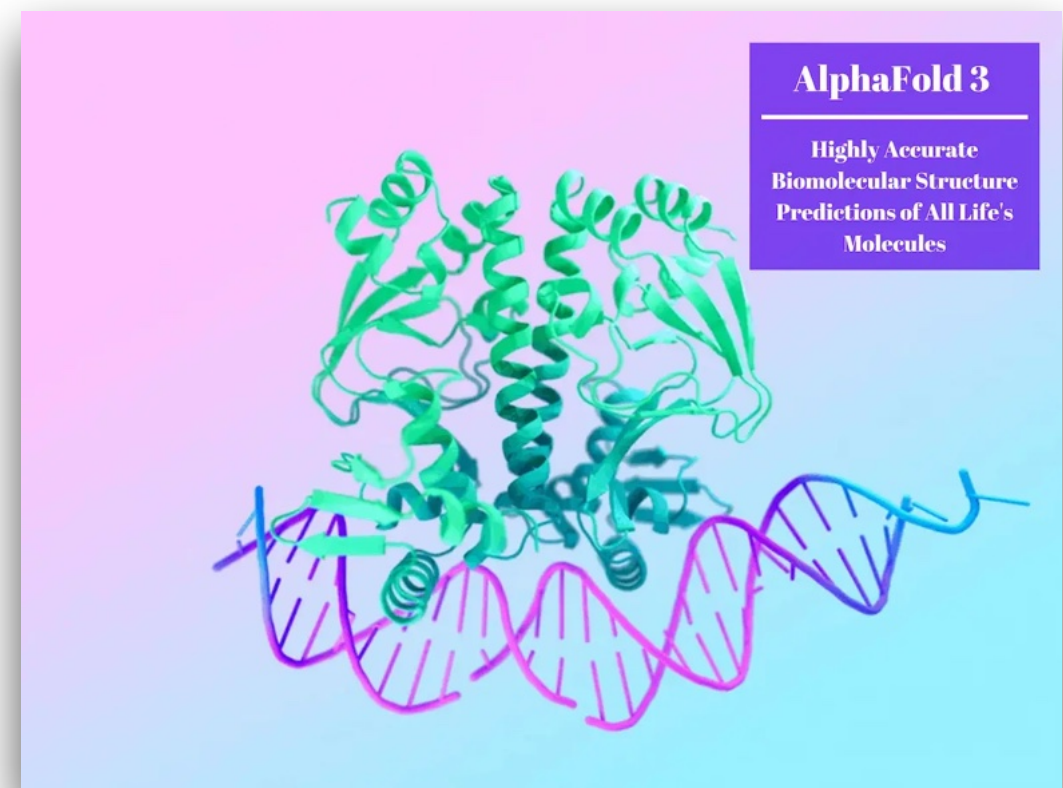
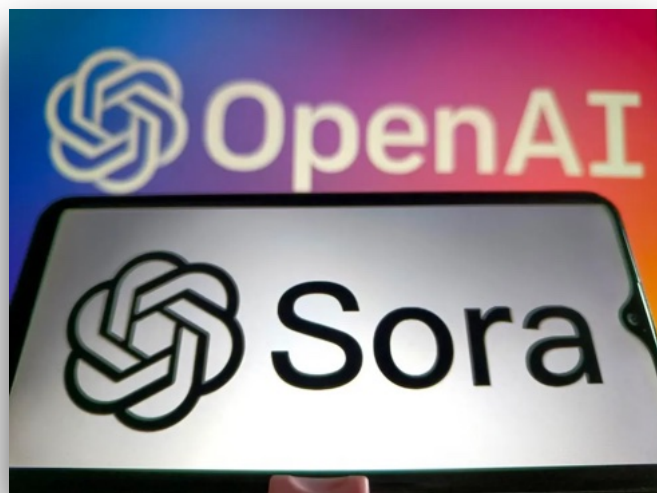
- What is Diffusion Model?
- Why DM?
- Physics of Diffusion Models
 - Stochastic Quantization
 - Effective Actions
- Generation of Lattice Fields
 - Scalar Fields
 - Gauge Fields
- Expandability, Exactness, Efficiency(3E)
- Summary and Outlook



What is Diffusion Model?

A Diffusion Model is a type of **generative model** used in machine learning to **create (or generate) data** — such as images, audio, or text — by **gradually transforming random noise into structured data**.

— GPT-5



Why Diffusion Models?

for Lattice simulations

Generative models

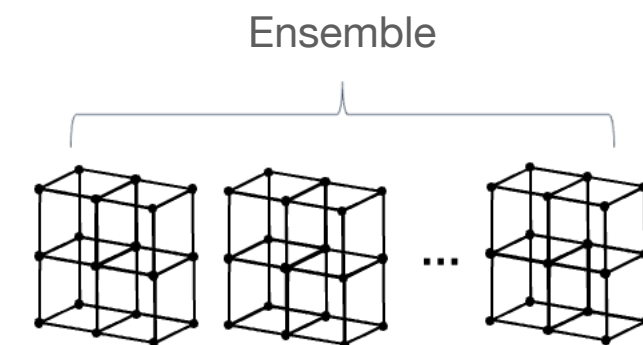
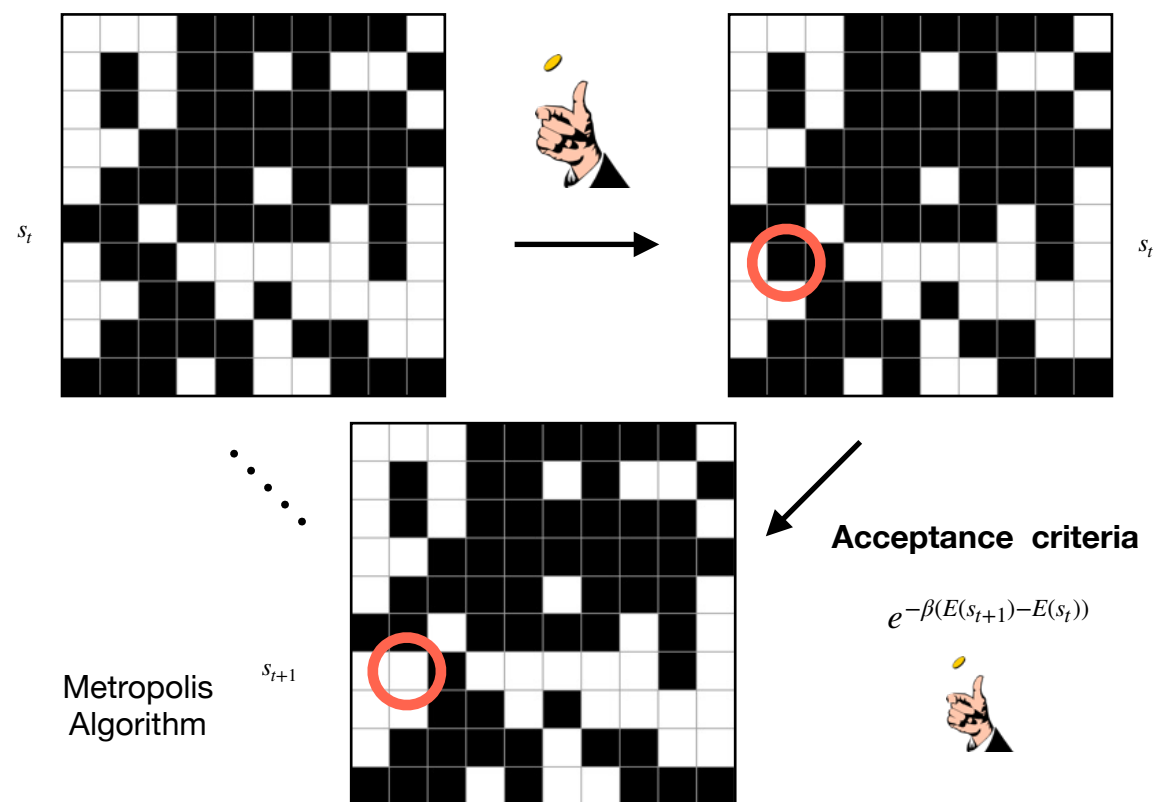
→ **underlying distributions** in data

Lattice simulations

→ **physical distributions**, sampling

$$p(\phi) = e^{-S(\phi)} / Z$$

$$\langle O \rangle \approx \frac{1}{N} \sum_i O_i$$



Universal and Accurate, but

- Local update, low-efficiency
- Critical Slowing Down

[U. Wolff, Nucl. Phys. B 17, 93 (1990)]

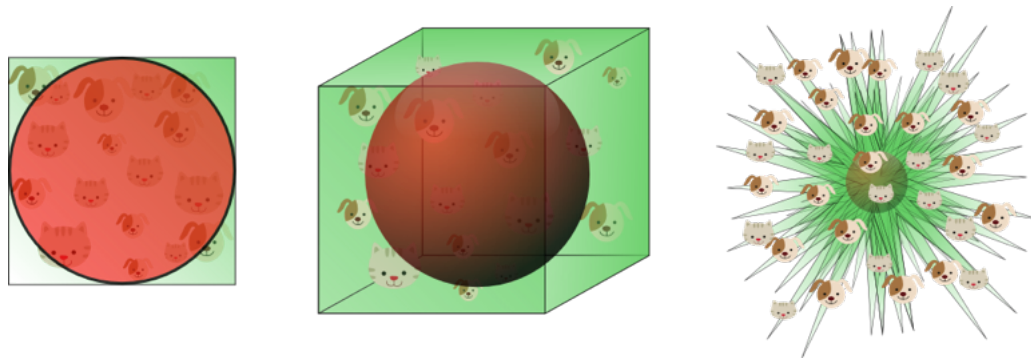
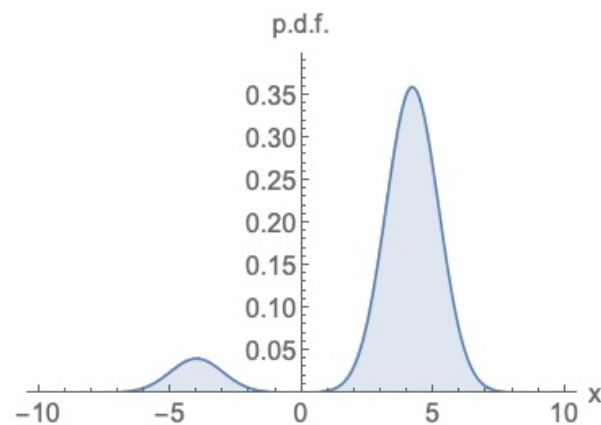
Why Diffusion Models?

for Lattice simulations

Generative models

$$q_{\theta}(\phi) \rightarrow p(\phi)$$

Physical distributions, sampling



Curse of Dimensionality

- **Implicit** Likelihood Estimation
(needs training data-set)

- VAEs and GANs

D. Giataganas, et al., New J. Phys. 24, 043040 (2022).

K. Zhou, et al., Phys. Rev. D 100, 011501 (2019).

J. M. Pawłowski and J. M. Urban, MLST 1, 045011 (2020).

J. Singh, et al., SciPost Phys. 11, 043 (2021).

- **Explicit** Likelihood Estimation

- Autoregressive models

D. Wu, et al., Phys. Rev. Lett. 122, 080602 (2019).

L. Wang, et al., CPL 39, 120502 (2022).

P. Białas, P. Korcyl, and T. Stebel, CPC 281, 108502 (2022).

- Flow-based models

M. Albergo, G. Kanwar, and P. Shanahan,

G. Kanwar, et al., Phys. Rev. Lett. 126,

K. A. Nicoli, et al., Phys. Rev. Lett. 126, 032001 (2021).

L. Del Debbio, et al., Phys. Rev. D 104, 094507 (2021).

M. Caselle, et al., J. High Energ. Phys. 2022, 15 (2022).

A. Singha, et al., Phys. Rev. D 107, 014512 (2023).

S. Chen, et al., Phys. Rev. D 107, 056001(2023).

R. Abbott, et al., Phys. Rev. D 109, 094514 (2024).

K. Cranmer, G. Kanwar, S. Racanière, D. J. Rezende, and P. E. Shanahan, Nat Rev Phys 1 (2023).

....

- **Score Matching [This Talk]**

**Gutej, Mathis, Fernando's
Talks on May 6**

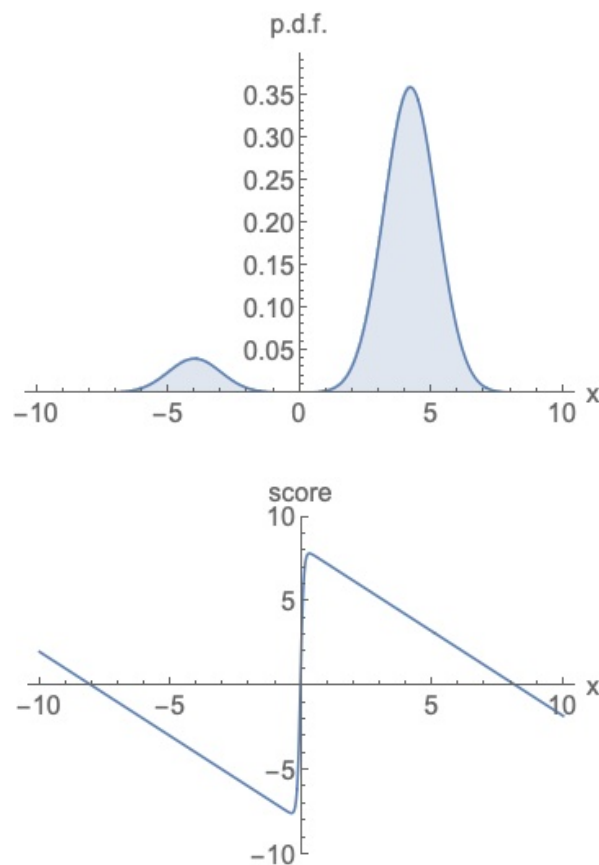
Why Diffusion Models?

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Generative models

$$q_{\theta}(\phi) \rightarrow p(\phi)$$

Physical distributions, sampling



$$p(\phi) = e^{-S(\phi)}/Z$$

$$q_{\theta}(\phi) = \frac{e^{-f_{\theta}(\phi)}}{Z_{\theta}}$$



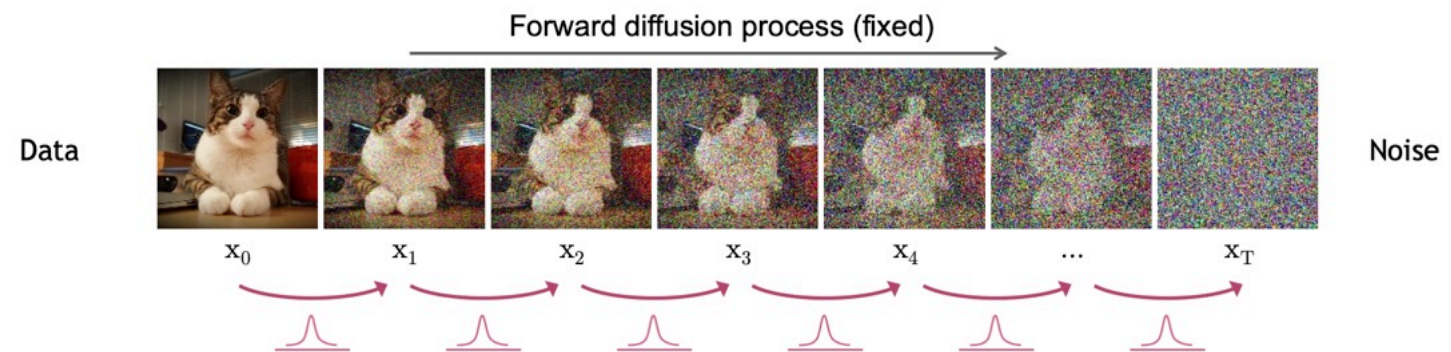
$$\nabla_{\phi} \log p(\phi)$$

$$s_{\theta}(\phi) \rightarrow \nabla_{\phi} \log p(\phi)$$

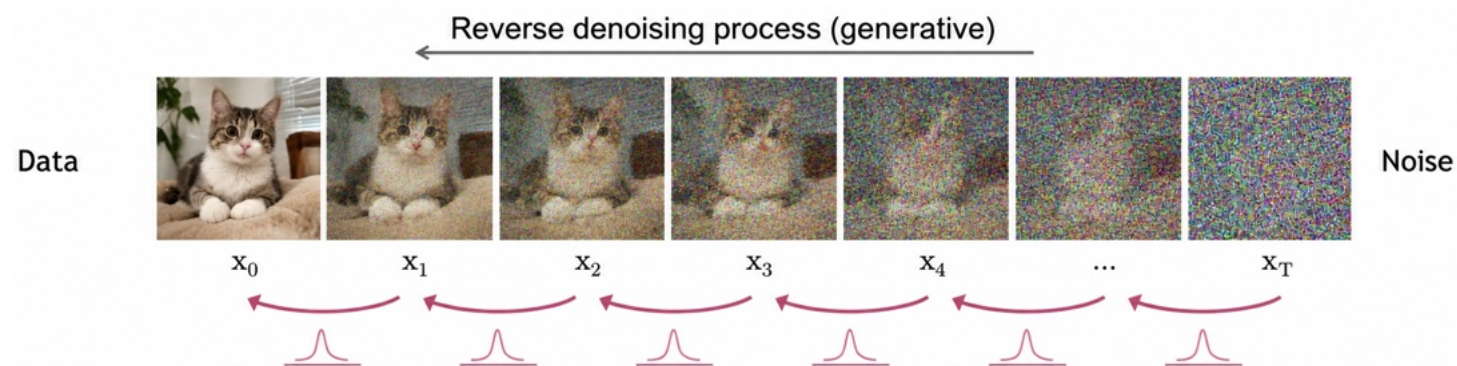
Diffusion Models

Practical Level

- ▶ **Forward** diffusion process **gradually adds noise** to input
- ▶ **Reverse** denoising process learns to **generate data by denoising**
- ▶ **Probabilistic models** to learn how to **denoise from a simple distribution to a target distribution**



Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS 2020



Diffusion Models

Forward diffusion

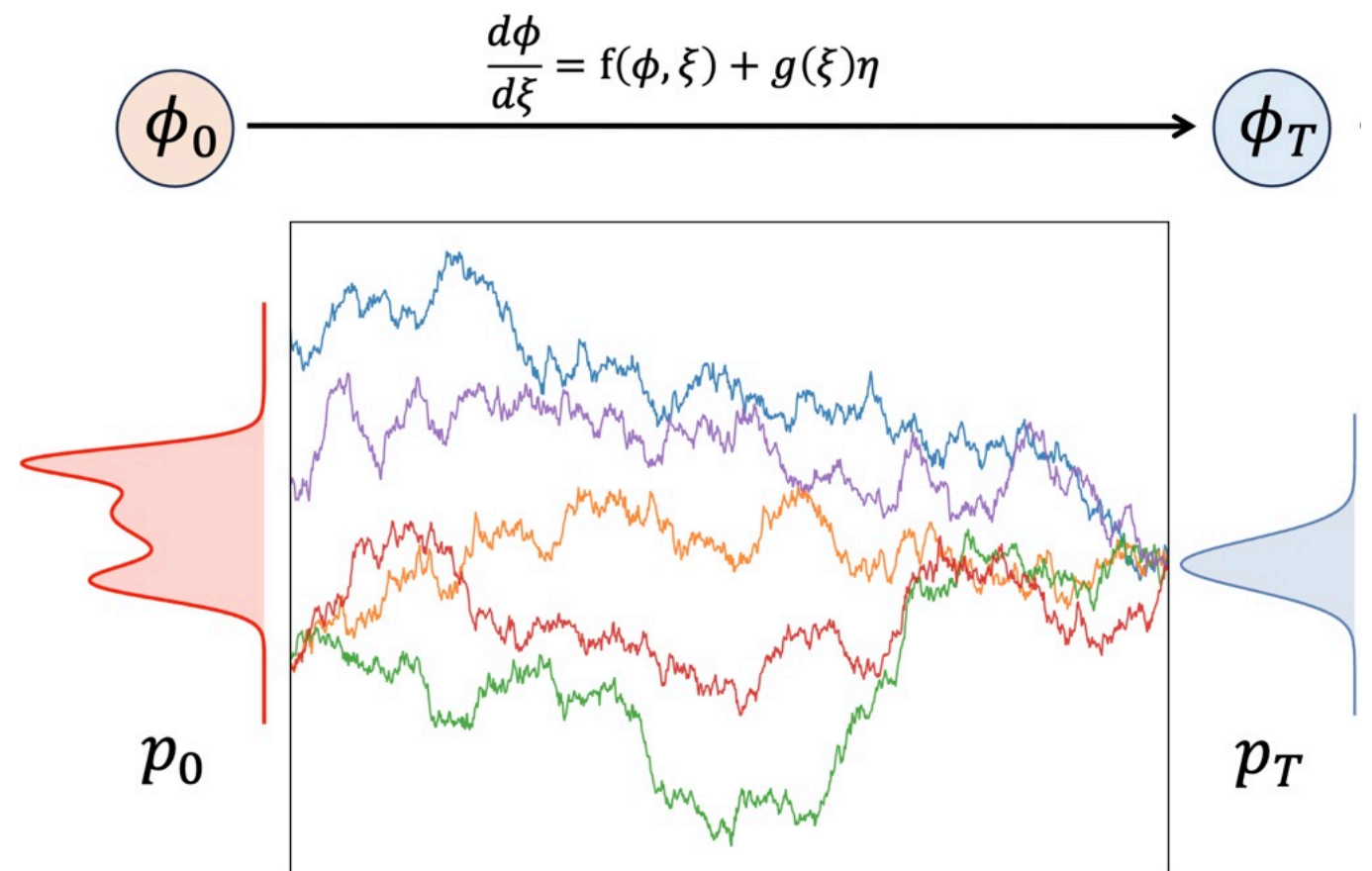
- Forward diffusion process is driven by a stochastic differential equation(SDE)

- Forward Diffusion SDE**

- Drift term**: pulls towards mode
- Diffusion term**: injects noise

$$\frac{d\phi}{d\xi} = \boxed{f(\phi, \xi)} + \boxed{g(\xi)\eta(\xi)}$$

- Reverse denoising process ?



Diffusion Models

Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, ICLR 2021

Anderson, in Stochastic Processes and their Applications, 1982

Reverse diffusion

Forward Diffusion SDE

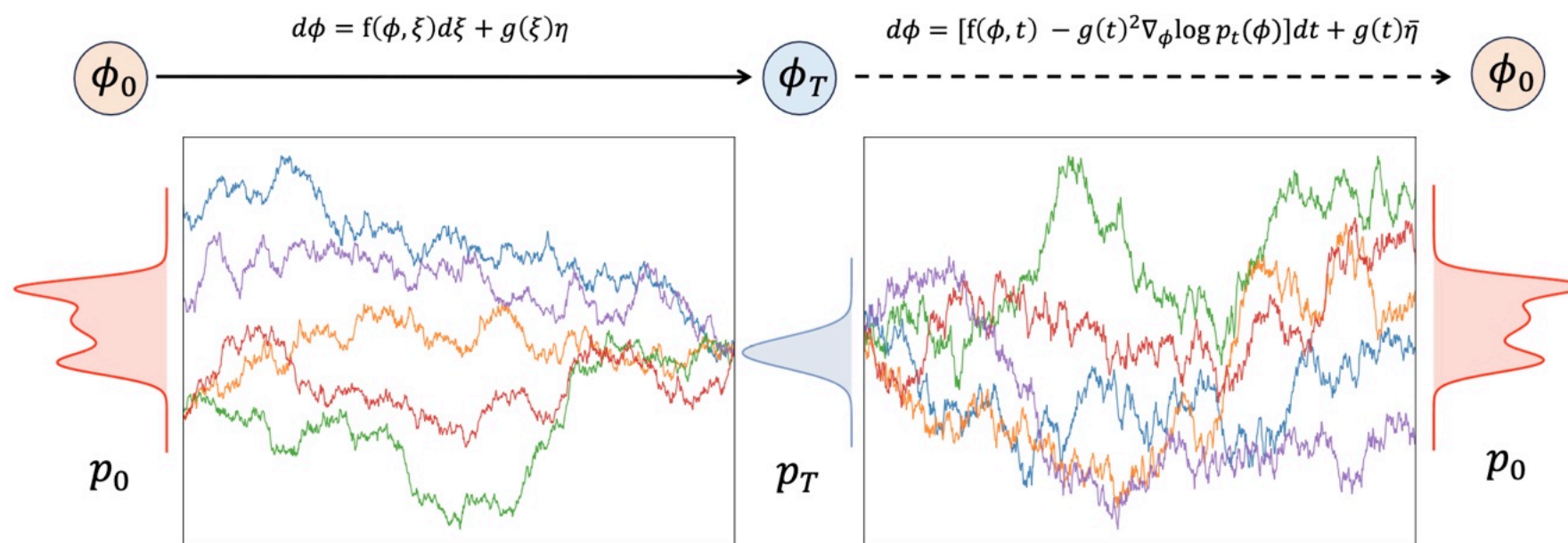
- Drift term: pulls towards mode
- Diffusion term: injects noise

$$\frac{d\phi}{d\xi} = f(\phi, \xi) + g(\xi)\eta(\xi)$$

Reverse Generative Diffusion SDE

- Drift term is adjusted with a “Score Function”
- But how to get the score function ?

$$\frac{d\phi}{dt} = \left[f(\phi, t) - g^2(t) \nabla_{\phi} \log p_t(\phi) \right] + g(t)\bar{\eta}(t)$$



Diffusion Models

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Anderson, in Stochastic Processes and their Applications, 1982

Reverse diffusion

Forward Diffusion SDE

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Reverse Generative Diffusion SDE

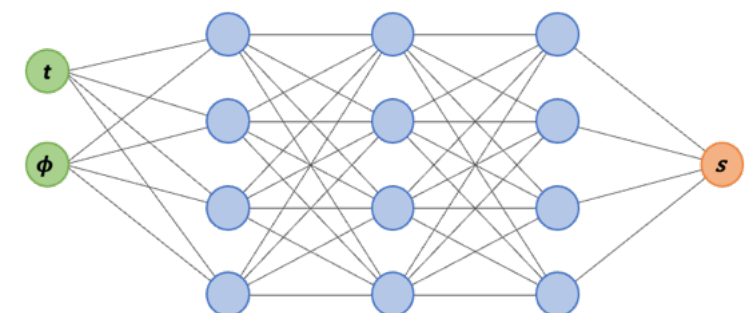
- Drift term is adjusted with a “Score Function”
- But how to get the score function ?

$$\frac{d\phi}{dt} = \left[f(\phi, t) - g^2(t) \nabla_{\phi} \log p_t(\phi) \right] + g(t)\bar{\eta}(t)$$

$$\frac{d\phi}{dt} = \left[f(\phi, t) - g^2(t) \mathbf{s}_{\hat{\theta}}(\phi, t) \right] + g(t)\bar{\eta}(t)$$

Universal Approximation Theorem (1989,1991)

A feed-forward network with a single hidden layer containing a finite number of neurons can approximate arbitrary continuous functions.



Stochastic Quantization

Parisi G. and Wu Y. S., Sci. China, A **24**, ASITP-80-004 (1980).

P. H. Damgaard and H. Hüffel, Stochastic Quantization, Phys. Rept. 152, 227 (1987).

M. Namiki, Basic Ideas of Stochastic Quantization, PTPS 111, 1 (1993).

$$\frac{\partial \phi(x, \tau)}{\partial \tau} = - \frac{\delta S_E[\phi]}{\delta \phi(x, \tau)} + \eta(x, \tau)$$

$$\langle \eta(x, \tau) \rangle = 0, \quad \langle \eta(x, \tau) \eta(x', \tau') \rangle = 2\alpha \delta(x - x') \delta(\tau - \tau')$$

τ : fictitious time, α : diffusion constant

► Fokker-Planck equation

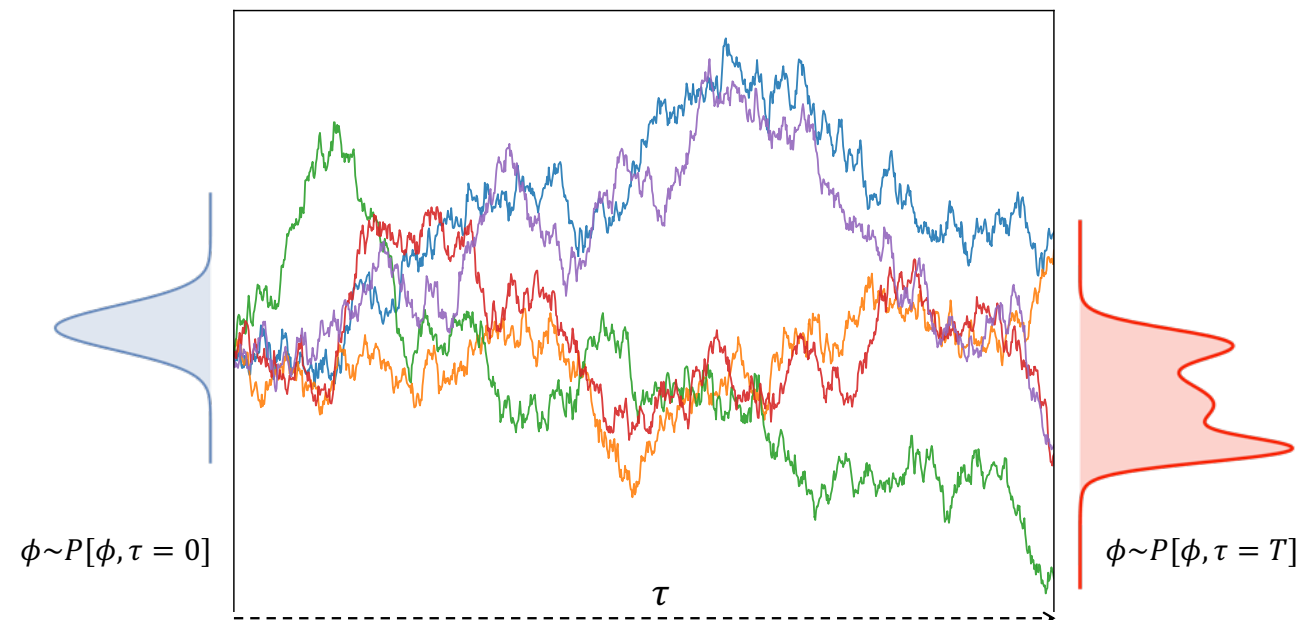
$$\frac{\partial P[\phi, \tau]}{\partial \tau} = \alpha \int d^n x \left\{ \frac{\delta}{\delta \phi} \left(\frac{\delta}{\delta \phi} + \frac{\delta S_E}{\delta \phi} \right) \right\} P[\phi, \tau]$$

Equilibrium solution (long-time limit),

$$P_{\text{eq}}[\phi] \propto e^{-\frac{1}{\alpha} S_E[\phi]}$$

► Set the diffusion constant as $\alpha = \hbar$

$$P_{\text{eq}}[\phi] \sim e^{-\frac{1}{\hbar} S_E[\phi]} = P_{\text{quantum}}[\phi]$$



**Thermal equilibrium limit
→ Quantum distribution**

- 1. No need gauge-fixing!**
- 2. Can handle fermionic fields naturally**
→ **(Complex Langevin method)**

Recent review: Gert Aarts, Dénes Sexty, [2604.24290](#) [hep-lat]

DM as SQ

► Diffusion models(Reverse SDE):

$$\frac{d\phi}{dt} = -g(t)^2 \nabla_{\phi} \log p_t(\phi) + g(t) \bar{\eta}$$

► Define: $\tau \equiv T - t (d\tau \equiv -dt)$

$$\frac{d\phi}{d\tau} = g_{\tau}^2 \nabla_{\phi} \log q_{\tau}(\phi) + g_{\tau} \bar{\eta}$$

$$\phi(\tau_{n+1}) = \phi(\tau_n) + g_{\tau}^2 \nabla_{\phi} \log q_{\tau_n}[\phi(\tau_n)] \Delta\tau + g_{\tau} \sqrt{\Delta\tau} \bar{\eta}(\tau_n)$$

introducing **Noise scale**: $\langle \bar{\eta}^2 \rangle \equiv 2\bar{\alpha}$, **time scale**: $g_{\tau}^2 \Delta\tau$

► FP equation

$$\frac{\partial p_{\tau}(\phi)}{\partial \tau} = \int d^n x \left\{ g_{\tau}^2 \bar{\alpha} \frac{\delta}{\delta \phi} \left(\frac{\delta}{\delta \phi} + \frac{1}{\bar{\alpha}} \nabla_{\phi} S_{\mathbf{DM}} \right) \right\} p_{\tau}(\phi)$$

$$\nabla_{\phi} S_{\mathbf{DM}} \equiv -\nabla_{\phi} \log q_{\tau}(\phi)$$

$$p_{eq}(\phi) \propto e^{-\frac{S_{DM}}{\bar{\alpha}}}$$

$$p_{\tau=T}(\phi) \rightarrow P[\phi, T]$$

$$O(\bar{\alpha}) \sim O(\hbar) \sim 1$$

The reverse mode of
a well-trained diffusion model
at $\tau \rightarrow T$ serves as
the stochastic quantization
for the input

SQ and DM

Aspect	Stochastic Quantisation (SQ)	Diffusion Models (DM)
Drift term	Fixed, determined from known action	Time-dependent, learned from data
Noise variance	Constant (can be generalised using kernels)	Time-dependent
Dynamics	Thermalisation followed by long-term equilibrium evolution	Evolution between $0 \leq \tau \leq T = 1$, many short runs
Learning objective	Based on known physical action $S(\phi)$	Learn probability evolution $P(\phi; \tau)$ from data
Equation	$\frac{\partial \phi(x, \tau)}{\partial \tau} = -\nabla_{\phi} S(\phi) + \sqrt{2} \eta(x, \tau)$	$\frac{\partial \phi(x, \tau)}{\partial \tau} = g^2(\tau) \nabla_{\phi} \log P(\phi; \tau) + g(\tau) \eta(x, \tau)$

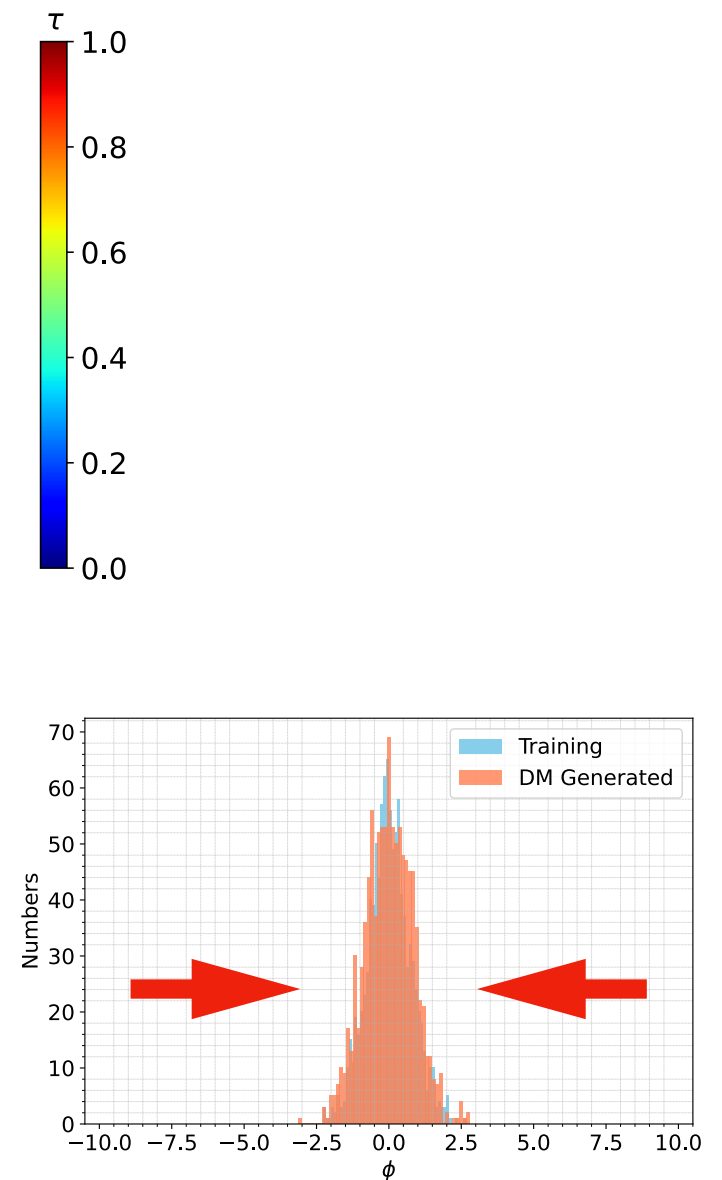
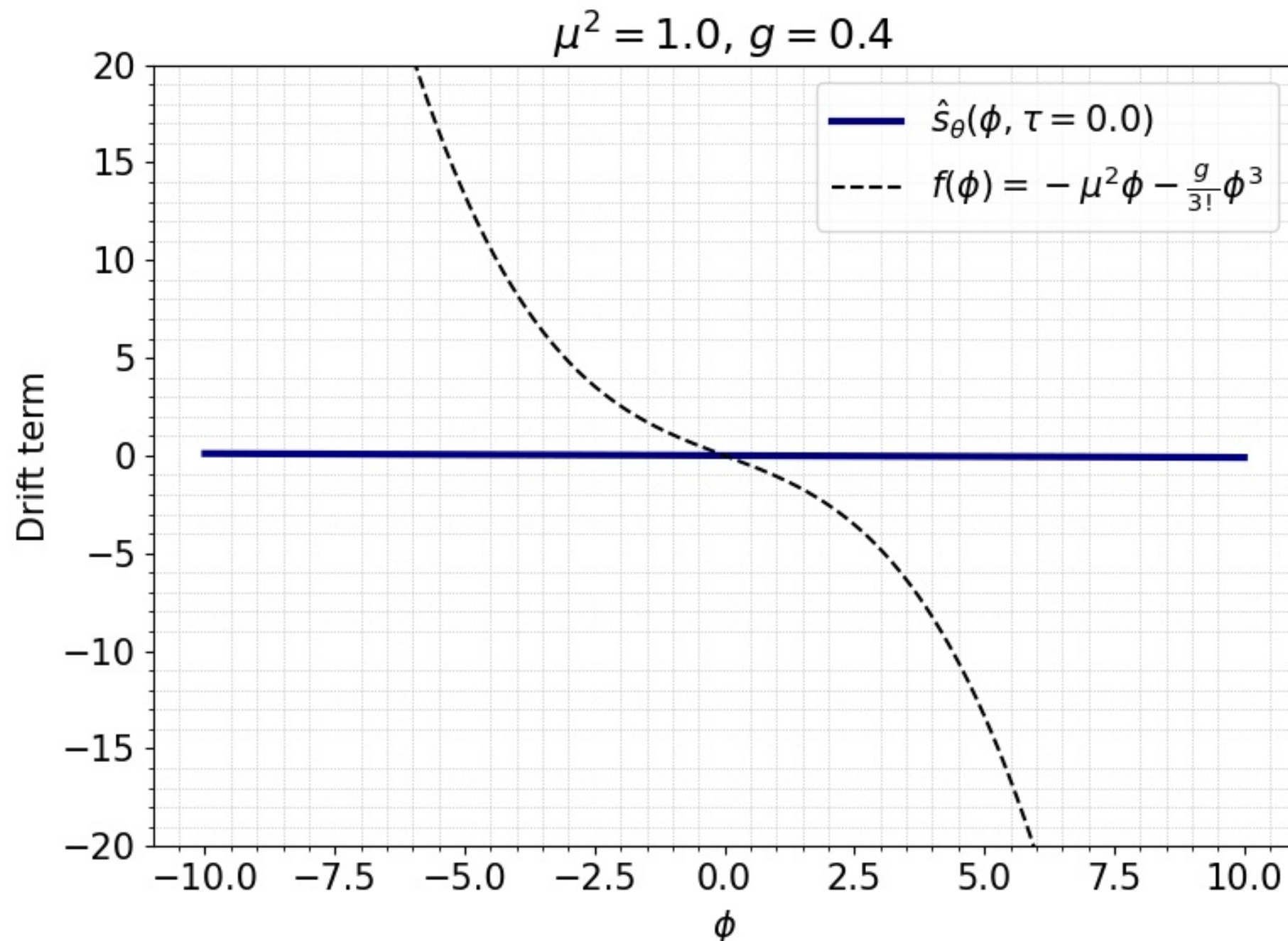
The reverse mode of
a well-trained diffusion model at $\tau \rightarrow T$ serves as
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for the input

Effective Action

Score function

$$S(\phi) = \frac{\mu^2}{2}\phi^2 + \frac{g}{4!}\phi^4,$$
$$f(\phi) = -\mu^2\phi - \frac{g}{3!}\phi^3$$

**0+0 dimensional FT
(1 variable)**

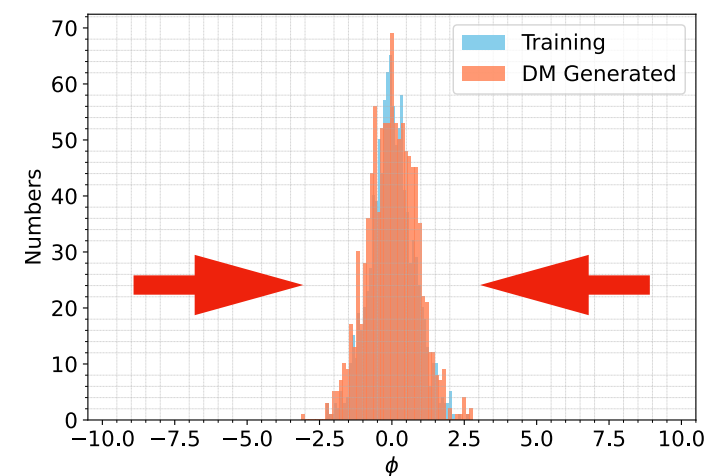
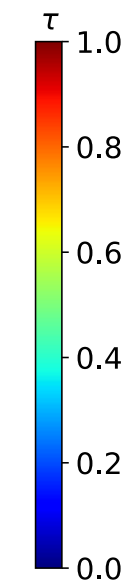
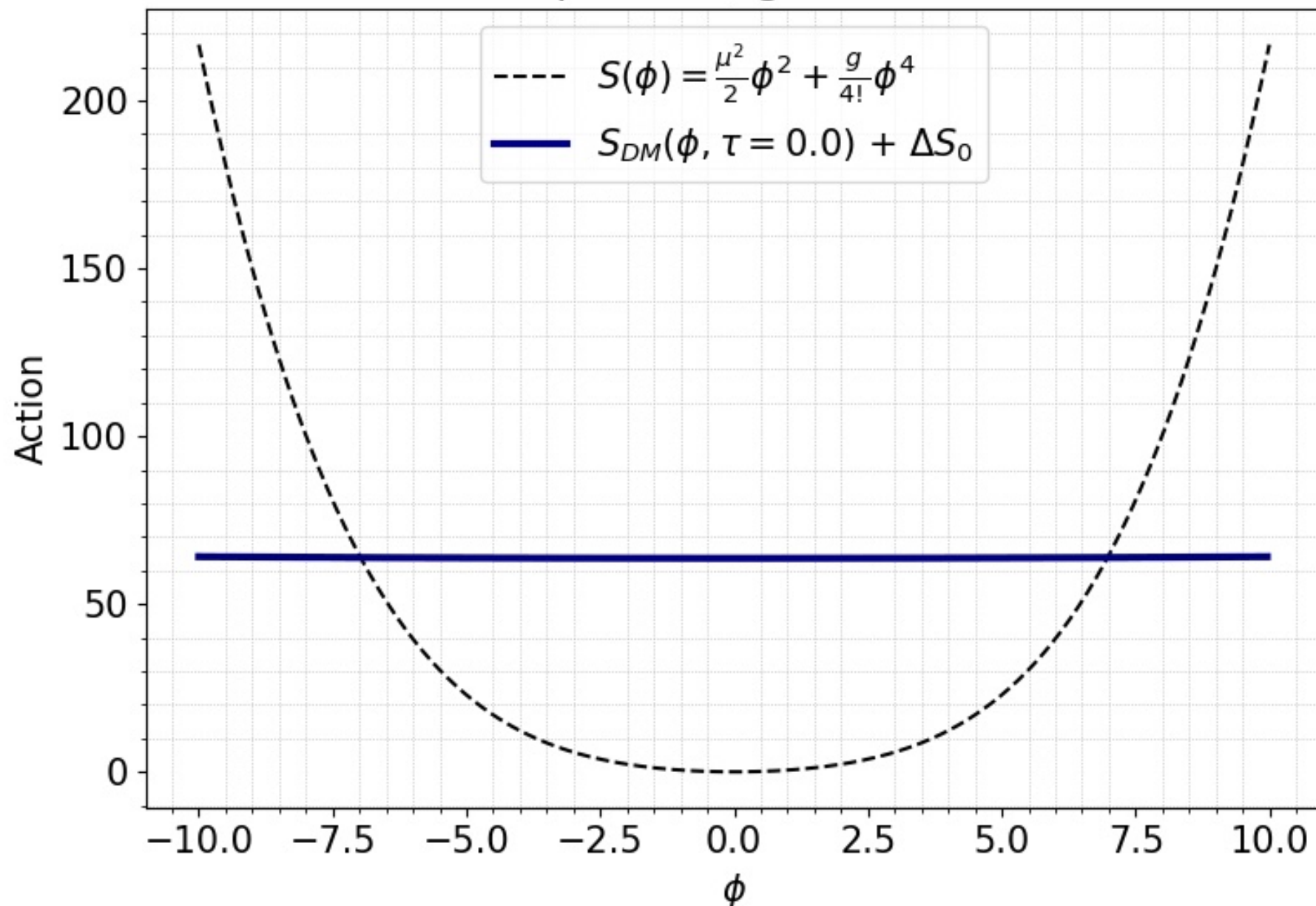


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**0+0 dimensional FT
(1 variable)**

$$\mu^2 = 1.0, g = 0.4$$



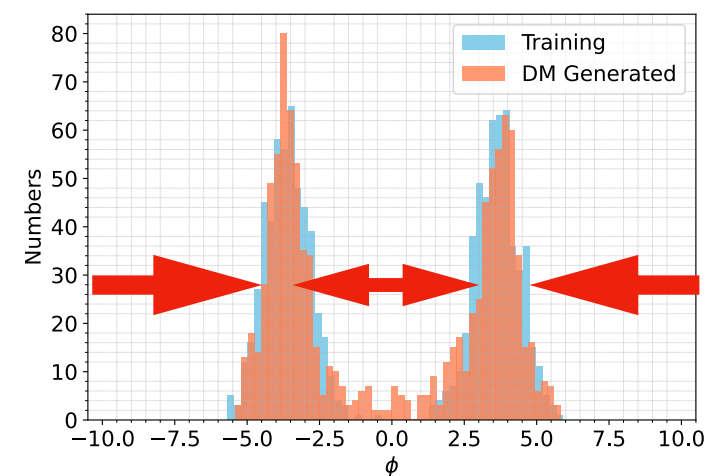
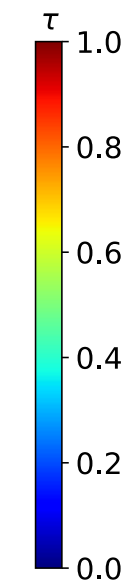
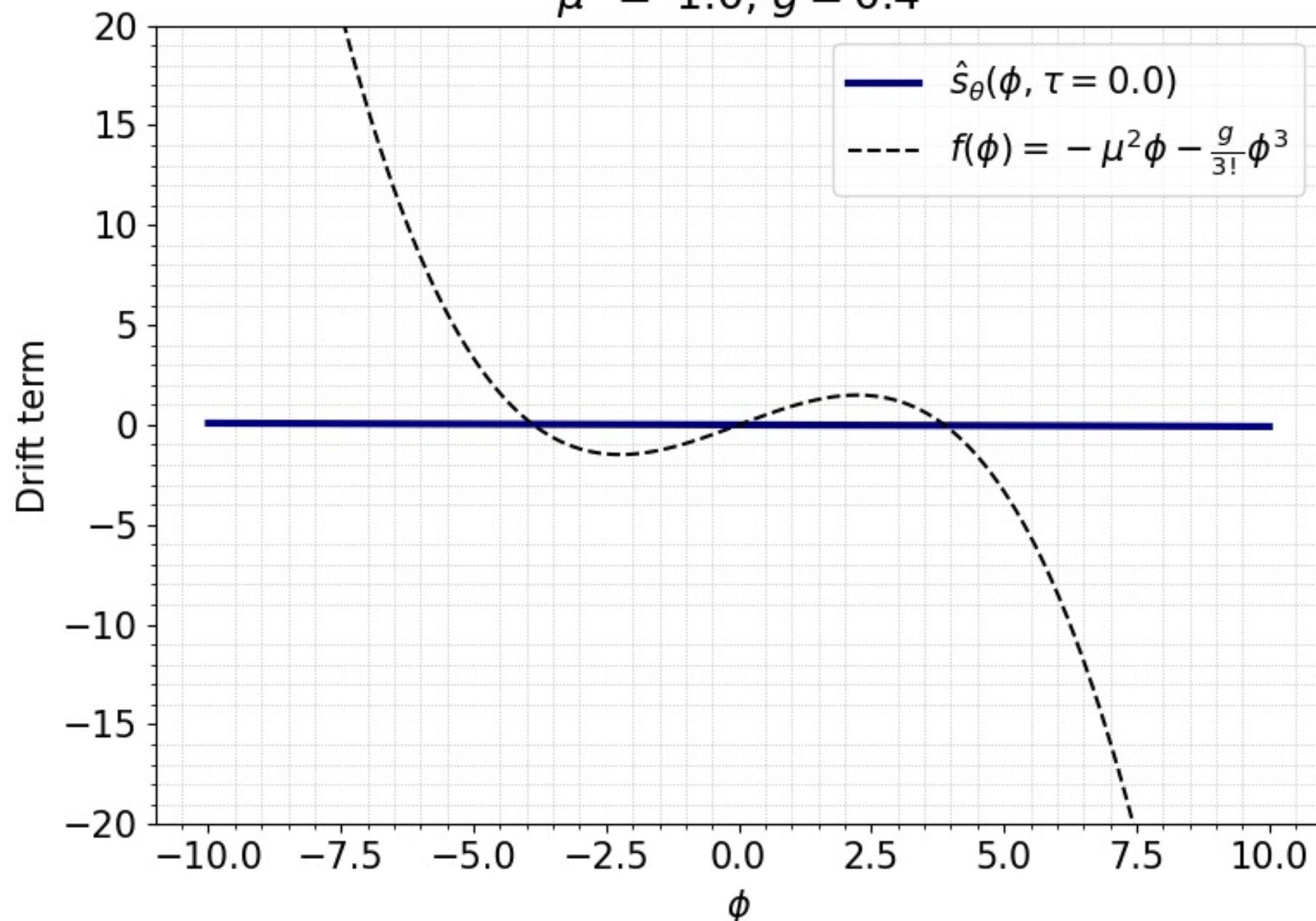
Effective Action

Score function

$$S(\phi) = \frac{\mu^2}{2}\phi^2 + \frac{g}{4!}\phi^4,$$
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**0+0 dimensional FT
(1 variable)**

$$\mu^2 = -1.0, g = 0.4$$

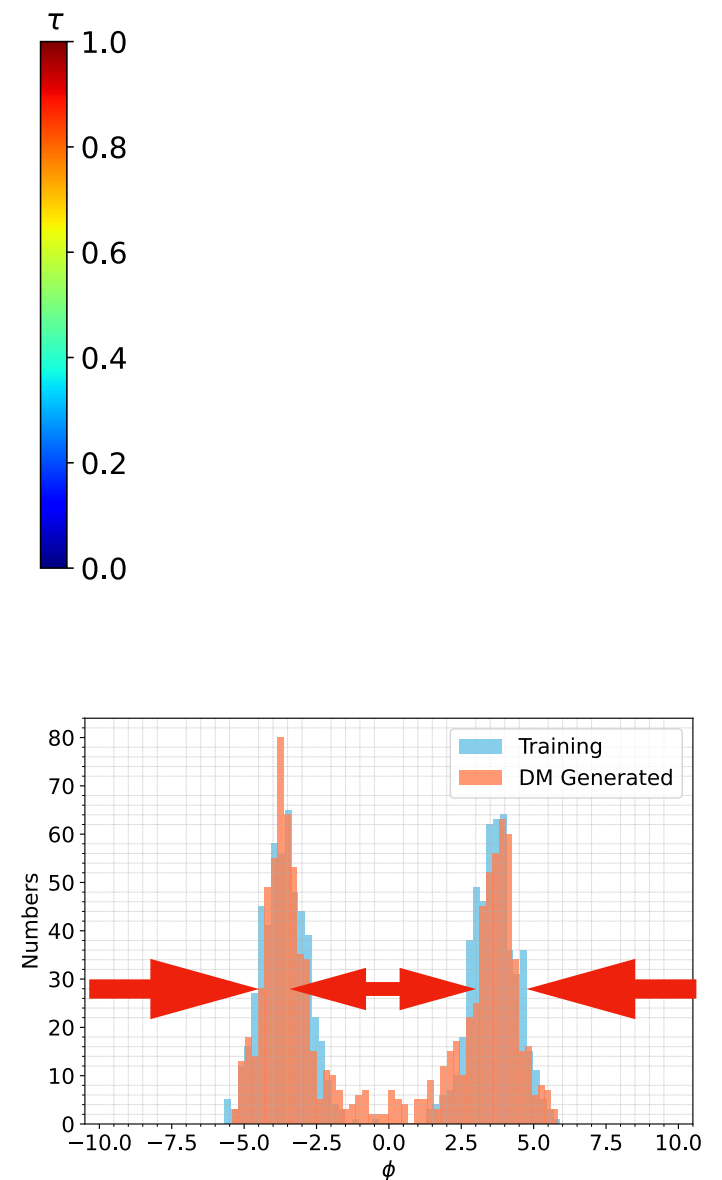
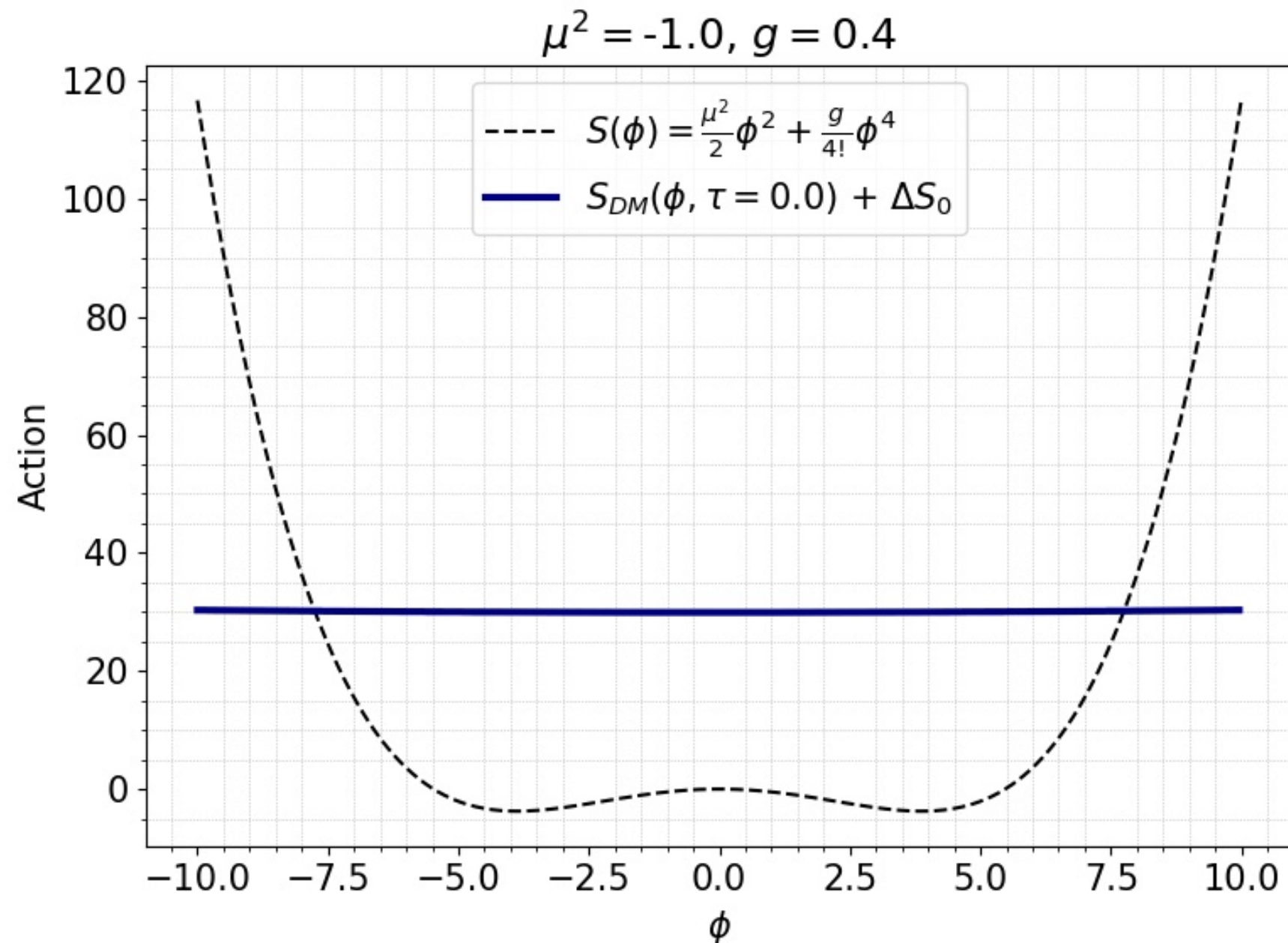


Effective Action

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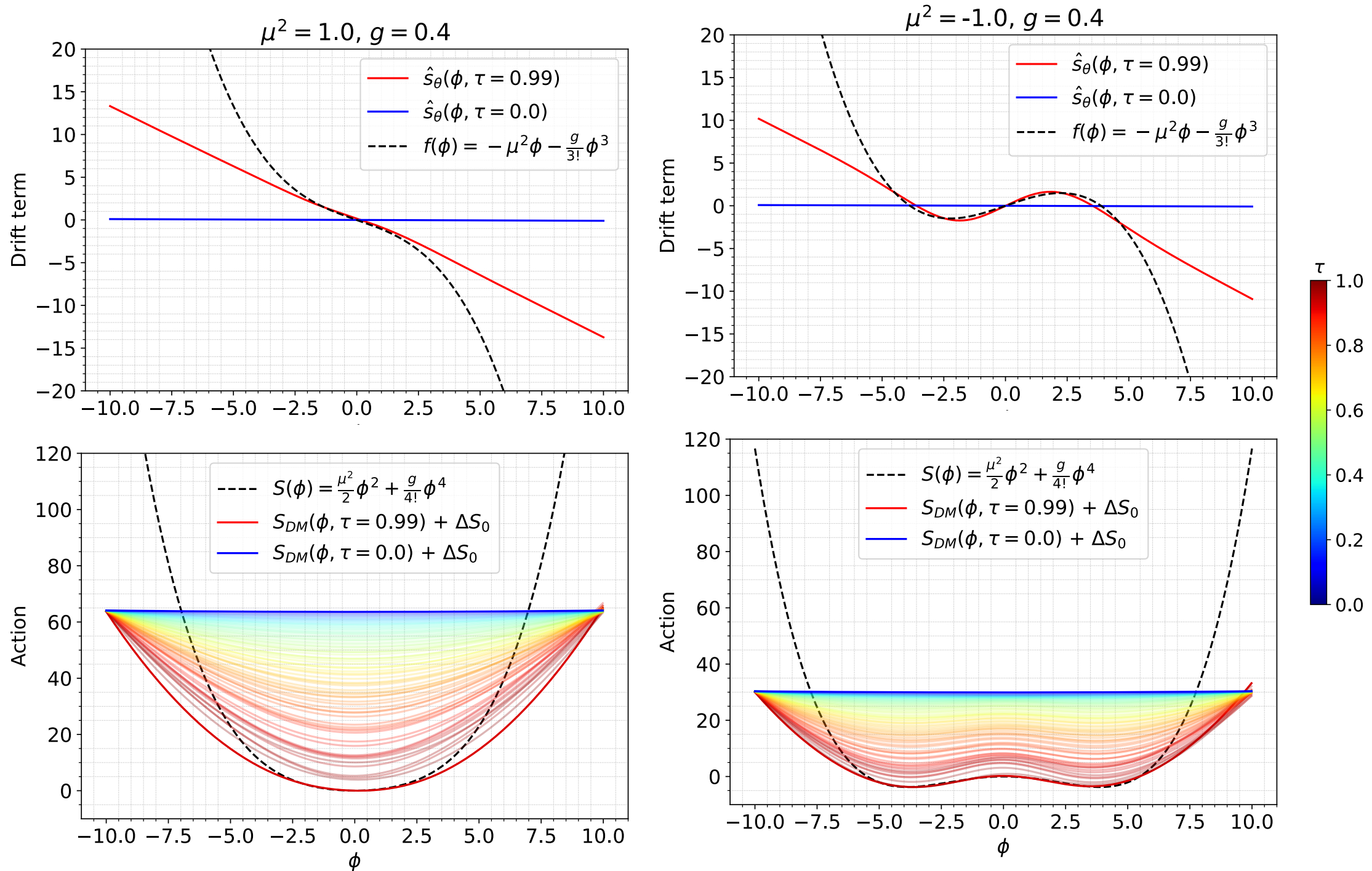
**0+0 dimensional FT
(1 variable)**



Effective Action

Flow

$$\frac{\partial S_{\mathbf{DM}}(\phi, \tau)}{\partial \tau} = \frac{\partial}{\partial \tau} \int^{\phi} \hat{s}_{\theta}(\tilde{\phi}, \tau) d\tilde{\phi}$$



Lattice Field Generations

What I will (or not) introduce

<https://github.com/lingxiao-mlphys/DMasSQ>
<https://github.com/zzzqt/DM4U1>
<https://dm-qft.github.io/homepage>

What I will introduce

- ▶ Numerical results for lattice **scalar** field in 2-dim and 3-dim, and **gauge** field simulations in 2-dim and 4-dim
- ▶ **Expandability** of diffusion models for different lattice **sizes** and **couplings**
- ▶ **Exactness** of generations → **Metroplis** algorithms
- ▶ **Efficiency** evaluations on acceptance rates and autocorrelation times

What I will not introduce, but can be found in Refs and GitHub pages

- ▶ Predetermined (Forward) Diffusion Schemes
Variance Expanding (e.g., Score-Based DM), **Variance Preserving** (e.g., DDPM), etc.
- ▶ Training Objective (Loss Function) and Set-Ups
Score-matching, $\mathcal{L}_\theta = \sum_{i=1}^N \sigma_i^2 \mathbb{E}_{p_0(\phi_0)} \mathbb{E}_{p_i(\phi_i|\phi_0)} \left[\left\| \mathbf{s}_\theta(\phi_i, \xi) - \nabla_{\phi_i} \log p_i(\phi_i|\phi_0) \right\|_2^2 \right]$
- ▶ Details of Neural Network for Score Functions
CNNs, but modified for different field theories, e.g., L-CNN for gauge fields.

Lattice Scalar Fields

Scalar ϕ^4 field

- **Euclidean action** of bare fields

$$S_E = \int d^d x d\tau \left(\frac{1}{2} (\partial^2 \phi_0^2 + m^2 \phi_0^2) - \frac{\lambda_0}{4!} \phi_0^4 \right)$$

- Action on discrete lattice

$$S_E = \sum_x a^d \left[\sum_{\mu=1}^d \frac{(\phi_0(x + a\hat{\mu}) - \phi_0(x))^2}{a^2} + \frac{m_0^2}{2} \phi_0^2 + \frac{\lambda_0}{4!} \phi_0^4 \right]$$

- Dimensionless form

$$S_E = \sum_x \left[-2\kappa \sum_{\mu=1}^d \phi(x) \phi(x + \hat{\mu}) + (1 - 2\lambda) \phi(x)^2 + \lambda \phi(x)^4 \right]$$

$$a^{\frac{d-2}{2}} \phi_0 = (2\kappa)^{1/2} \phi$$

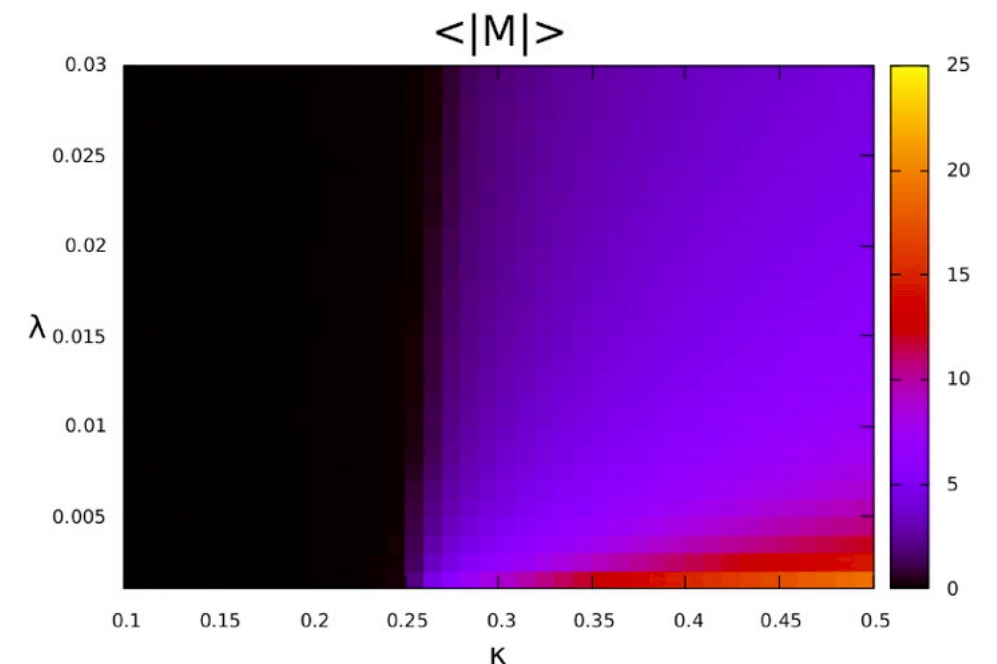
$$(am_0)^2 = \frac{1 - 2\lambda}{\kappa} - 2d, \quad a^{-d+4} \lambda_0 = \frac{6\lambda}{\kappa^2}$$

- Hopping parameter κ , Coupling constant λ

Phase Transition

$\mathbb{Z}_2$ symmetry spontaneously broken above the critical point

Order parameter: magnetization



Phase diagram at $d = 2$ @Julian Urban

Lattice Scalar Fields

LW, G Aarts, K Zhou, JHEP(2024)

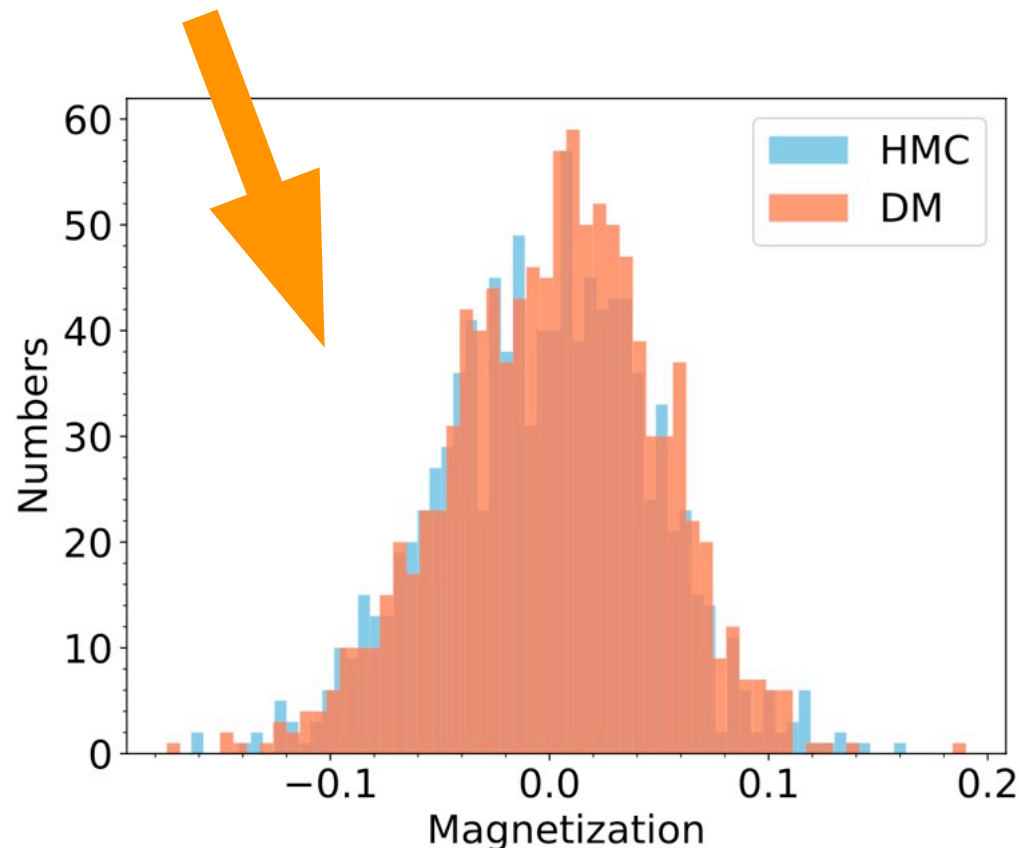
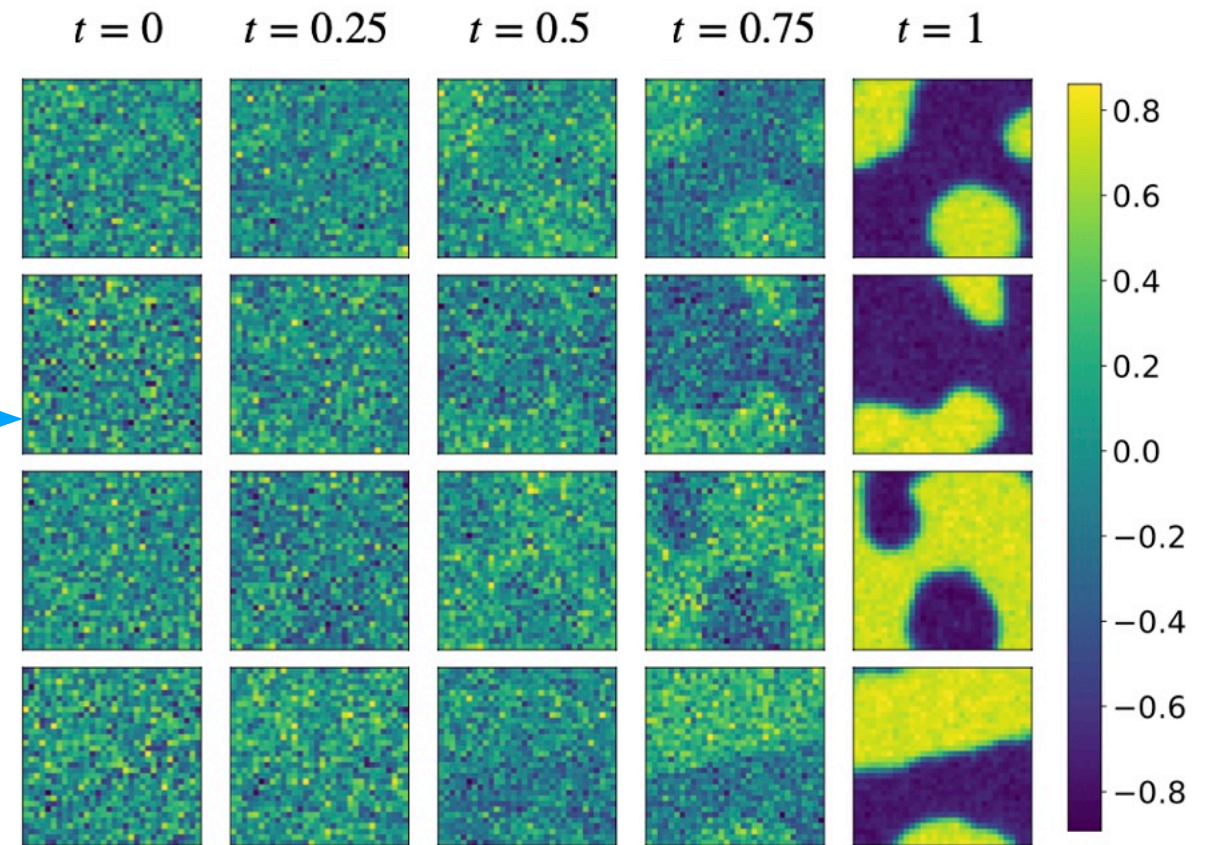
Different phases

Diffusion models

- Variance Expanding with $T = 1.0, \sigma = 25$

Data generation

- 2-d 32×32 lattice; Hamiltonian Monte Carlo(HMC); 5120 configurations for training.
- Broken phase: $\kappa = 1.0, \lambda = 0.022$
- Symmetric phase: $\kappa = 0.21, \lambda = 0.022$



data-set	$\langle M \rangle$	χ_2	U_L
Training(HMC)	0.0012 ± 0.0007	2.5160 ± 0.0457	0.1042 ± 0.0367
Testing(HMC)	0.0018 ± 0.0015	2.4463 ± 0.1099	-0.0198 ± 0.1035
Generated(DM)	0.0017 ± 0.0015	2.4227 ± 0.1035	0.0484 ± 0.0959

Lattice Scalar Fields

Criticality

▸ Data-preparation

▸ Wolff cluster flips: $\phi_x \rightarrow -\phi_x$

▸ Fourier-accelerated(**FA**)-**HMC**

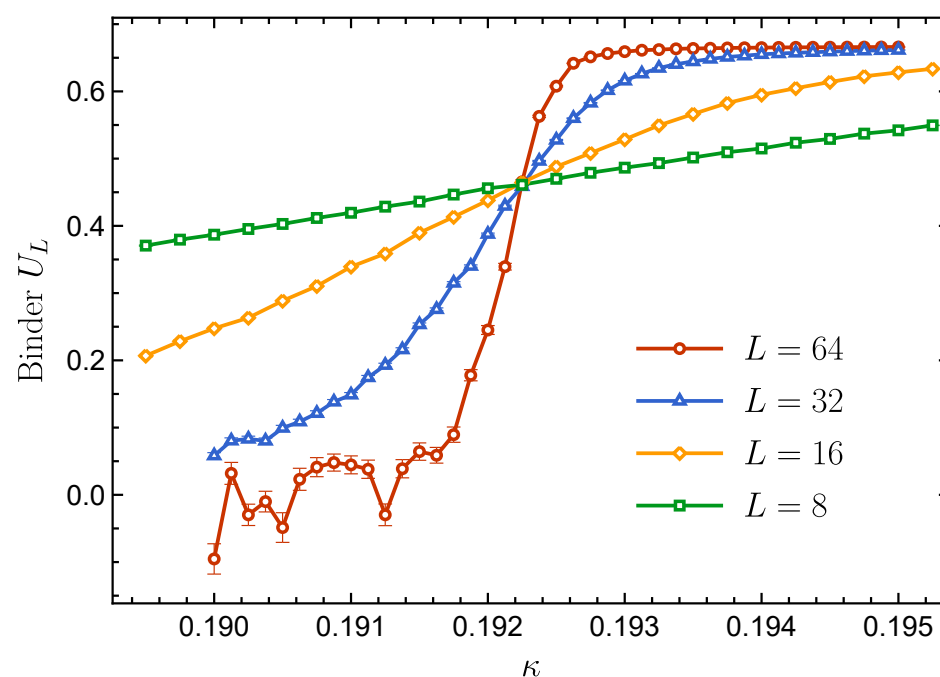
$$M(\hat{p}) = \hat{p}^2 + m_{\text{eff}}^2, \quad \hat{p}^2 = \sum_{\mu=1}^D 4 \sin^2\left(\frac{\pi k_\mu}{L}\right)$$

▸ 2D Critical Point

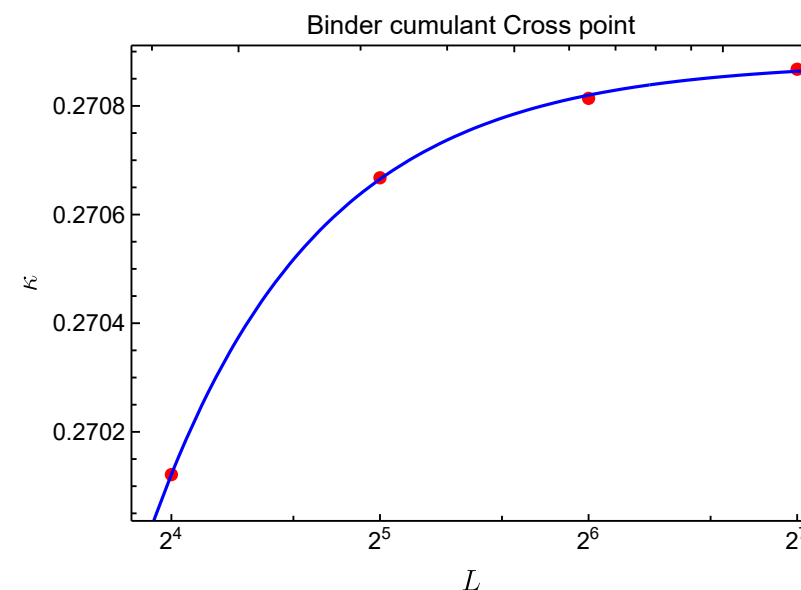
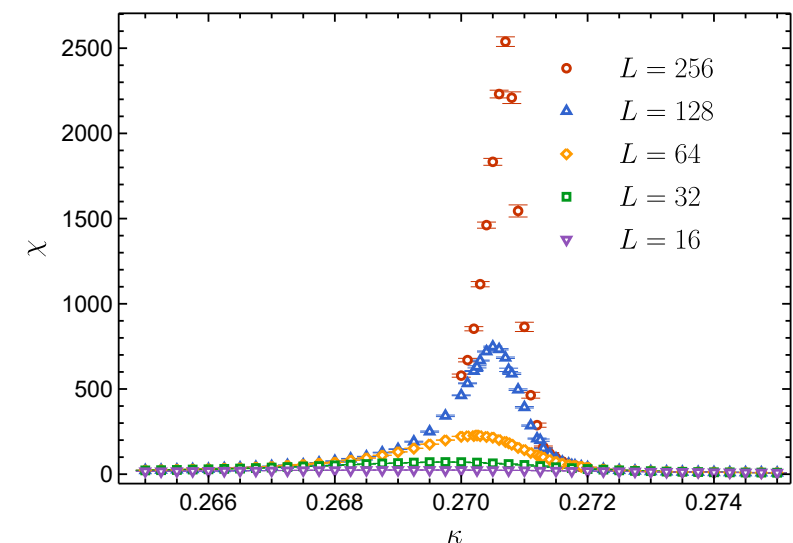
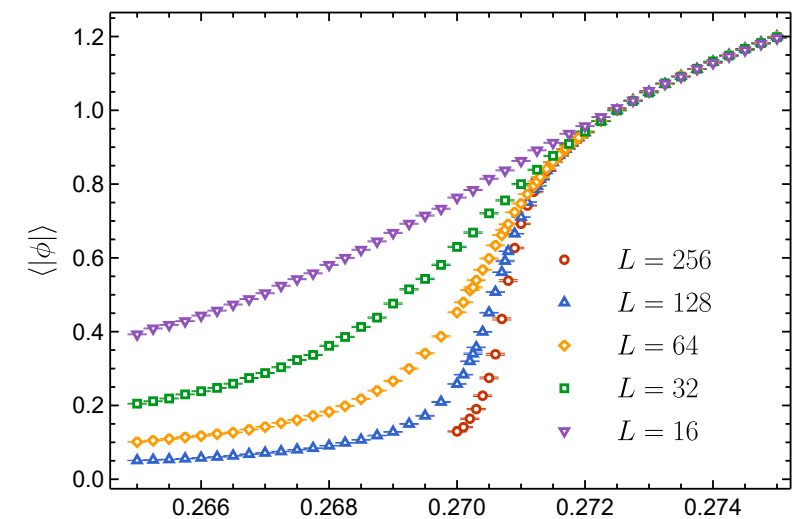
$$\kappa_c(\lambda = 0.022) \approx 0.27088$$

▸ 3D Critical Point

$$\kappa_c(\lambda = 0.9) \approx 0.19225$$



Y Tang, LW, etc., arXiv:2606.xxx



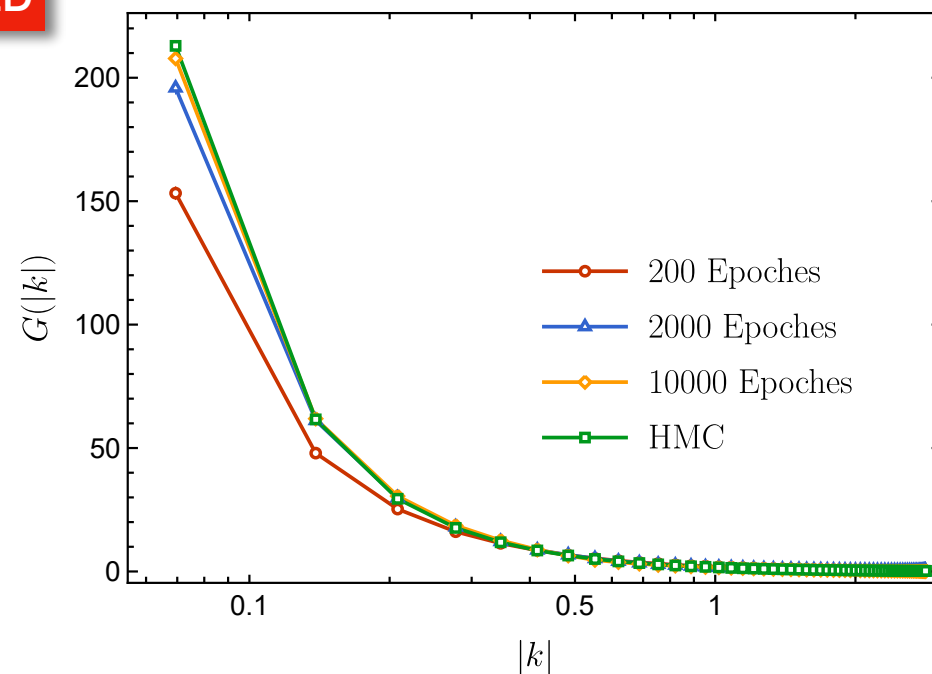
Lattice Scalar Fields

Y Tang, LW, etc., arXiv:2606.xxx

Criticality

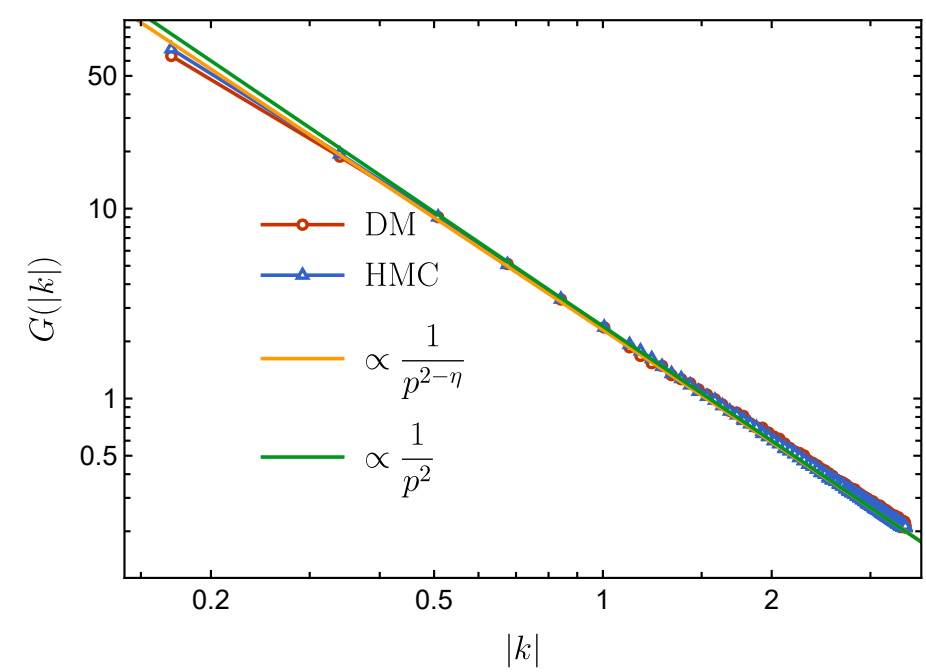
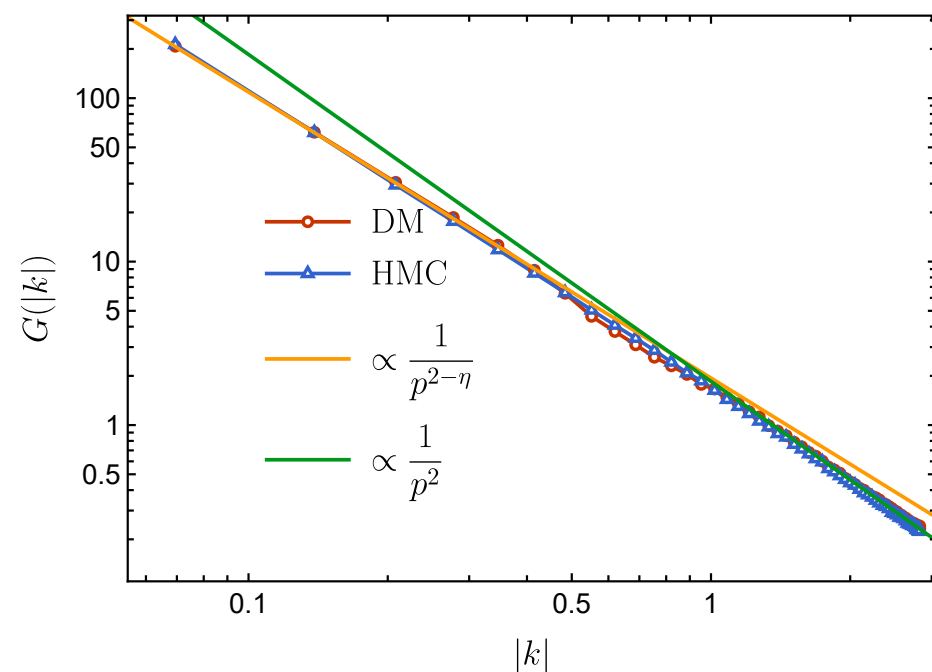
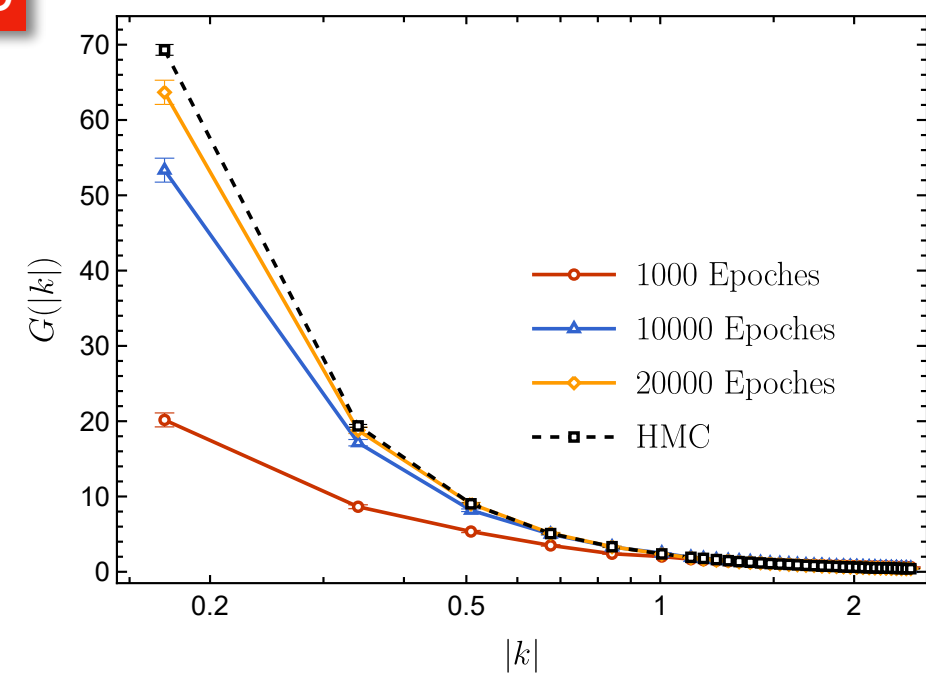
2D

$L = 128, \lambda = 0.022, \kappa \approx \kappa_c = 0.2705$



3D

$L = 64, \lambda = 0.9, \kappa \approx \kappa_c \approx 0.1923$



Lattice Gauge Fields

Q Zhu, G Aarts, W Wang, K Zhou, **LW**, JHEP(2026)

2D U(1)

- **Euclidean action** on lattice

$$S = \beta \sum_{\square} \left(1 - \text{Re}(U_{\square}) \right)$$

- Defined on links as **plaquette**

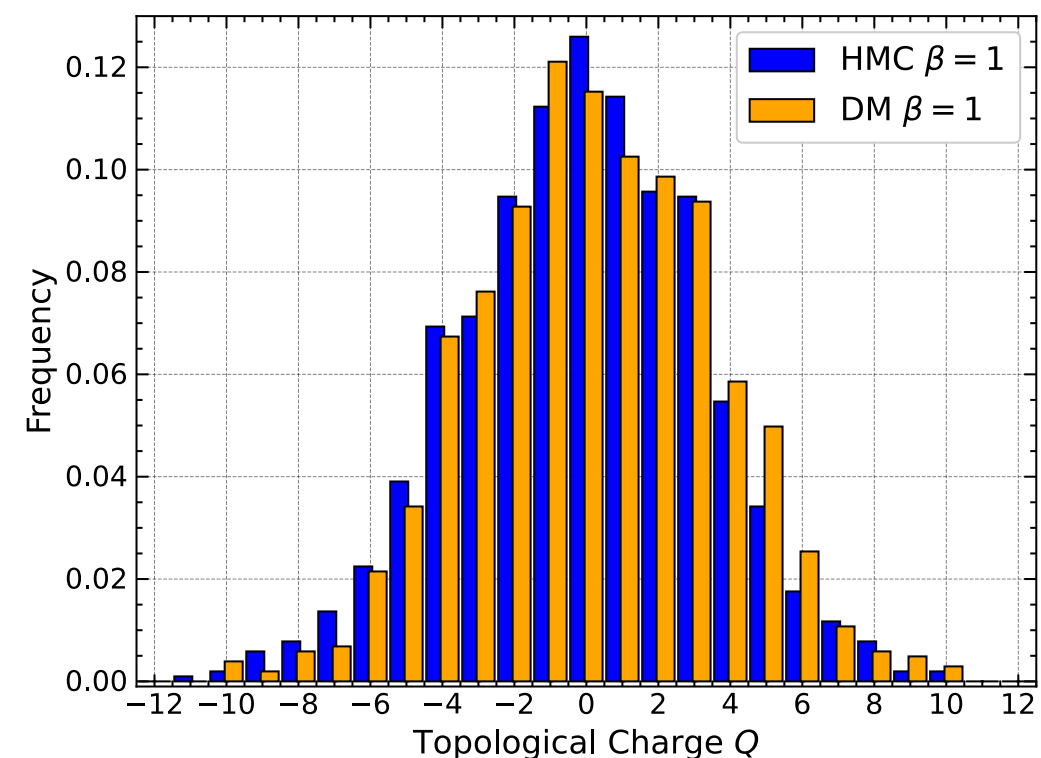
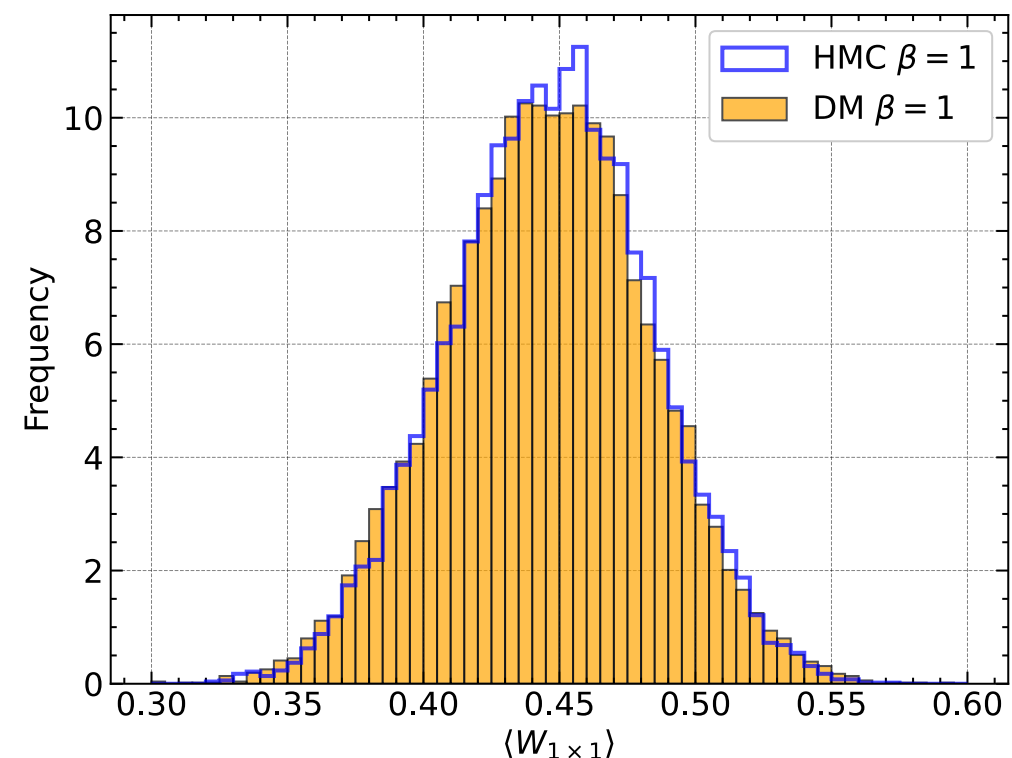
$$U_{\square} = U_{x,\mu} U_{x+\hat{\mu},\nu} U_{x+\hat{\nu},\mu}^{\dagger} U_{x,\nu}^{\dagger}$$

- Observables

Wilson loop, $W(C) = \text{Tr} \left(\prod_{(x,\mu) \in C} U_{x,\mu} \right)$

Topological charge, $Q = \frac{1}{2\pi} \sum_x F_{01}(x)$

Learned at $\beta = 1$ with **10,240** configurations, $L = 16$
Generated 1024 configs for testing



Lattice Gauge Fields

2D U(N)/SU(N)

- **Euclidean action** on lattice

$$S = -\frac{\beta}{N} \sum_{\mu < \nu} \text{Re}(P_{\mu\nu})$$

- Defined on links as **plaquette**

$$P_{\mu\nu} \equiv U_{x,\mu} U_{x+\hat{\mu},\nu} U_{x+\hat{\nu},\mu}^\dagger U_{x,\nu}^\dagger$$

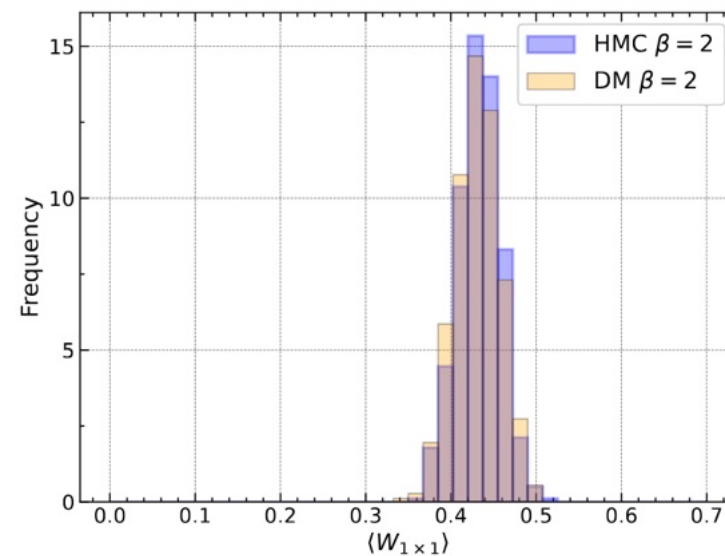
$$U_{x,\mu} = e^{i\theta_{a,x,\mu} T^a}, \quad T^a \text{ are the generators}$$

- **Group-preserving diffusion**

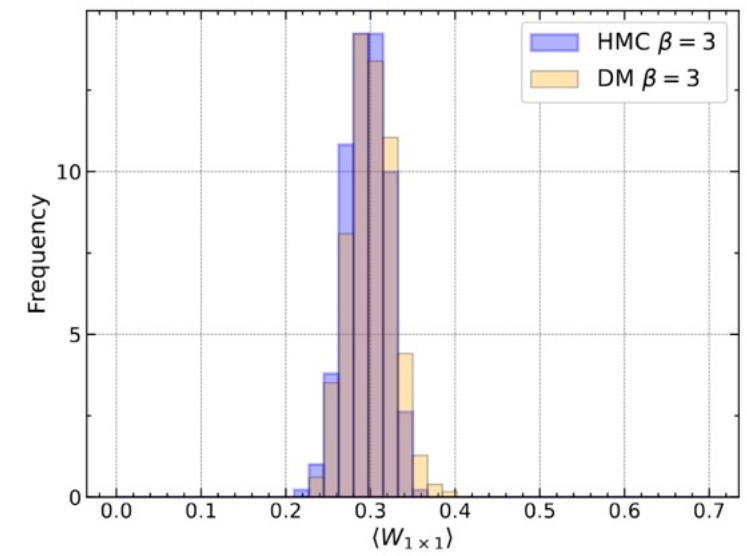
$$U_{x,\mu}^{(t)} = \exp\left[i\sigma(t) \sum_a T^a \eta_{x,\mu}^a\right] U_{x,\mu}^{(0)}$$

Learned with **10,240** configurations, $L = 16$

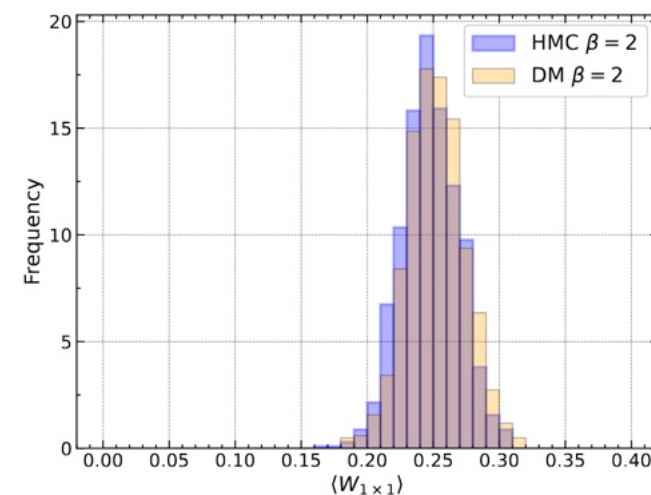
Generated 1024 configs for testing



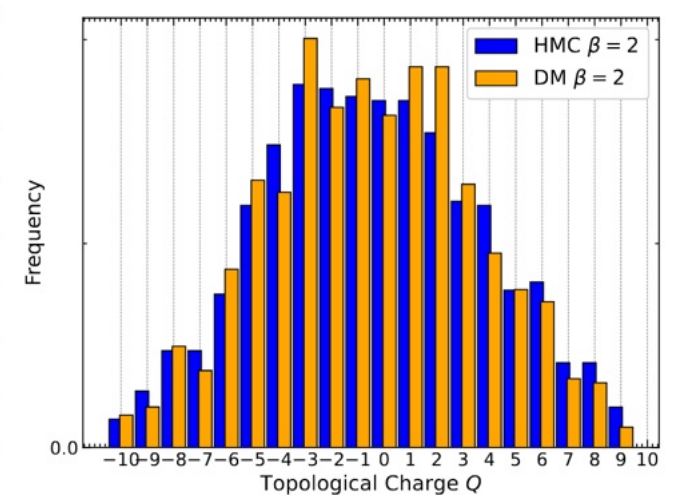
SU(2)



SU(3)



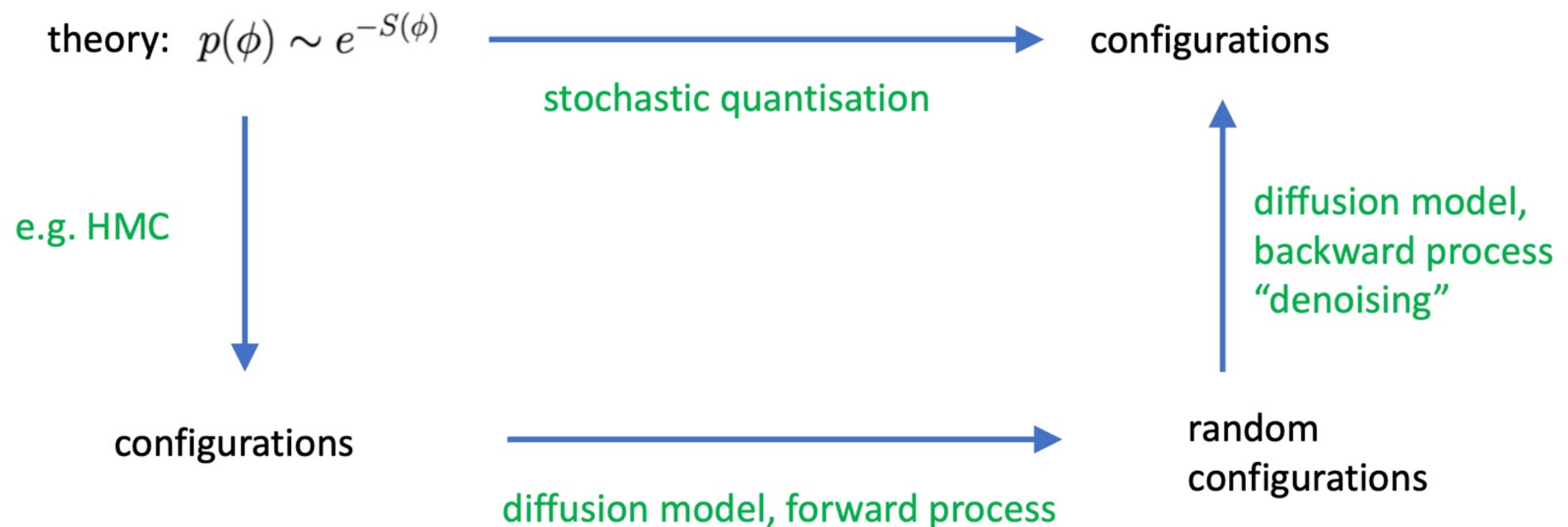
U(2)



G Aarts, D Habibi, A Ipp, D, Müller, T Ranner, **LW**, W Wang, Q Zhu, arXiv:2606.xxx

Expandability

Did we take a detour?



Learn to Sample
Learn once, Sample all ?!



Expandability

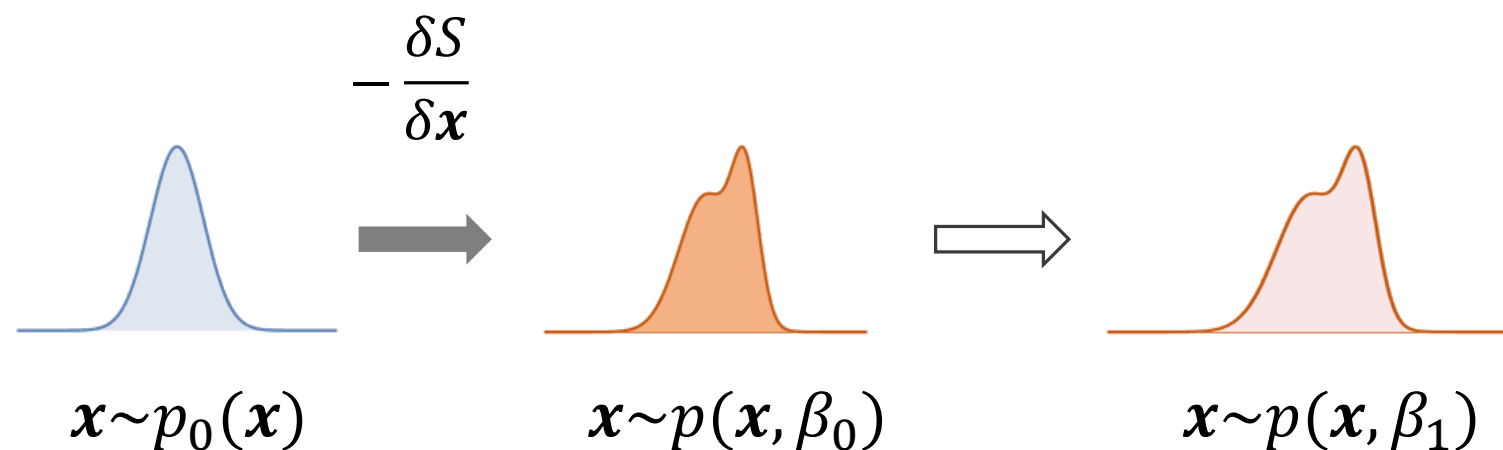
Did we take a detour?

$$\frac{\partial \phi(x, \tau)}{\partial \tau} = - \boxed{\frac{\delta S_E[\phi]}{\delta \phi(x, \tau)}} + \eta(x, \tau)$$

Drift Term

$$\frac{d\phi}{dt} = \boxed{-g^2(t) \mathbf{s}_{\hat{\theta}}(\phi, t)} + g(t) \bar{\eta}(t)$$

Score Function



e.g., pure gauge

$$S = \beta \sum_{\square} \left(1 - \mathbf{Re}(U_{\square}) \right)$$

$$\tilde{\mathbf{s}}_{\hat{\theta}}(\phi, t) \equiv \beta \mathbf{s}_{\hat{\theta}}(\phi, t)$$

Expandability

Q Zhu, G Aarts, W Wang, K Zhou, LW, JHEP(2026)

Did we take a detour?

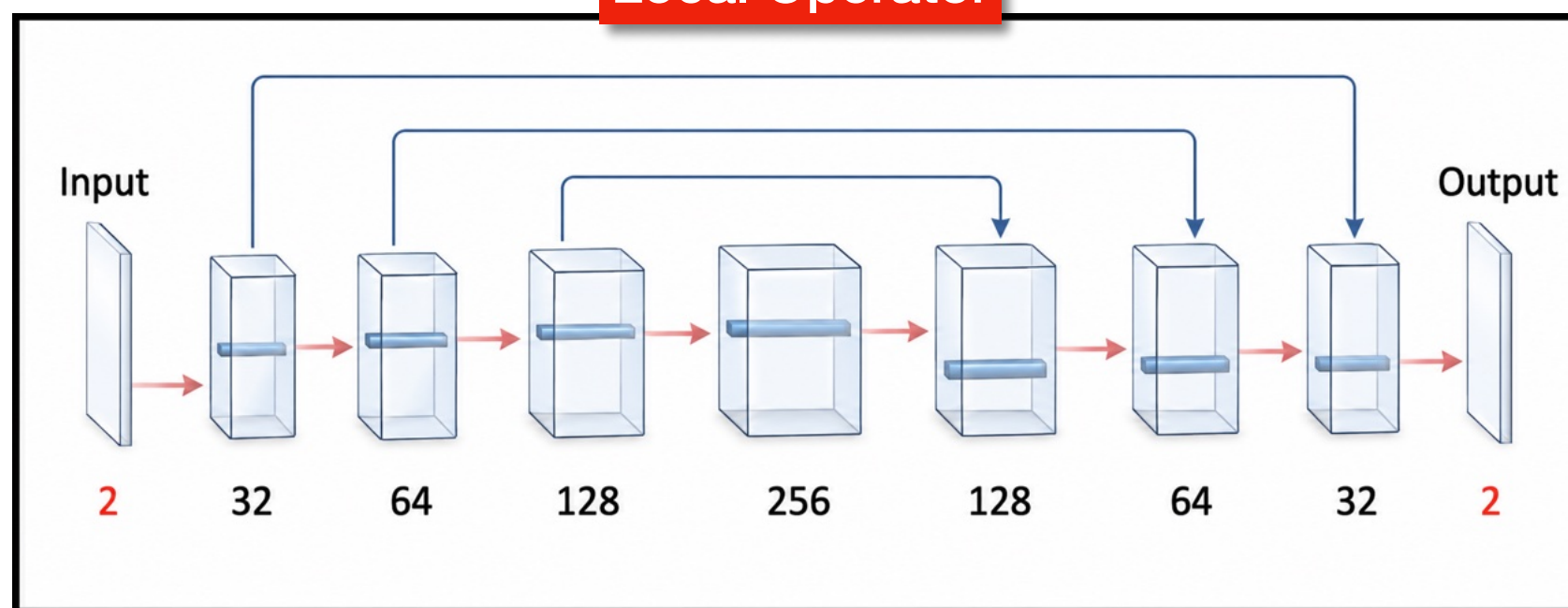
$$\frac{\partial \phi(x, \tau)}{\partial \tau} = - \frac{\delta S_E[\phi]}{\delta \phi(x, \tau)} + \eta(x, \tau)$$

Drift Term

$$\frac{d\phi}{dt} = - g^2(t) \mathbf{s}_{\hat{\theta}}(\phi, t) + g(t) \bar{\eta}(t)$$

Score Function

Local Operator



e.g., pure gauge

$$S = \beta \sum_{\square} \left(1 - \mathbf{Re}(U_{\square}) \right)$$

fully-convolutional networks

$$(f_{\theta}u)(x) = F_{\theta}(\{ u(x + \delta) \mid \delta \in \Omega \})$$

Expandability

Did we take a detour?

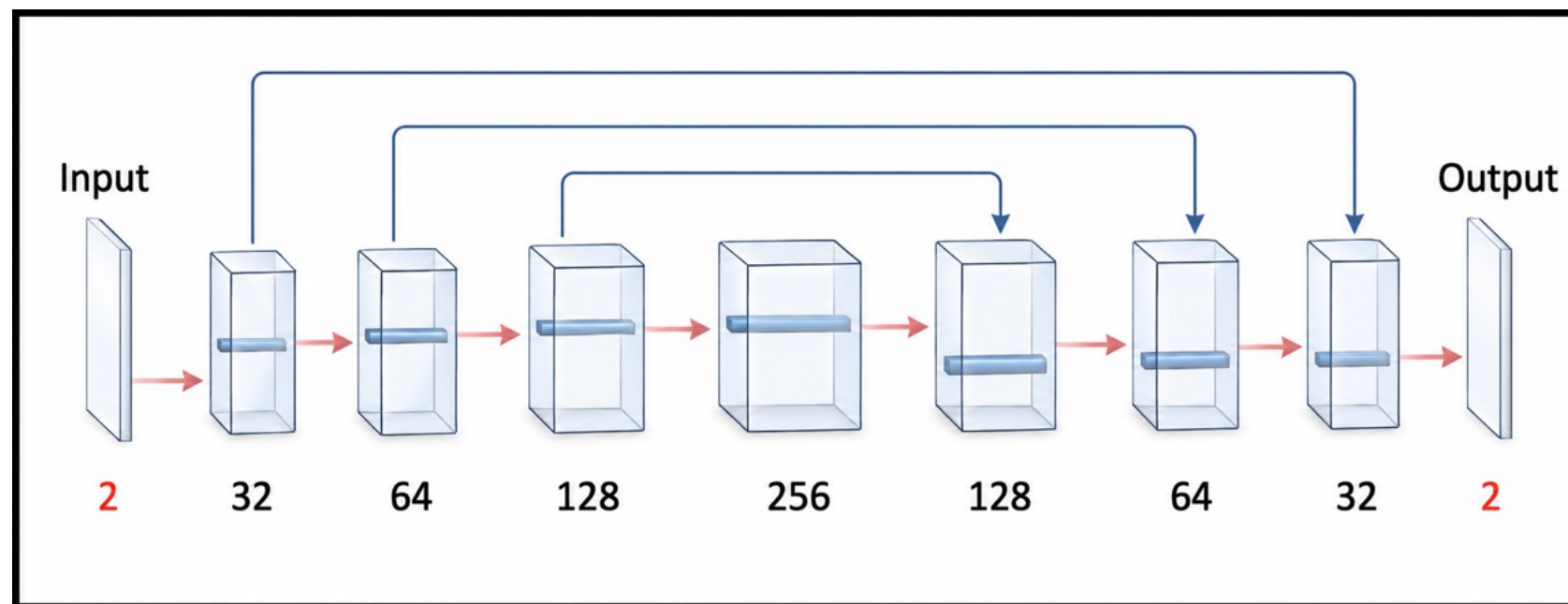
$$\frac{\partial \phi(x, \tau)}{\partial \tau} = - \boxed{\frac{\delta S_E[\phi]}{\delta \phi(x, \tau)}} + \eta(x, \tau)$$

Drift Term

$$\frac{d\phi}{dt} = \boxed{-g^2(t) \mathbf{s}_{\hat{\theta}}(\phi, t)} + g(t) \bar{\eta}(t)$$

Score Function

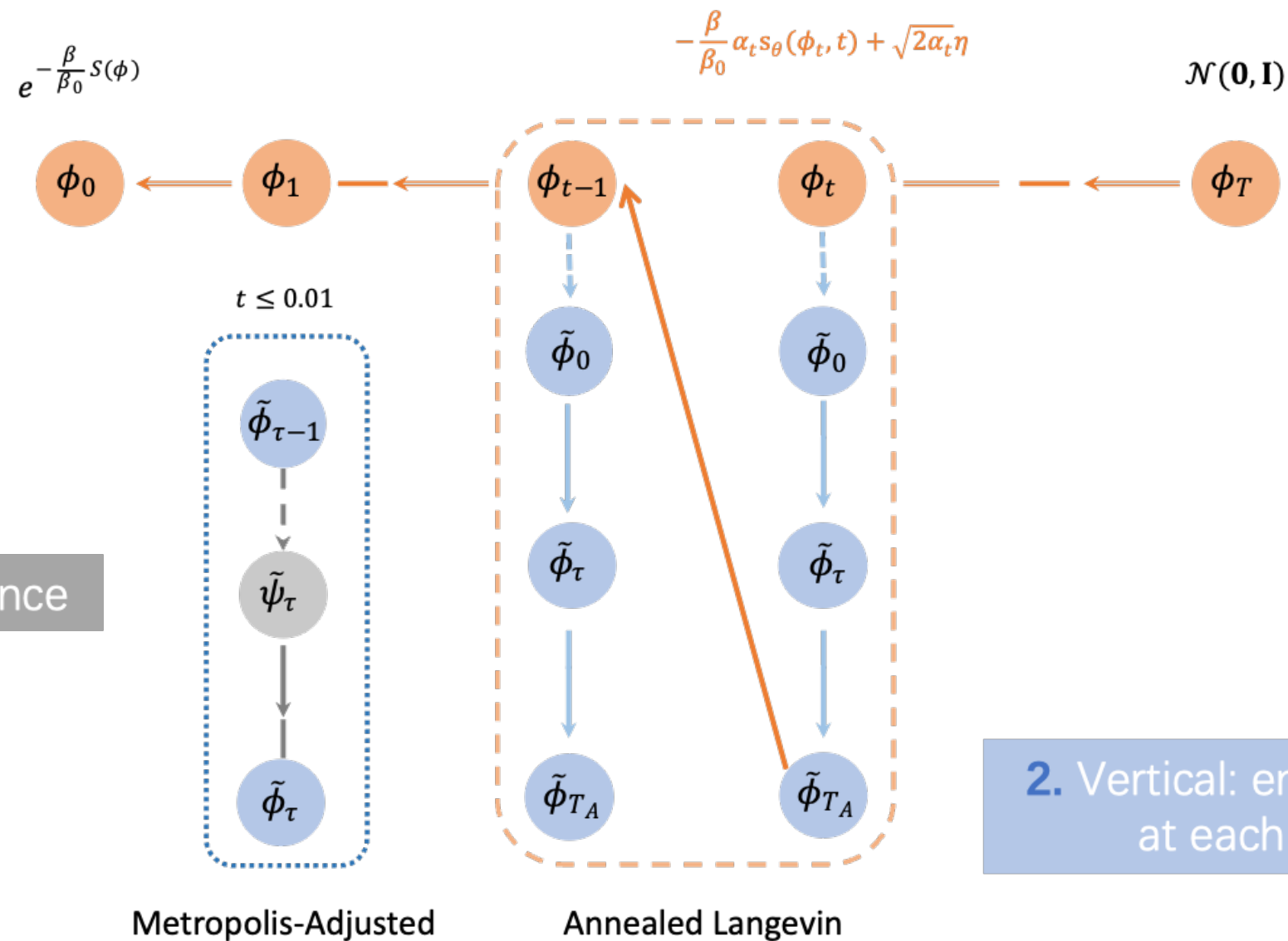
Different Lattice Sizes, Inverse Coupling Constants



G. Aarts, K. Fukushima, T. Hatsuda, A. Ipp, S. Shi, **L. Wang***, and K. Zhou,
Physics-Driven Learning for Inverse Problems in Quantum Chromodynamics,
Nature Reviews Physics volume 7, pages154–163 (2025)

How can one trust DM generation?

1. Horizontal : Euler-Maruyama scheme gradually decrease noise



Qianteng Zhu

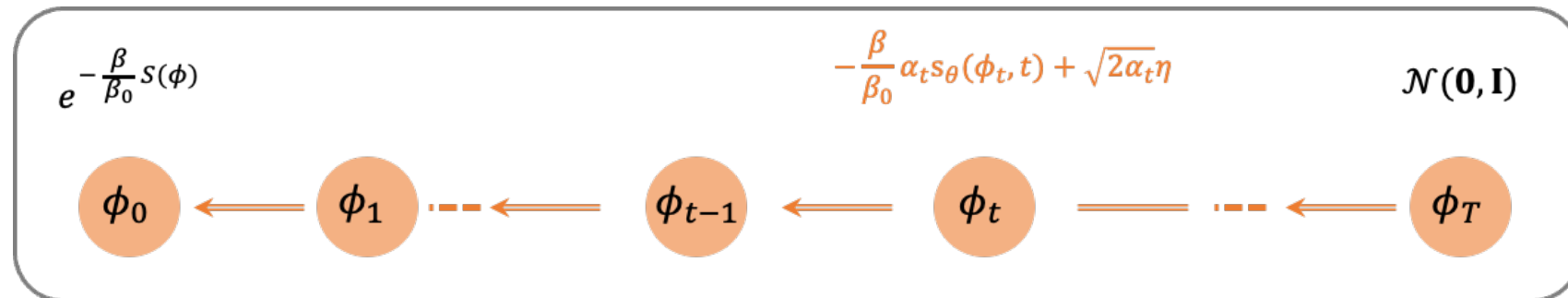
3. Detailed balance

2. Vertical: ensure equilibrium at each noise scale

Metropolis-Adjusted Annealed Langevin Algorithm(MAALA)

Exactness

1. SDE Generation

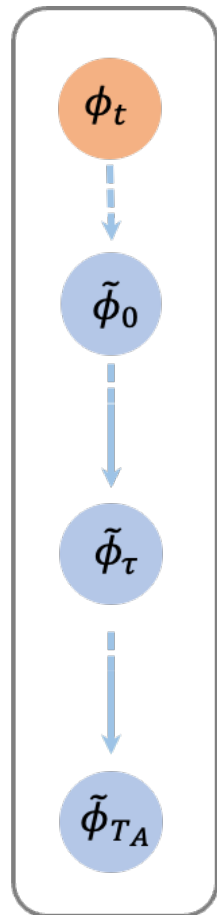


- ▶ α_t noise level $\sim$ time-step size, e.g., 0.9^t
- ▶ β target inverse coupling constant
- ▶ β_0 initial inverse coupling constant
- ▶ $S(\phi)$ separated physics action (w/o coupling)
- ▶ θ parameters of neural networks for modelling the score function
- ▶ ϕ_t field generated at time-step t
- ▶ T final time-step for forward process, but initial point for generations

$$\min_{\theta} \mathcal{L}_{\theta} = \sum_{i=1}^N \sigma_i^2 \mathbb{E}_{p_0(\phi_0)} \mathbb{E}_{p_i(\phi_i|\phi_0)} \left[\left\| \mathbf{s}_{\theta}(\phi_i, \xi) - \nabla_{\phi_i} \log p_i(\phi_i|\phi_0) \right\|_2^2 \right]$$

Exactness

2. Annealing



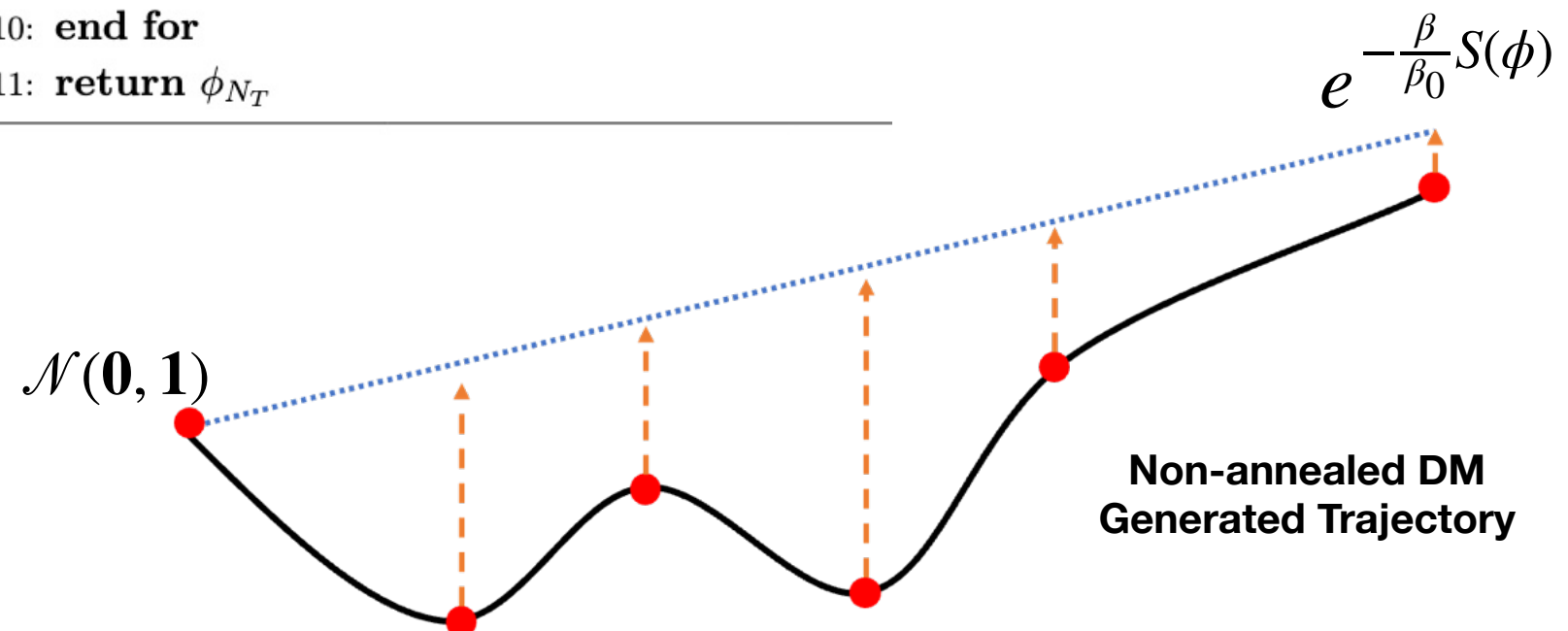
Algorithm 1 Annealed Langevin Sampler

Require: $\{t_i\}_{i=1}^{N_T}, \epsilon, N_A$

- 1: Initialize ϕ_0
 - 2: **for** $i \leftarrow 1$ to N_T **do**
 - 3: $\alpha_i \leftarrow \epsilon \cdot g \cdot t_i^2 / t_{N_T}^2$
 - 4: $\tilde{\phi}_0 \leftarrow \phi_{i-1}$
 - 5: **for** $\tau \leftarrow 1$ to N_A **do**
 - 6: Draw $\eta \sim \mathcal{N}(0, I)$
 - 7: $\tilde{\phi}_\tau \leftarrow \tilde{\phi}_{\tau-1} + \frac{\beta}{\beta_0} \alpha_i s_\theta(\tilde{\phi}_{\tau-1}, t_i) + \sqrt{2\alpha_i} \eta$
 - 8: **end for**
 - 9: $\phi_i \leftarrow \tilde{\phi}_{N_A}$
 - 10: **end for**
 - 11: **return** ϕ_{N_T}
-

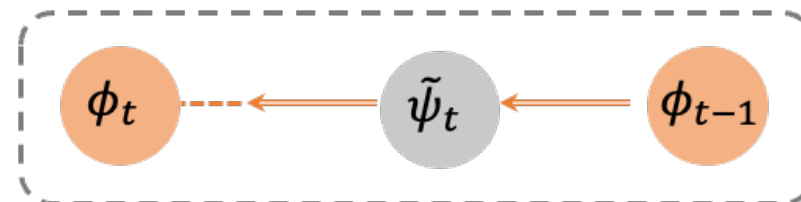
$$p_{eq}(\phi) \propto e^{-\frac{S_{DM}}{\bar{\alpha}}}$$

**Equilibrate at
Each Noise Scale**



Exactness

3. Metropolis-Adjusted



Accept/Reject Step

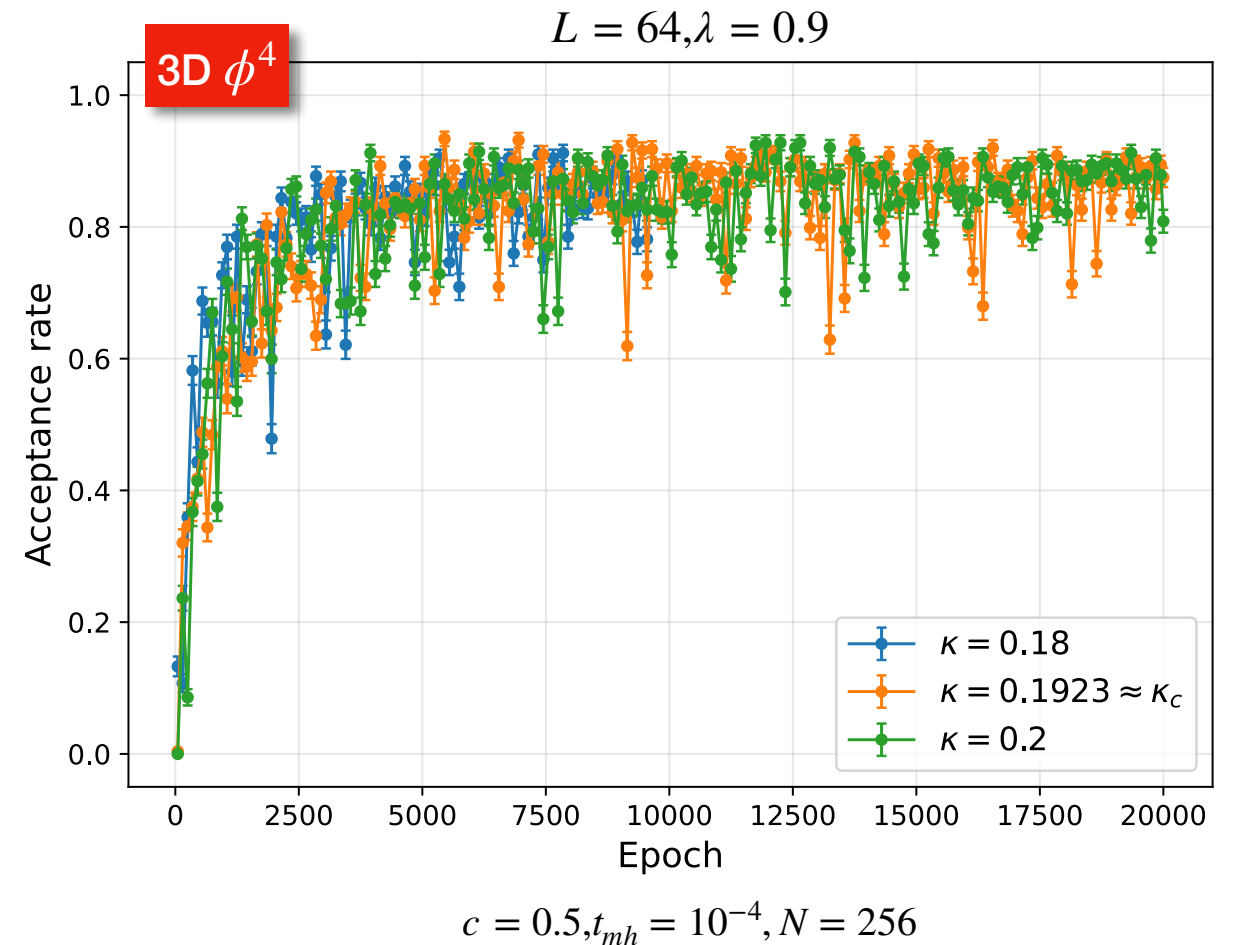
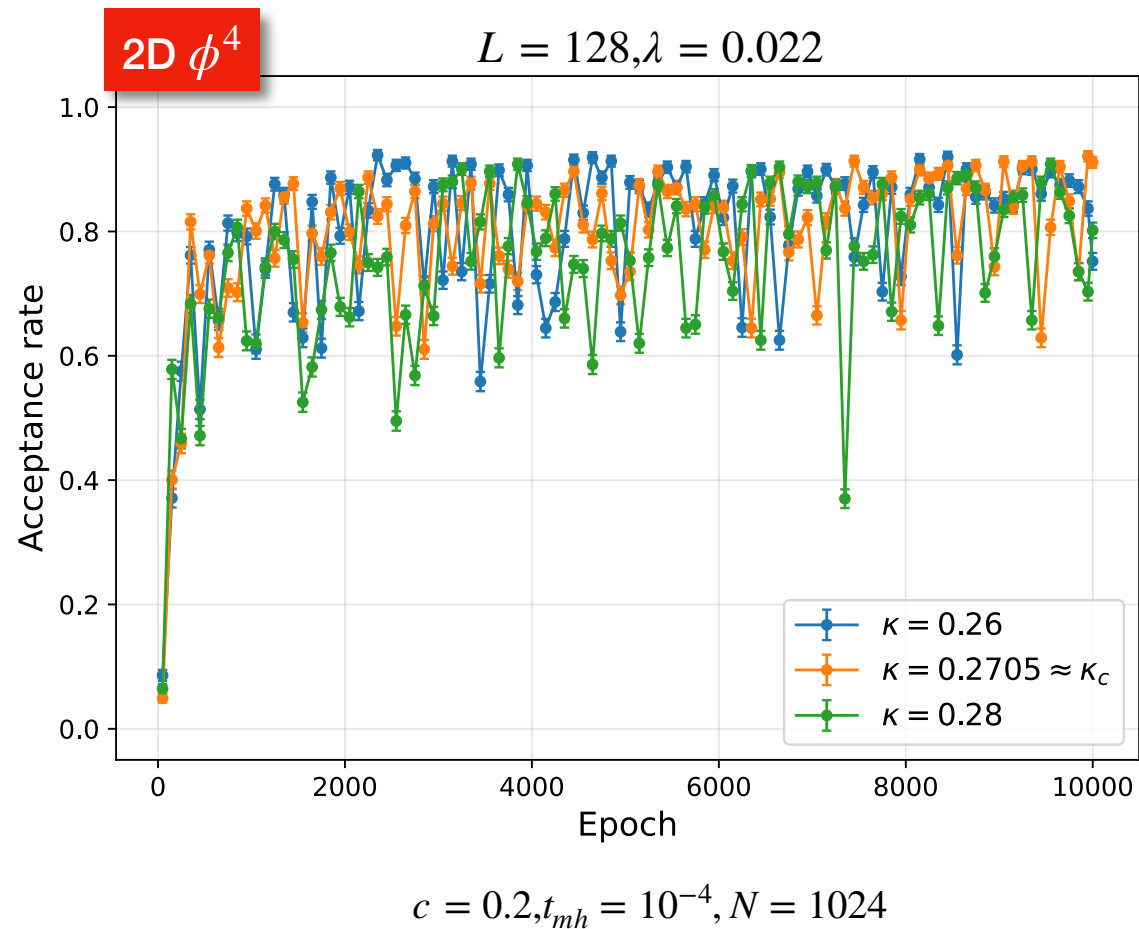
$$\phi_{\tau+1} = \begin{cases} \psi_{\tau+1} & \text{with probability } \min \left\{ 1, \frac{p(\psi_{\tau+1})q(\phi_{\tau}|\psi_{\tau+1})}{p(\phi_{\tau})q(\psi_{\tau+1}|\phi_{\tau})} \right\} \\ \phi_{\tau} & \text{with the remaining probability} \end{cases}$$

$$\psi_{\tau+1} = \phi_{\tau} + \alpha_i f(\phi_{\tau}, \tau) + \boxed{\sqrt{2\alpha_i}\eta} \quad \text{Gaussian noise}$$

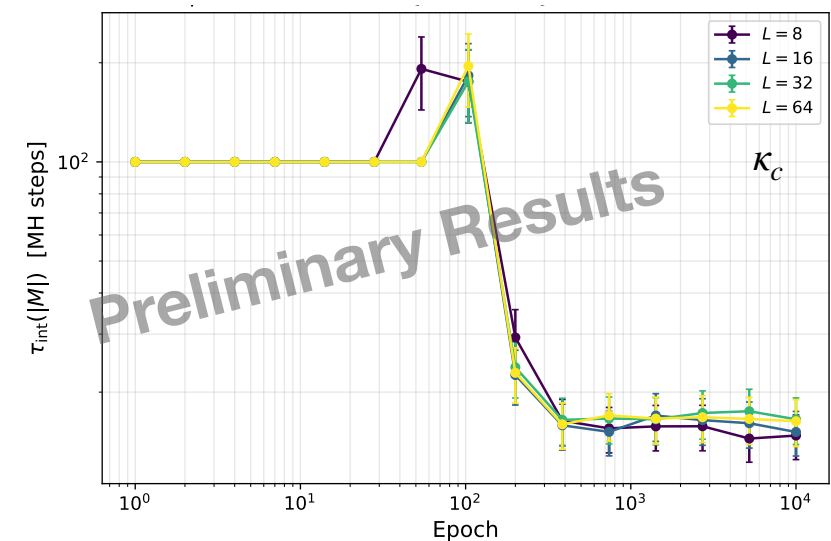
$$q(\phi_{\tau}|\psi_{\tau+1}) = \frac{1}{(4\pi\alpha_i)^{n/2}} \exp \left(-\frac{1}{4\alpha_i} \left\| \phi_{\tau} - (\psi_{\tau+1} + \alpha_i f(\psi_{\tau+1}, \tau + 1)) \right\|_2^2 \right)$$

α_i is a redefined noise-scale parameter, $p(\phi)$ is computed from the known physical distribution, n number of *d.o.f.*

G.O. Roberts and J.S. Rosenthal, Journal of the Royal Statistical Society: Series B 60 (1998) 255



Good Global Sampling Can Always Reduce The Auto-Correlation Time



3E for Scalar Fields

Y Tang, LW, etc., arXiv:2606.xxx

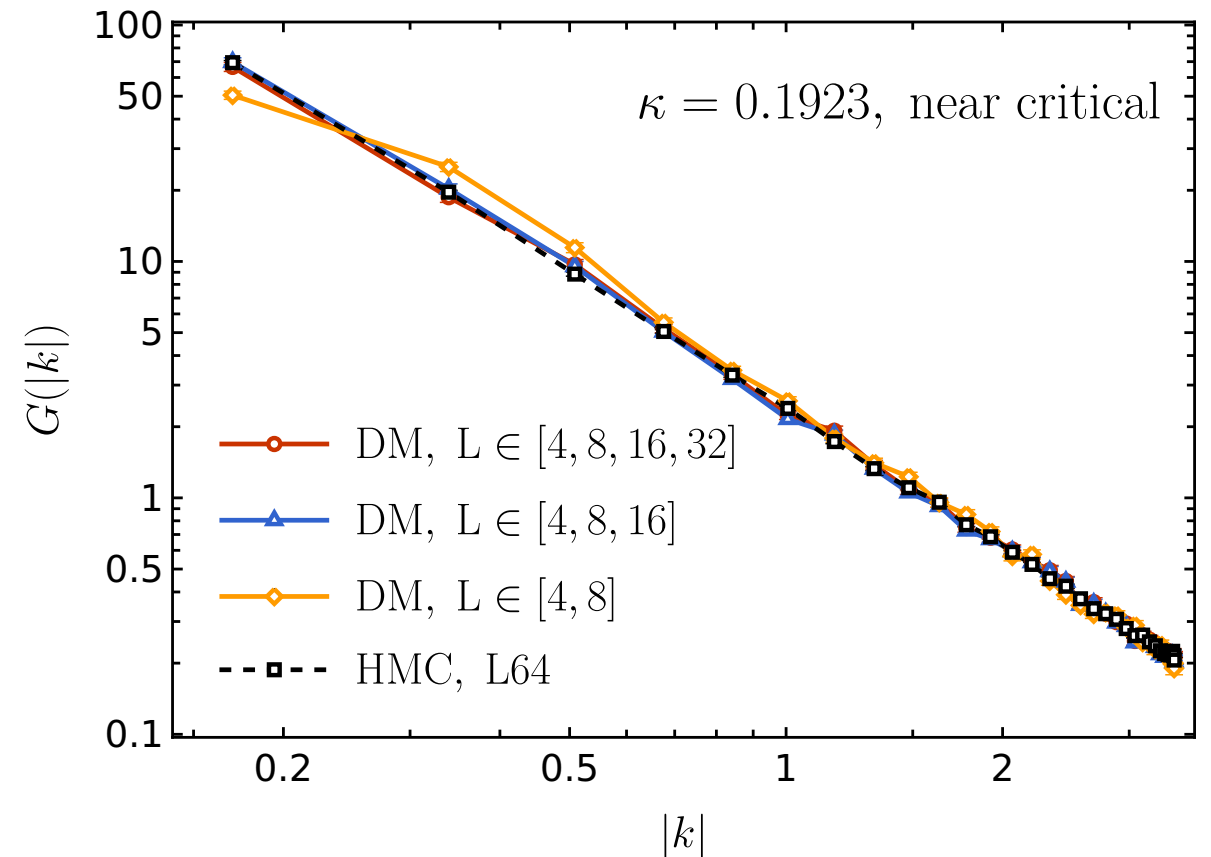
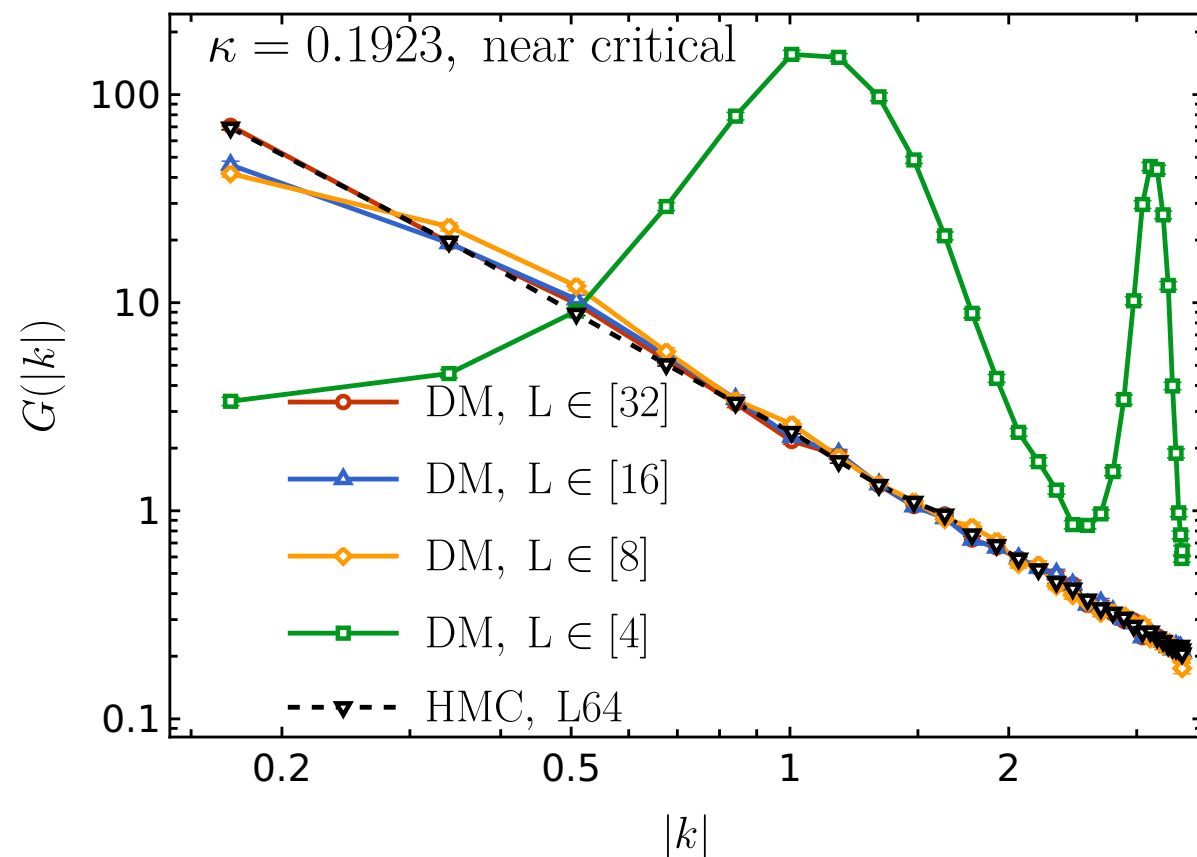


Yang-Yang Tan

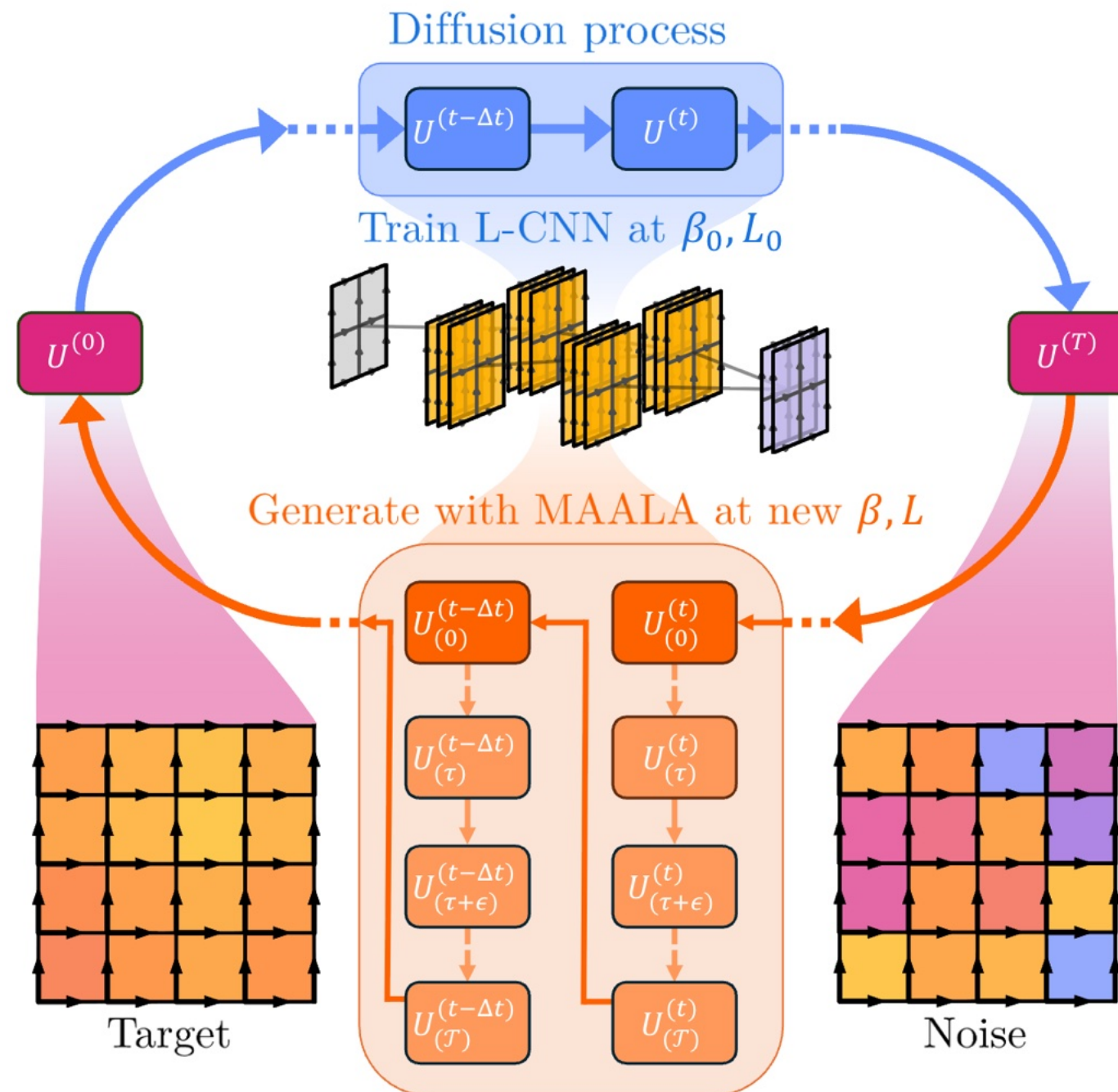
κ	source	$\langle \bar{\phi} \rangle$	χ	U_4	$\langle \phi \rangle$	$\kappa_2^{(s)}$	$\kappa_4^{(s)}$	$\kappa_6^{(s)}$	$\kappa_8^{(s)}$	$U_4^{(s)}$
0.18	HMC	0.00571(6)	5.0(1)	-0.02(2)	0.66846(2)	0.58604(2)	-0.40066(4)	1.3686(2)	-10.172(2)	0.38887(2)
	HMC ($N=512$)	0.0059(1)	5.3(3)	-0.03(5)	0.66845(4)	0.58603(5)	-0.40062(9)	1.3683(5)	-10.169(5)	0.38884(4)
	DM {4, 8}	0.0062(1)	6.0(4)	-0.09(7)	0.66525(3)	0.58116(5)	-0.39175(9)	1.3250(5)	-9.742(5)	0.38663(4)
	DM {4, 8, 16}	0.0073(2)	6.8(4)	0.09(4)	0.66672(4)	0.58361(6)	-0.3956(1)	1.3434(6)	-9.922(5)	0.38712(4)
	DM {4, 8, 16, 32}	0.0128(3)	14.6(6)	0.31(2)	0.67334(9)	0.5939(1)	-0.4135(2)	1.436(1)	-10.84(1)	0.39075(4)
0.1923	HMC	0.1165(7)	620(10)	0.516(3)	0.68999(3)	0.61815(4)	-0.46111(8)	1.6968(5)	-13.560(5)	0.40225(2)
	HMC ($N=512$)	0.112(2)	636(23)	0.503(7)	0.68987(7)	0.61797(10)	-0.4607(2)	1.695(1)	-13.54(1)	0.40217(5)
	DM {4, 8}	0.0287(9)	94(5)	0.24(3)	0.68054(4)	0.60564(6)	-0.4343(1)	1.5467(7)	-11.973(7)	0.39467(4)
	DM {4, 8, 16}	0.064(2)	446(24)	0.27(3)	0.68416(8)	0.6090(1)	-0.4449(2)	1.608(1)	-12.62(1)	0.39992(6)
	DM {4, 8, 16, 32}	0.100(2)	694(30)	0.44(1)	0.6914(1)	0.6204(2)	-0.4655(3)	1.722(2)	-13.83(2)	0.40314(5)
	DM $L=64$	0.156(3)	$1.33(35) \times 10^3$	-0.1(5)	0.7044(9)	0.645(2)	-0.498(1)	1.919(9)	-15.94(8)	0.400(3)
0.2	HMC	0.51623(9)	4.9(1)	0.666574(2)	0.74510(3)	0.70104(4)	-0.6438(1)	2.8272(7)	-26.836(9)	0.43663(2)
	HMC ($N=512$)	0.5163(2)	4.6(3)	0.666579(6)	0.74516(6)	0.70114(9)	-0.6440(2)	2.829(1)	-26.86(2)	0.43668(4)
	DM {4, 8}	0.372(3)	$1.05(16) \times 10^3$	0.644(3)	0.7146(3)	0.6545(5)	-0.538(1)	2.151(7)	-18.62(8)	0.4184(2)
	DM {4, 8, 16}	0.5072(2)	7.6(5)	0.666516(10)	0.74041(9)	0.6934(2)	-0.6276(3)	2.721(2)	-25.49(3)	0.43504(4)
	DM {4, 8, 16, 32}	0.5202(3)	13.1(8)	0.66642(2)	0.7508(1)	0.7114(3)	-0.6651(5)	2.970(4)	-28.65(5)	0.43806(5)
	DM $L=64$	0.5291(5)	45(4)	0.66583(8)	0.7580(3)	0.7255(6)	-0.689(1)	3.134(8)	-30.8(1)	0.43665(4)

Acceptance Rate

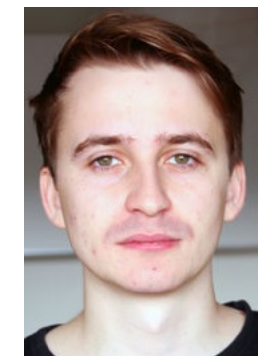
training set	$\kappa = 0.18$	$\kappa \approx 0.1923$	$\kappa = 0.2$
{4, 8}	0.358(84)	0.326(130)	0.334(76)
{4, 8, 16}	0.548(135)	0.562(47)	0.531(130)
{4, 8, 16, 32}	0.745(38)	0.733(161)	0.618(114)



3E for U(N)/SU(N) Gauge Fields



Thomas R. Ranner



David Müller



Qianteng Zhu

G Aarts, D Habibi, A Ipp, D, Müller, T Ranner, **LW**, W Wang, Q Zhu [[2601.19552](#) [hep-lat]]

3E for U(N)/SU(N) Gauge Fields

Group-Preserving Forward Process

T^a group generators, a color indices

$$dU_{x,\mu}^{(t)} = i \sum_a T^a \left[K_{x,\mu}^a(\mathbf{U}^{(t)}) dt + g(t) dW_{x,\mu}^{(t),a} \right] \circ U_{x,\mu}^{(t)}$$

$$K_{x,\mu}^a = 0$$

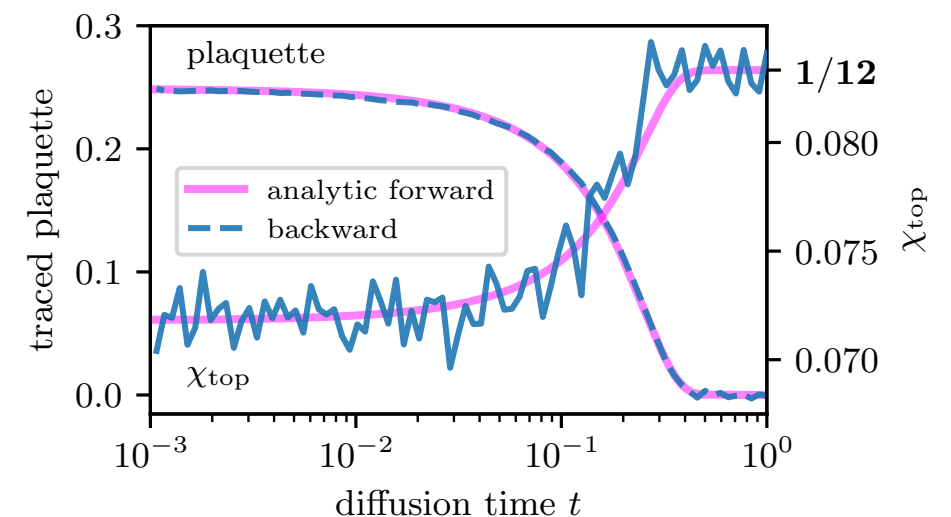
$$\eta_{x,\mu}^a \sim \mathcal{N}(0,1)$$

$$\sigma(t) = \sqrt{(s^{2t} - 1)/(2 \ln s)}$$

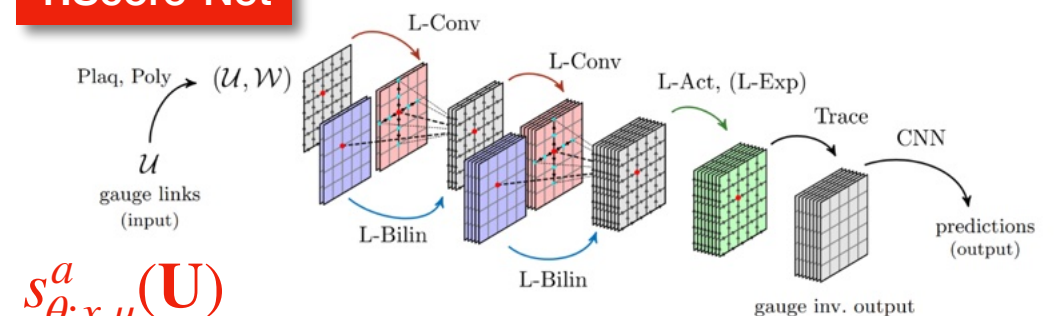
$$U_{x,\mu}^{(t)} = \exp \left[i \sigma(t) \sum_a T^a \eta_{x,\mu}^a \right] U_{x,\mu}^{(0)}$$

Loss Function

$$\int_0^T dt \mathbb{E}_{p_0(\mathbf{U}^{(0)})} \left[\sum_{x,\mu,a} \left(s_{\theta;x,\mu}^a(\mathbf{U}^{(t)}, t) + \frac{\eta_{x,\mu}^a}{\sigma(t)} \right)^2 \right]$$



1.Score-Net



Favoni et al., "Lattice Gauge Equivariant Convolutional Neural Networks"

2.MAALA

$$U_{(\tau+\epsilon)}^{(t)} = \exp \left[i \left(\epsilon \frac{\beta}{\beta_0} \hat{s}_{\theta}(\mathbf{U}_{(\tau)}^{(t)}, t) + \sqrt{2\epsilon} \hat{\eta} \right) \right] U_{(\tau)}^{(t)}$$

$$p_{\text{accept}} = \min \left\{ 1, \frac{p(\tilde{\mathbf{U}}_{(\tau)}^{(t)}) q(\mathbf{U}_{(\tau)}^{(t)} | \tilde{\mathbf{U}}_{(\tau)}^{(t)})}{p(\mathbf{U}_{(\tau)}^{(t)}) q(\tilde{\mathbf{U}}_{(\tau)}^{(t)} | \mathbf{U}_{(\tau)}^{(t)})} \right\}$$

3E for U(N)/SU(N) Gauge Fields

Group-Preserving Forward Process

T^a group generators, a color indices

$$dU_{x,\mu}^{(t)} = i \sum_a T^a \left[K_{x,\mu}^a(\mathbf{U}^{(t)}) dt + g(t) dW_{x,\mu}^{(t),a} \right] \circ U_{x,\mu}^{(t)}$$

$$K_{x,\mu}^a = 0$$

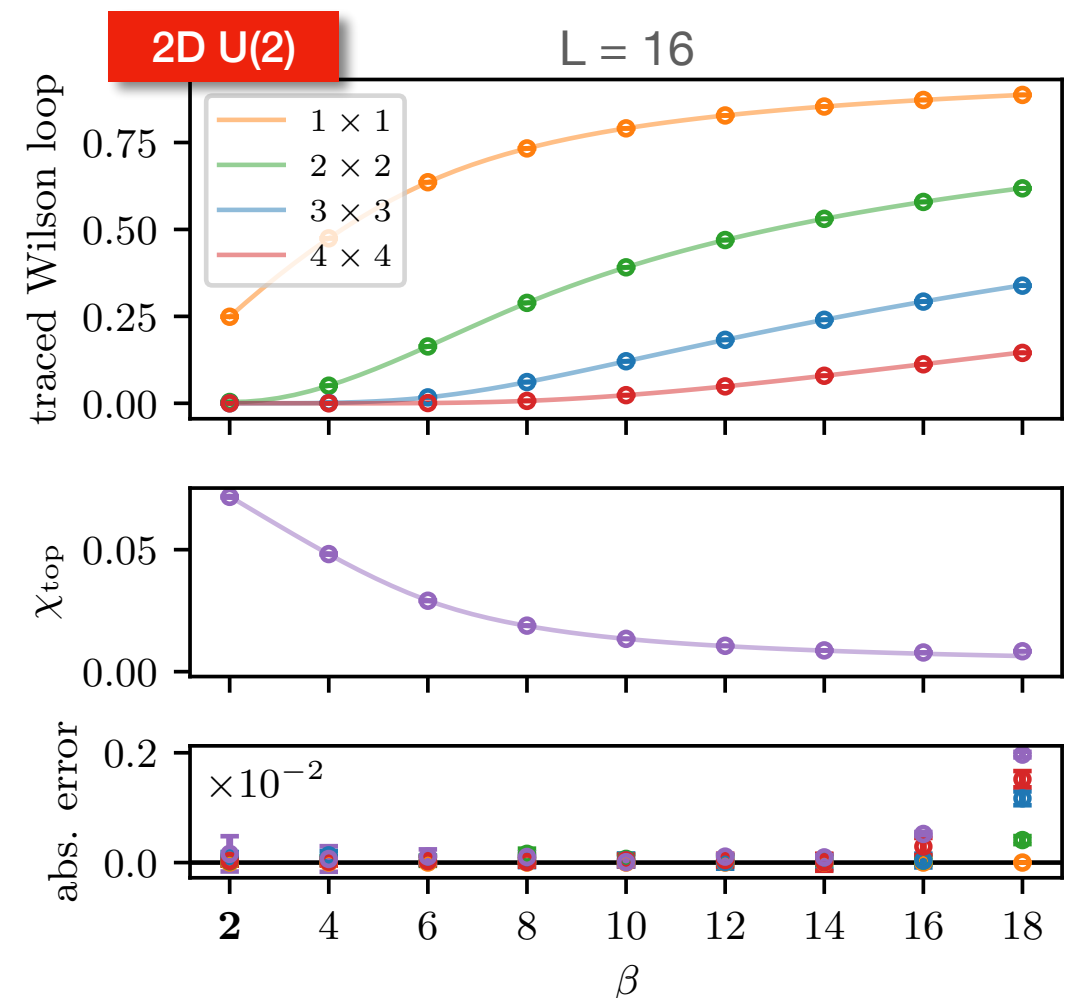
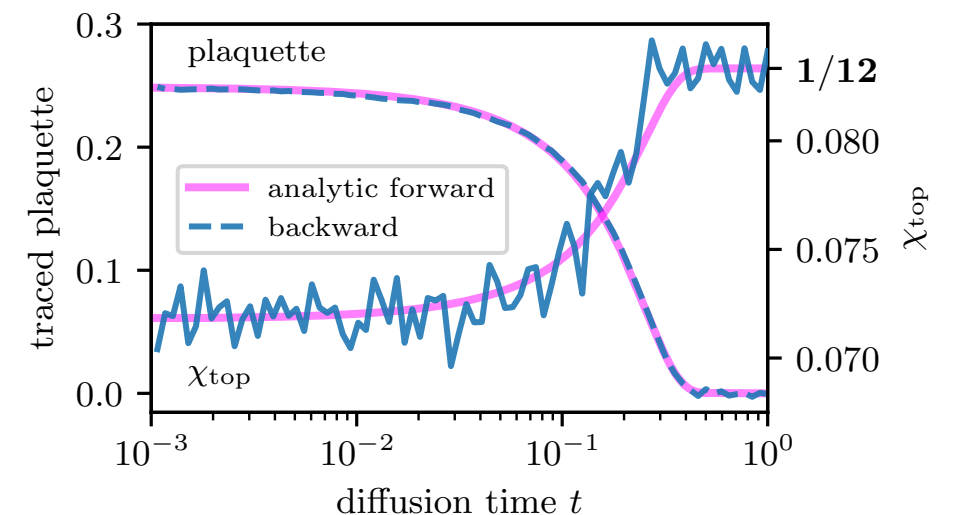
$$\eta_{x,\mu}^a \sim \mathcal{N}(0,1)$$

$$\sigma(t) = \sqrt{(s^{2t} - 1)/(2 \ln s)}$$

$$U_{x,\mu}^{(t)} = \exp \left[i \sigma(t) \sum_a T^a \eta_{x,\mu}^a \right] U_{x,\mu}^{(0)}$$

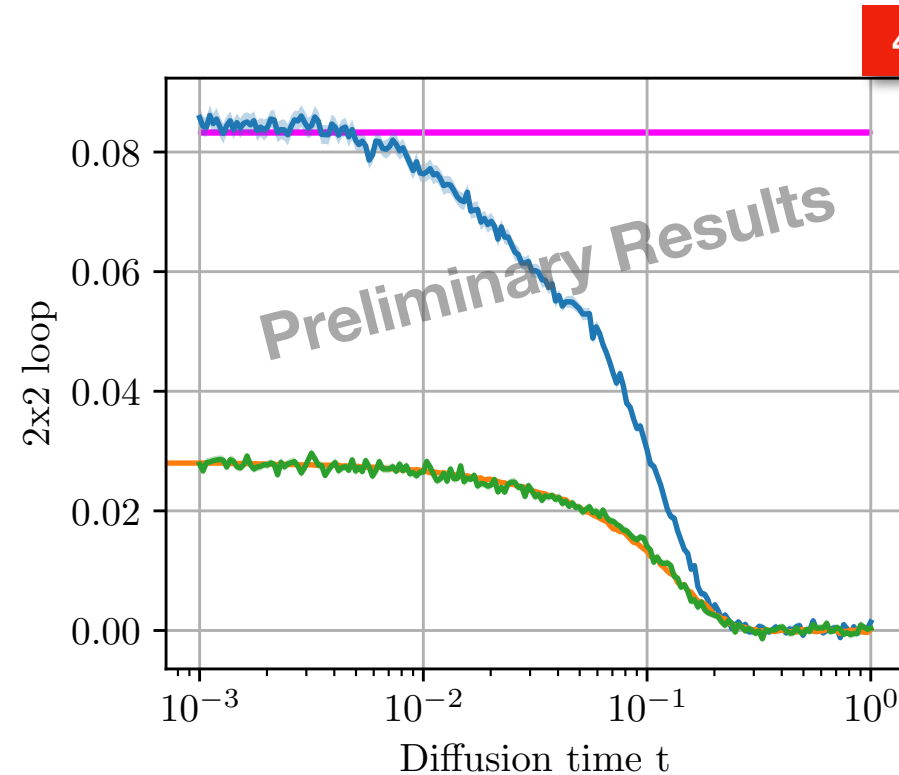
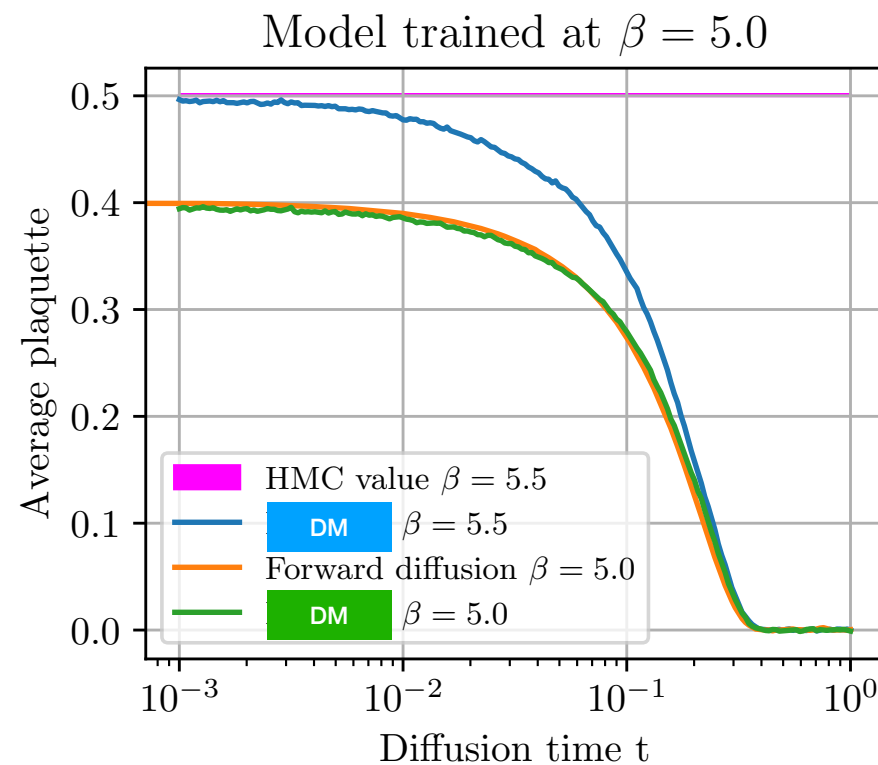
Loss Function

$$\int_0^T dt \mathbb{E}_{p_0(\mathbf{U}^{(0)})} \left[\sum_{x,\mu,a} \left(s_{\theta;x,\mu}^a(\mathbf{U}^{(t)}, t) + \frac{\eta_{x,\mu}^a}{\sigma(t)} \right)^2 \right]$$



3E for U(N)/SU(N) Gauge Fields

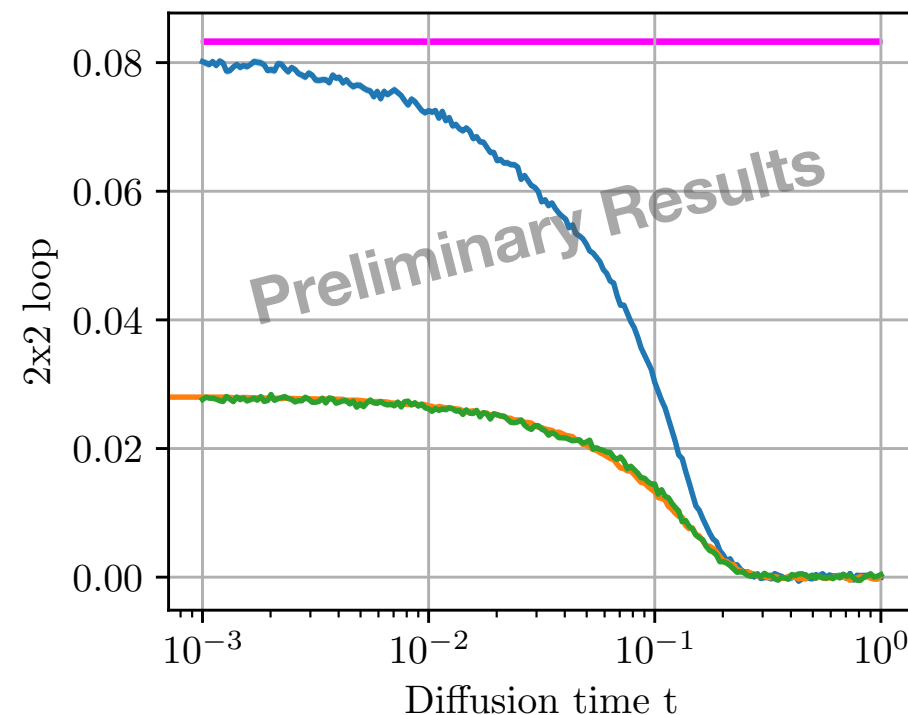
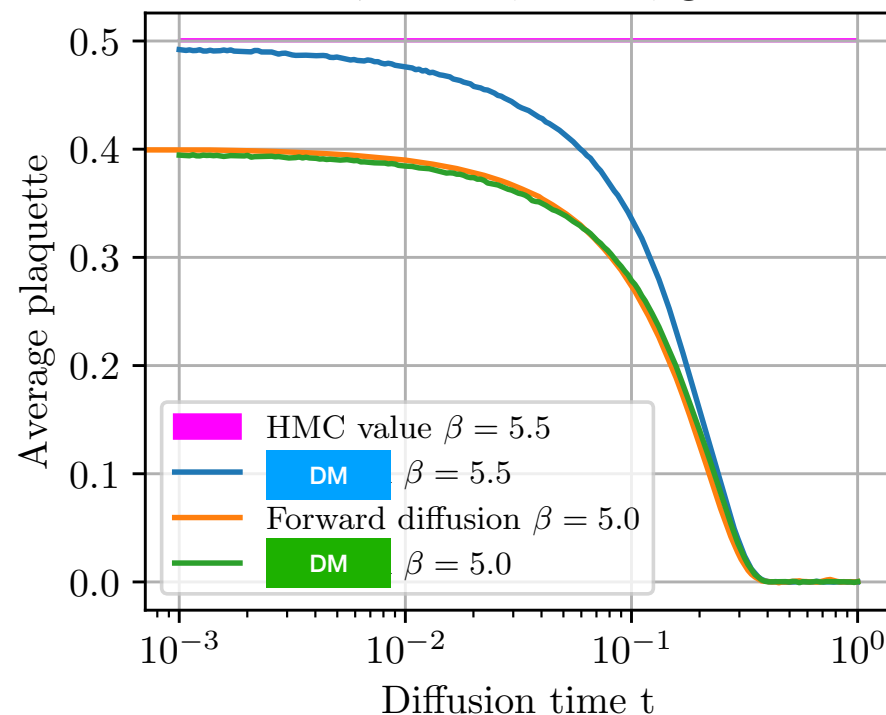
DM-QFT [260x.xxx [hep-lat]]



$$V = 4^4, \beta_0 = 5$$

$$V = 4^4, \beta = 5.5$$

Model trained at $\beta = 5.0, L = 4$, generate at $L = 6$

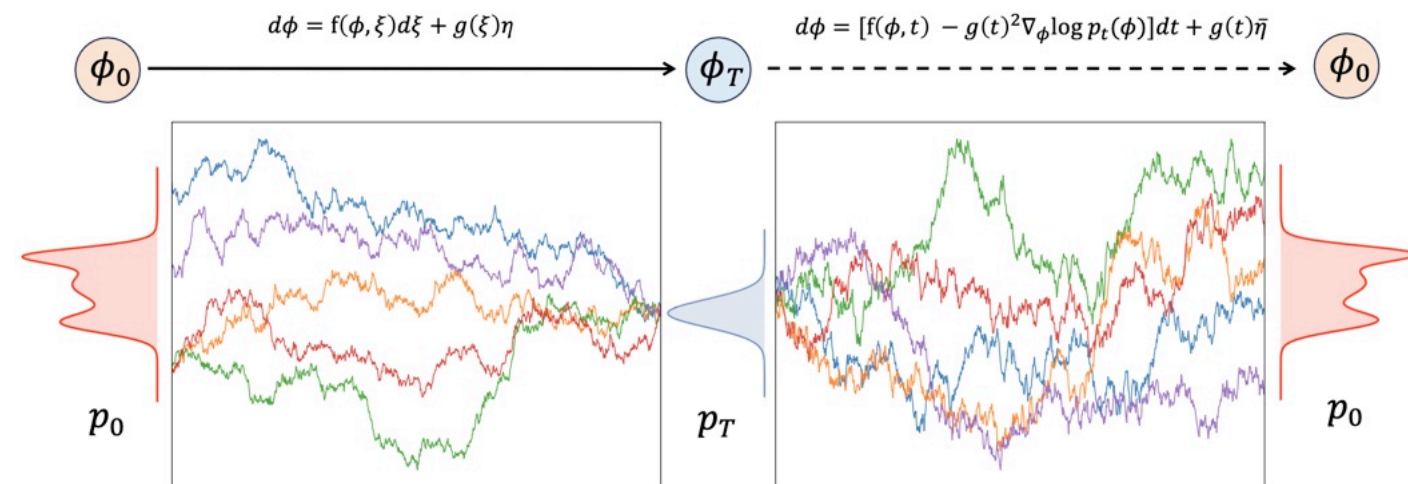


$$V = 6^4, \beta = 5.5$$

Summary

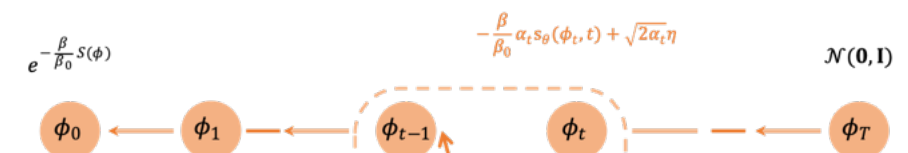
Where we are now

- ▶ **Physicists can understand DM very well**
 - ▶ Langevin process
 - ▶ Diffusion models as SQ
- ▶ **Lattice field theories can be well-simulated**
 - ▶ Scalar fields
 - ▶ Gauge fields
- ▶ **Learn-to-Sample**
 - ▶ Starting from existing configs
 - ▶ **Expandable**
 - ▶ **Exact**
 - ▶ **Efficient**

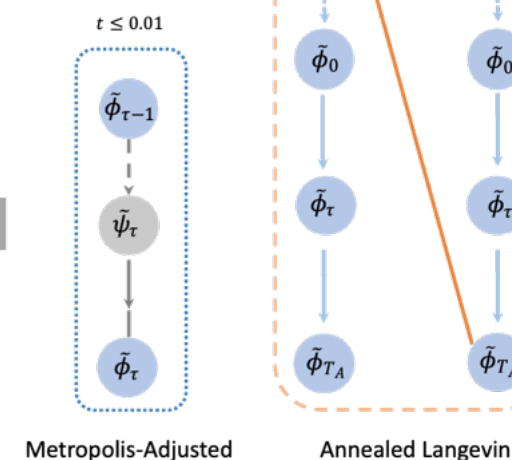


$$\frac{\partial \phi(x, \tau)}{\partial \tau} = - \frac{\delta S_E[\phi]}{\delta \phi(x, \tau)} + \eta(x, \tau)$$

1. Horizontal : Euler-Maruyama scheme gradually decrease noise



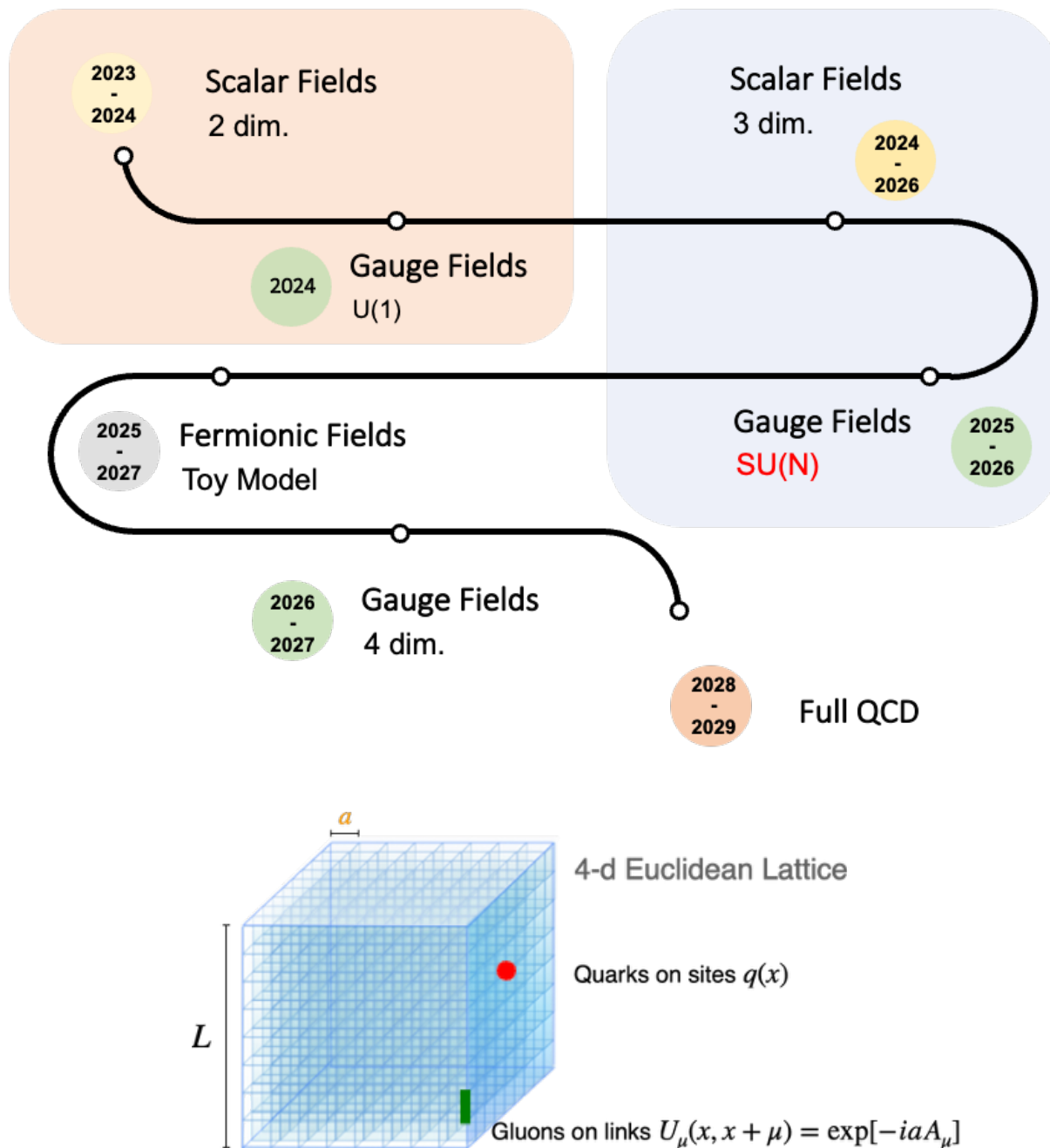
3. Detailed balance



2. Vertical: ensure equilibrium at each noise scale

Roadmap toward full QCD

Where we are now



2023-2024

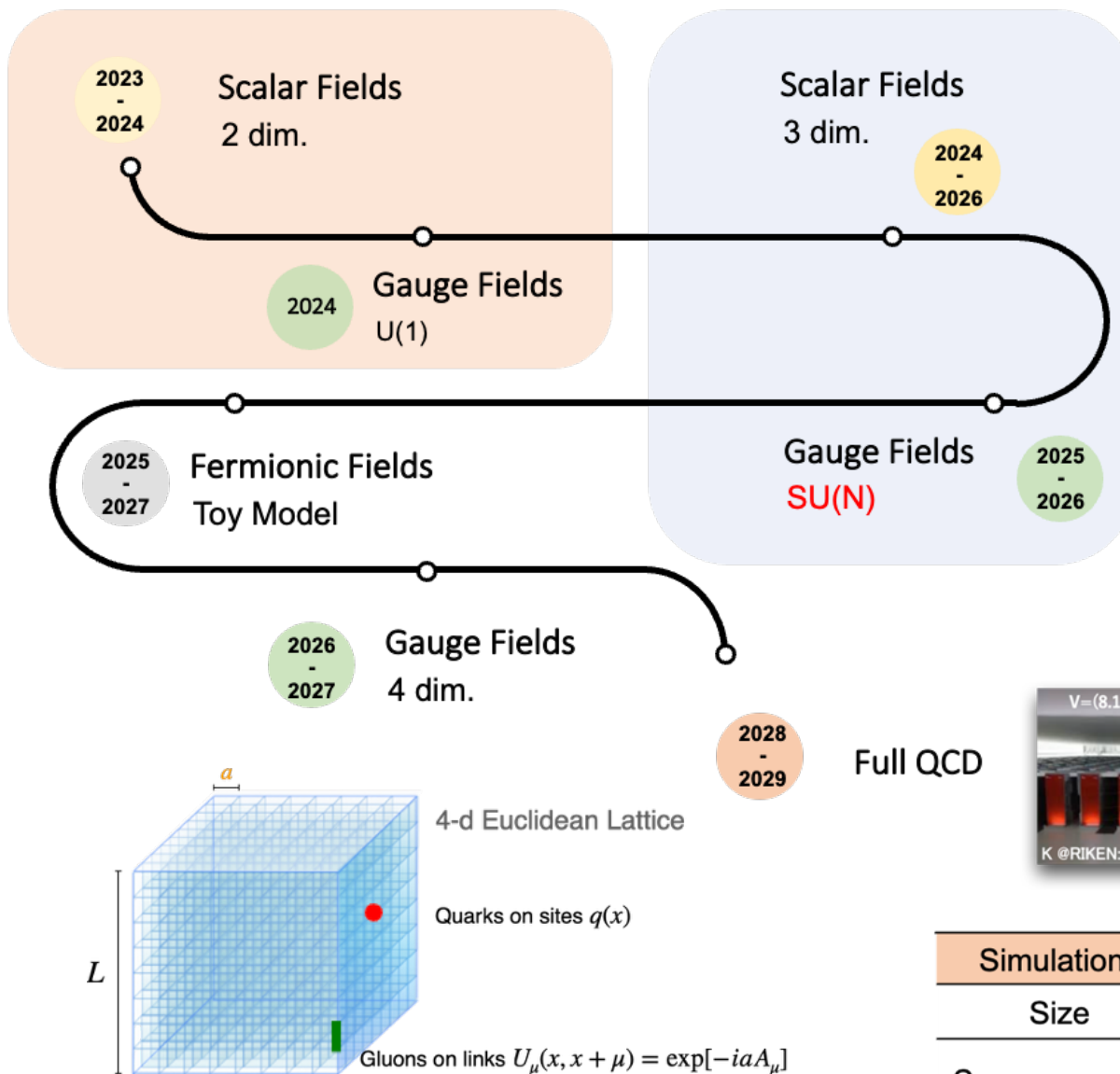
- ▶ **LW**, G Aarts, K Zhou, JHEP 05 (2024) 060 [[2309.17082](#) [hep-lat]]
- ▶ G Aarts, D Habibi, **LW**, K Zhou, Mach.Learn.Sci.Tech. 6 (2025) 2, 025004 and NeurIPS 2024 ML4PS [[2410.21212](#) [hep-lat]] ([Best "Physics for AI" Paper](#))

2024-2025

- ▶ Q Zhu, G Aarts, W Wang, K Zhou, **LW**, JHEP03(2026)111 [[2502.05504](#) [hep-lat]]
- ▶ G Aarts, K Fukushima, T Hatsuda, A Ipp, S Shi, **LW**, K Zhou, Nature Rev.Phys. 7 (2025) 3, 154-163, [[2501.05580](#) [hep-lat]]
- ▶ G Aarts, D Habibi, A Ipp, D Mueller, T Ranner, **LW**, W Wang, Q Zhu, [2601.19552](#) [hep-lat]

Outlook

Where shall we go



► Systematical 4D Gauge Field Simulations

► Evaluations and Improvements

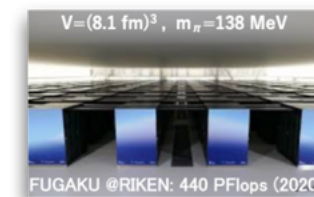
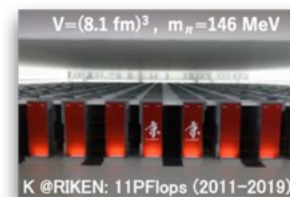
► Faster Generation

► Fermions

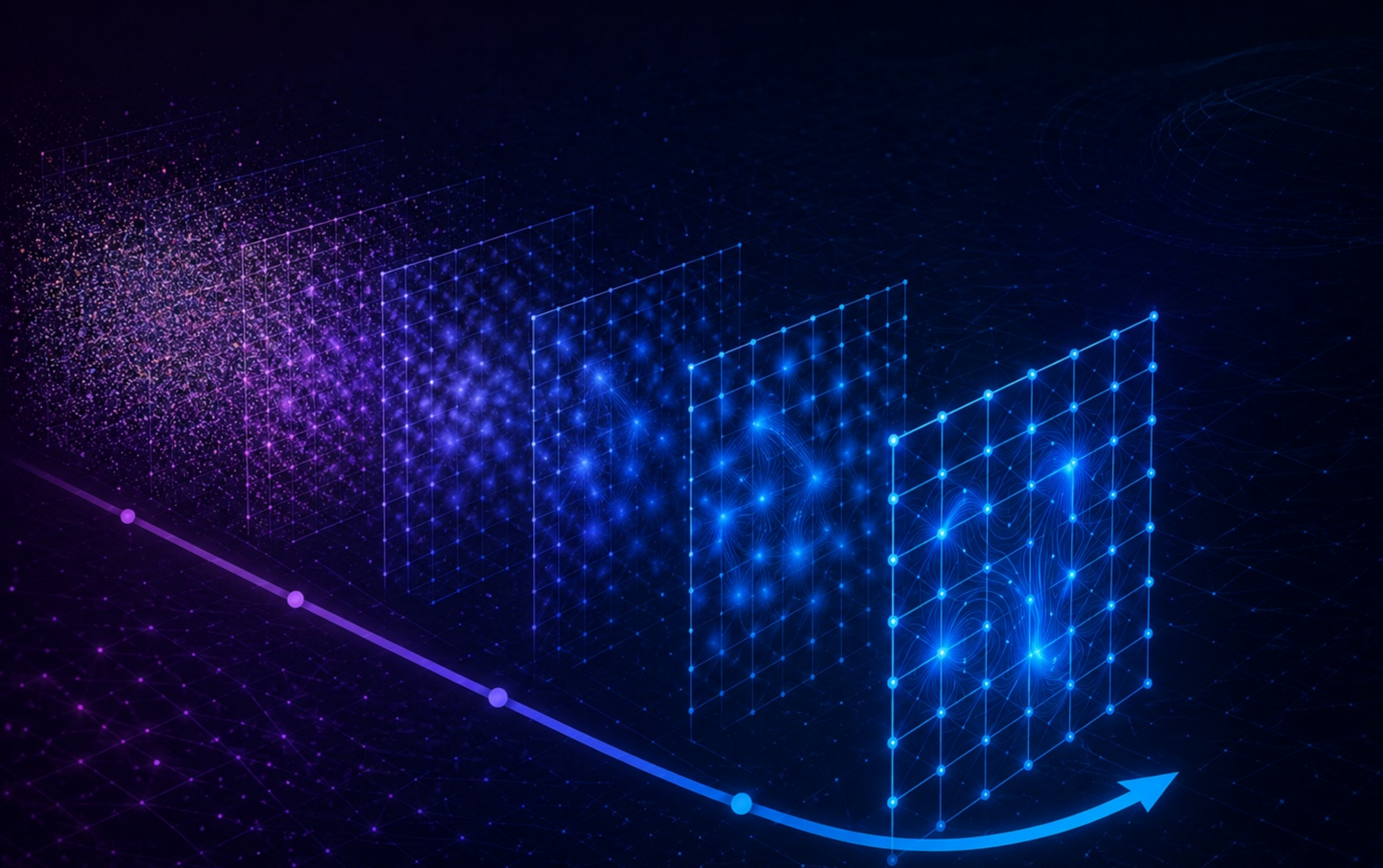
► Pseudo-Fermion, ect.

► Sign Problem

► Complex Langevin Method



Simulations	Lattice QCD	City-Scale Atmosphere Simulation
Size	96×96×96×96	6 hours, ~ 1 km
Supercomputer	Fugaku	GPU Clusters
Resource Consumption	512 cores	~ 1000 cores
Time Consumption	~ 1 day/configuration	~ 5 hours



DM meets LQFT, opportunities and challenges

NSCN++ for Scalar Fields

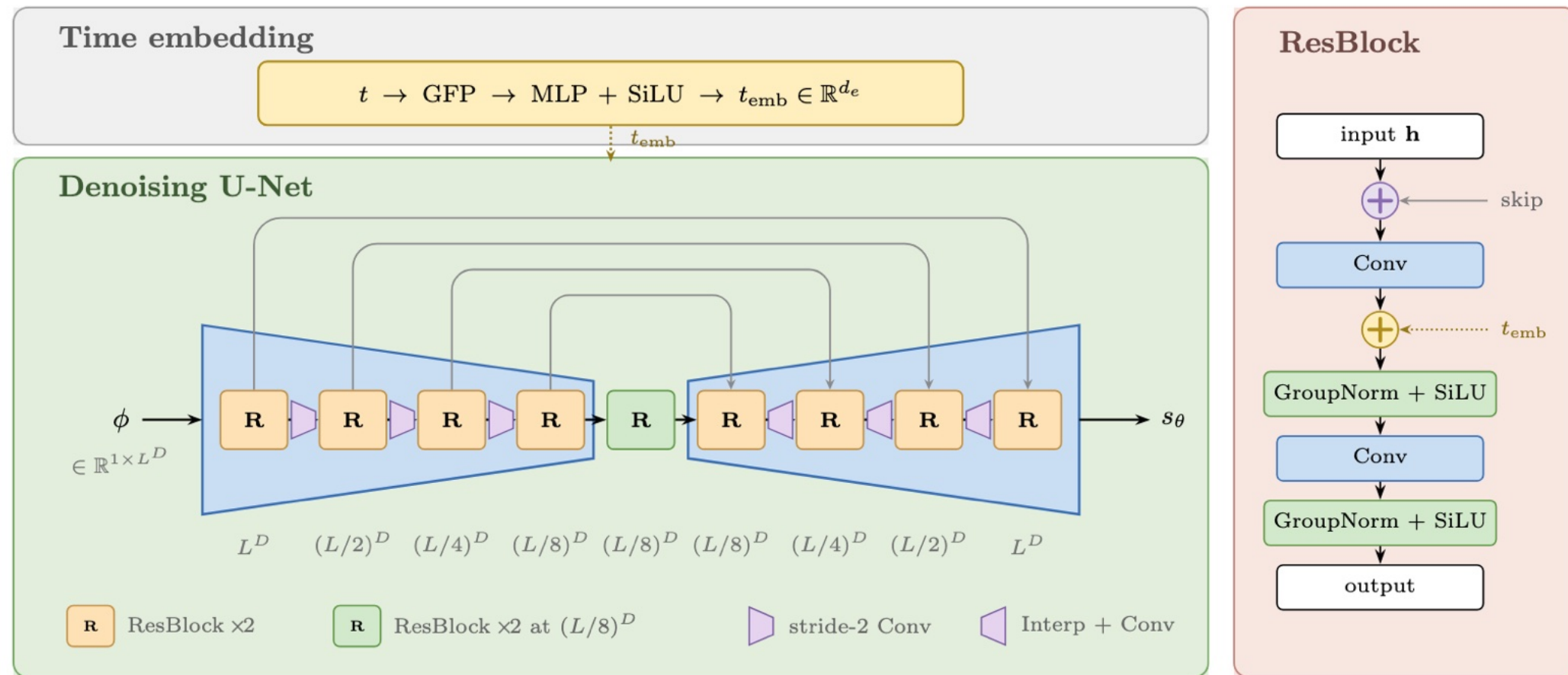


FIG. 9. Architecture of the NSCN++ score network used for both 2D and 3D ϕ^4 lattice field theory. (a) U-Net overview. The encoder (blue) consists of four ResBlock pairs at progressively coarser resolutions, separated by stride-2 convolutions (purple) that halve the spatial dimensions. The decoder (orange) mirrors this structure, upsampling via nearest-neighbor interpolation followed by a convolution (purple). Each encoder level provides two skip connections (from both sub-blocks) to the corresponding decoder level via channel concatenation (“concat $\times 2$ ”). The diffusion time t is encoded by Gaussian Fourier projection (GFP) and a learned MLP, then injected additively into every ResBlock. The final output is divided by $\sigma(t)$. (b) Internal structure of a single ResBlock; no residual connection is used. All convolutions use circular padding to enforce periodic boundary conditions. For 2D fields all convolutions are Conv2d; for 3D they become Conv3d ($D = 3$).

Architecture

L-CNN for Gauge Fields

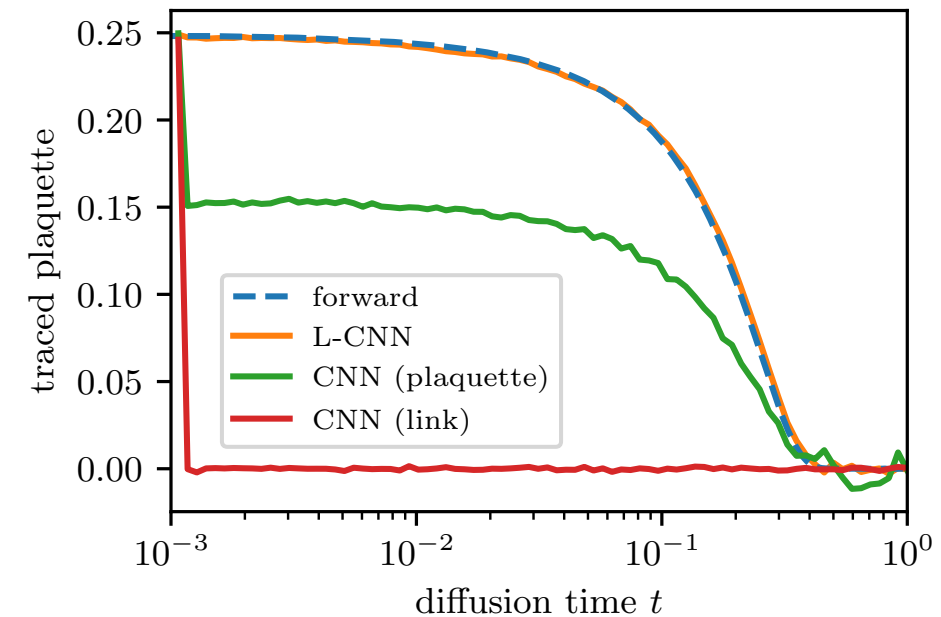
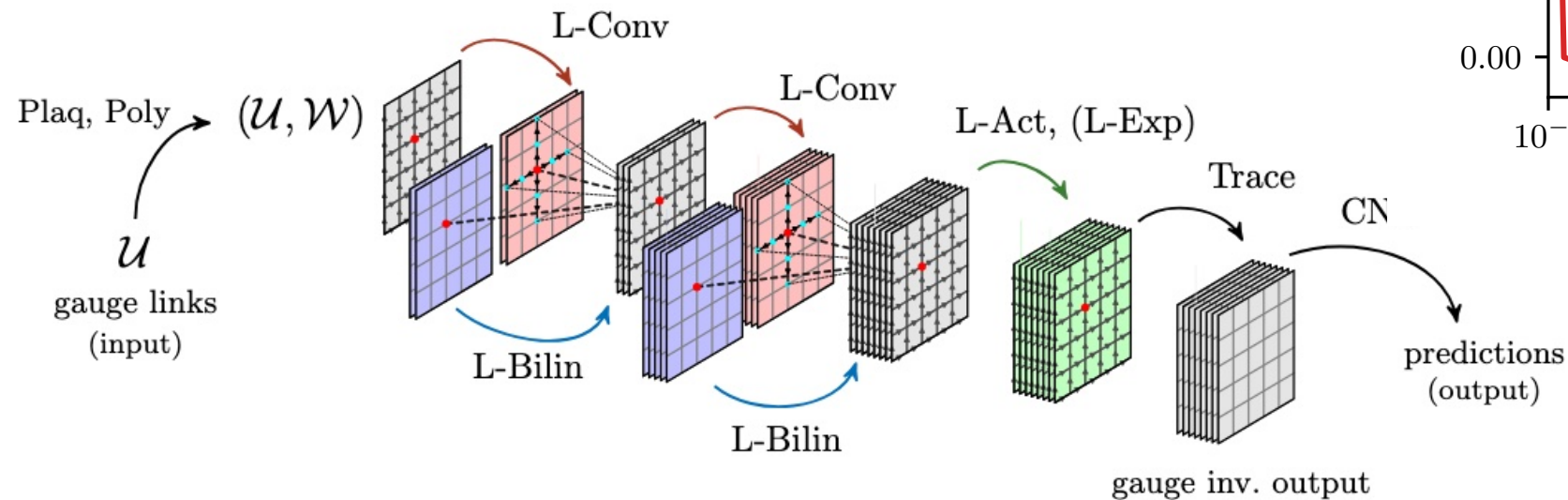


Figure 2. Schematic representation of a lattice gauge-equivariant convolutional neural network (L-CNN). This architecture processes data defined on a lattice, representing quantum field theory in discretized space-time. Each layer in the L-CNN is carefully constructed to preserve gauge symmetry, a property crucial for ensuring physically consistent predictions in quantum field theory. The network first processes the input lattice data given by gauge links $\mathcal{U}$ using simple Wilson loops on the lattice such as plaquettes (Plaq) or Polyakov loops (Poly) to create objects $\mathcal{W}$ that transform locally. These objects are then combined into progressively more complex Wilson loops while maintaining the symmetry through specialized convolutional (L-Conv) and bilinear (L-Bilin) operations. Additional gauge equivariant activation functions (L-Act) or exponentiation layers (L-Exp) can modify the local fields in a gauge equivariant manner. Finally, the network generates gauge-invariant outputs through a trace layer (Trace) that can be processed by standard convolutional layers to produce the desired physical predictions. Unlike conventional CNNs, this design is robust to random and adversarial gauge transformations, making it essential for simulations of fundamental physics. Image from³³.

G. Aarts, K. Fukushima, T. Hatsuda, A. Ipp, S. Shi, **L. Wang***, and K. Zhou,
Physics-Driven Learning for Inverse Problems in Quantum Chromodynamics,
 Nature Reviews Physics volume 7, pages154–163 (2025)

Metropolis-Adjusted Annealed Langevin Sampler

U(1) in two dim.

- Autocorrelation time

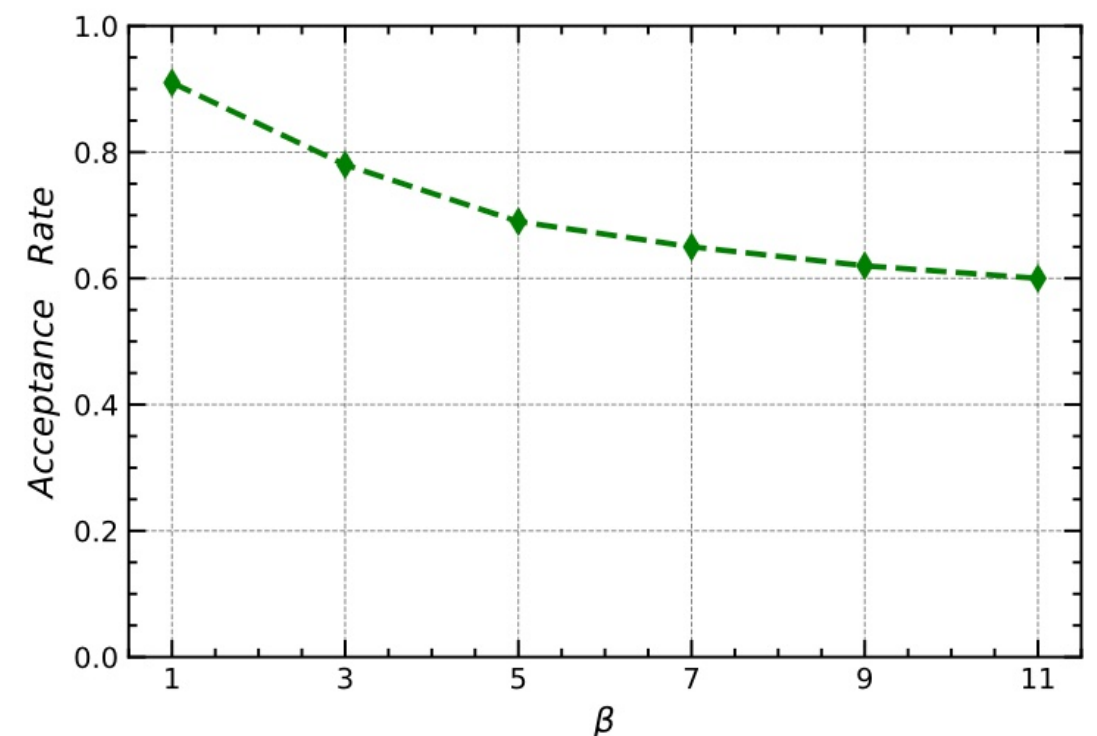
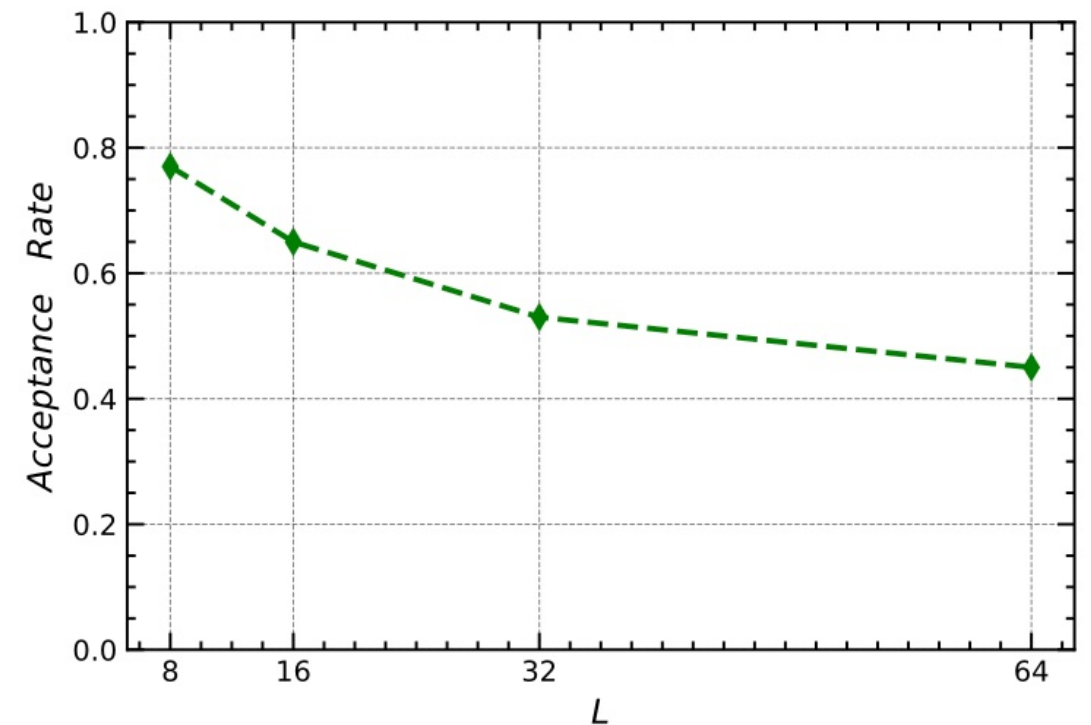
$$C_O(t) = \langle O_{t_0} O_{t_0+t} \rangle - \langle O_{t_0} \rangle \langle O_{t_0+t} \rangle$$

- Integrated autocorrelation time

$$\Gamma_O(t) \equiv \frac{C_O(t)}{C_O(0)}$$

$$\tau_O^{int} = \frac{1}{2} + \sum_{t=1}^N \Gamma_O(t)$$

J. Statist. Phys. 50 (1988) 109.



Metropolis-Adjusted Annealed Langevin Sampler

U(1) in two dim.

- Autocorrelation time

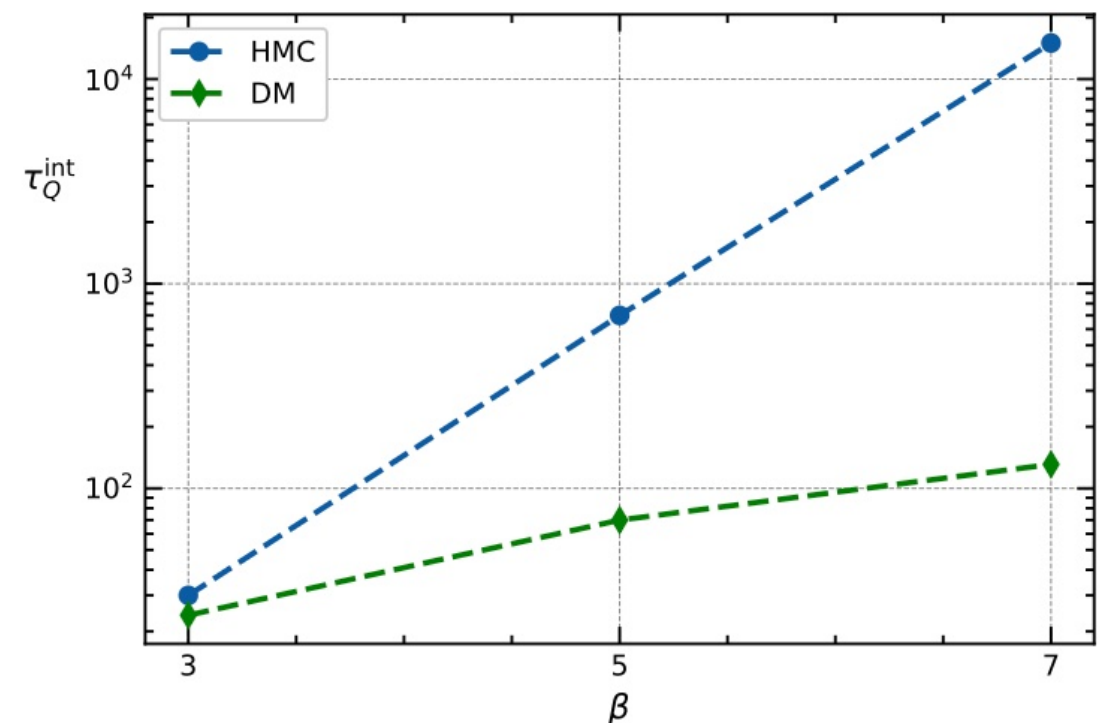
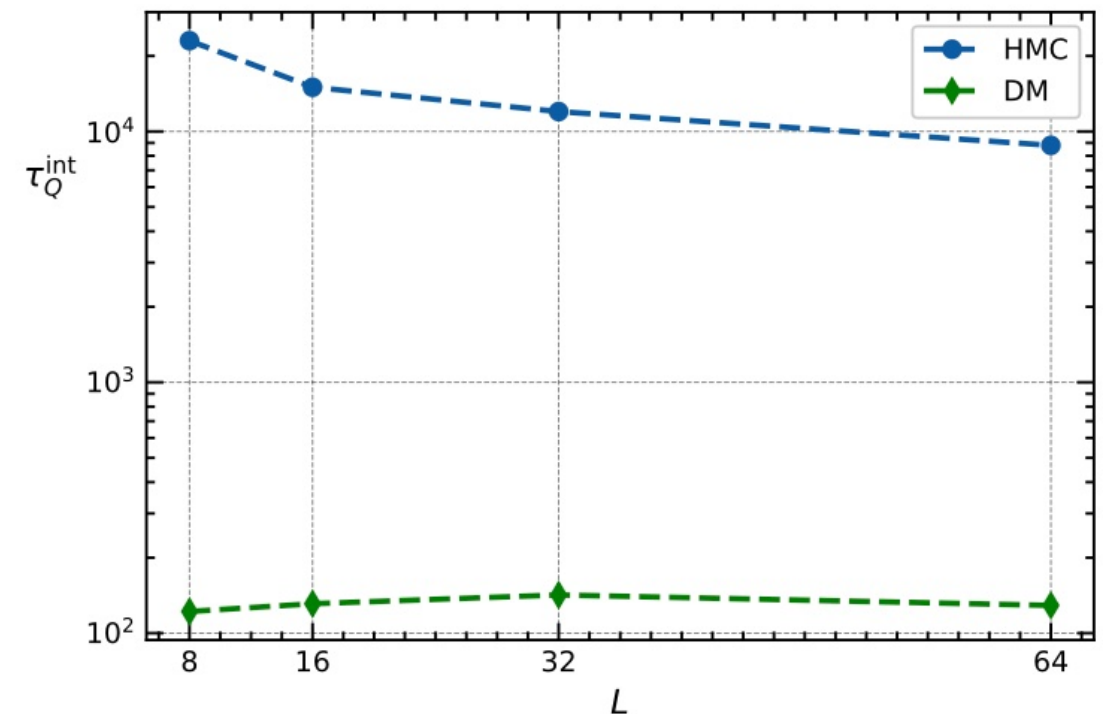
$$C_O(t) = \langle O_{t_0} O_{t_0+t} \rangle - \langle O_{t_0} \rangle \langle O_{t_0+t} \rangle$$

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J. Statist. Phys. 50 (1988) 109.



3E for U(1) Gauge Field

Q Zhu, G Aarts, W Wang, K Zhou, **LW**, JHEP(2026)

Comprehensive Comparision for 2D U(1)

Learned at $\beta = 1$ with **30k** cofigurations, $L = 16$

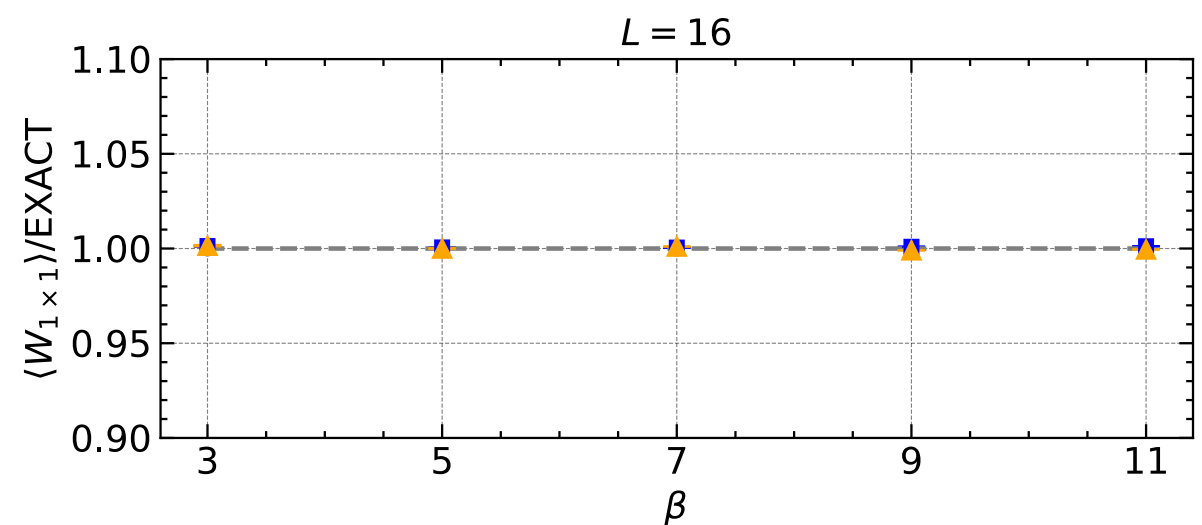
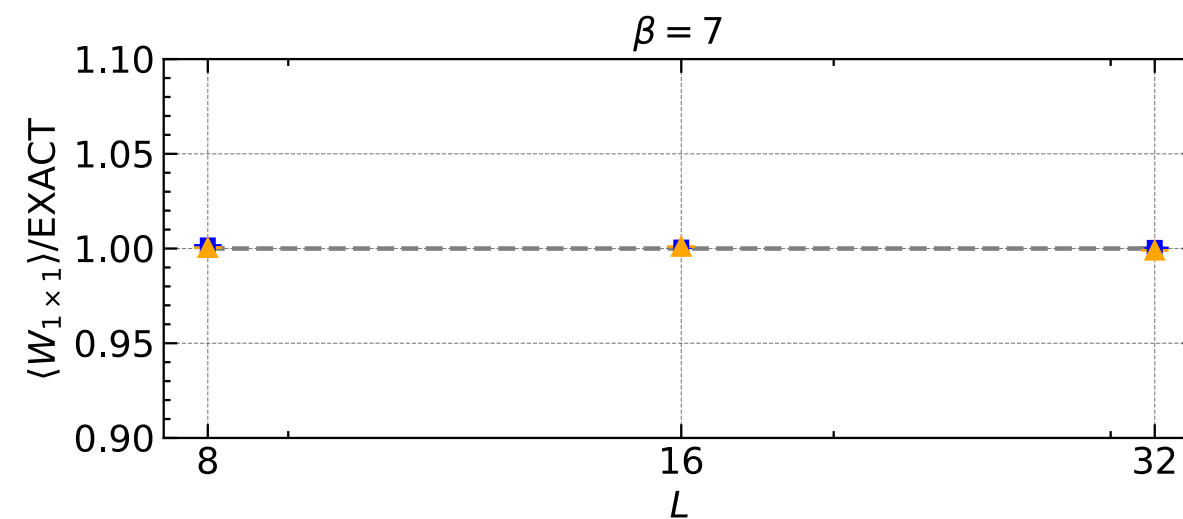
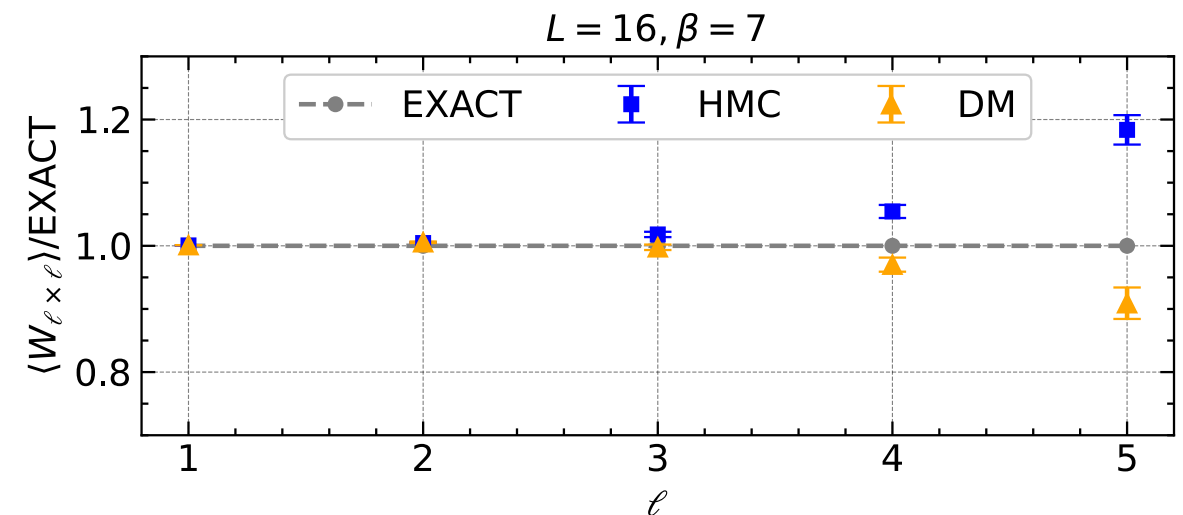
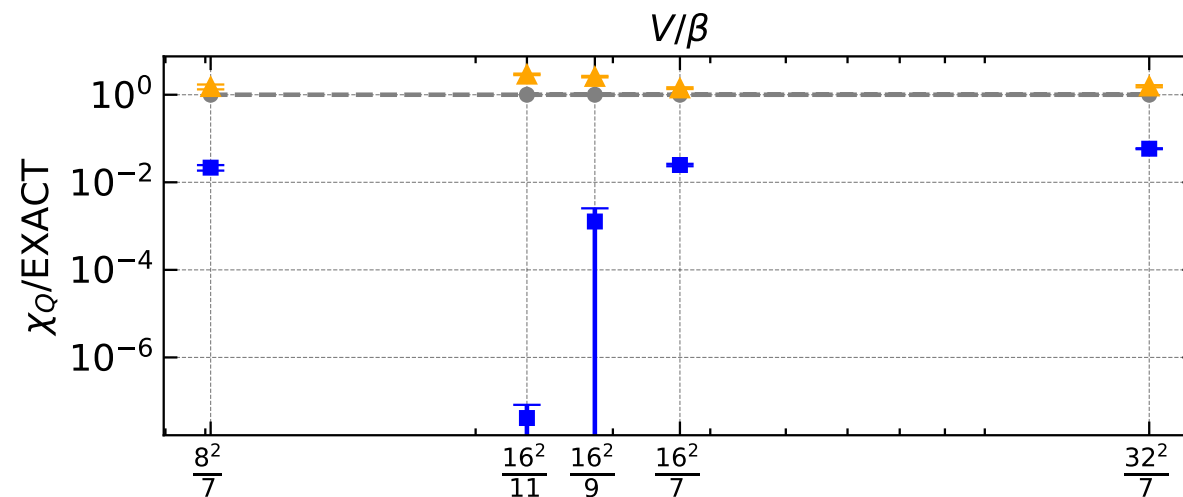


Table 1. Comparison of observables for $\beta = 1$ at different lattice sizes

Lattice Size (L)	1 $\times$ 1 Wilson Loop				Topological Susceptibility			
	HMC	DM	Langevin	Exact	HMC	DM	Langevin	Exact
8	0.447(72)	0.445(74)	0.443(80)	0.446	0.0402(17)	0.0413(18)	0.0418(18)	0.0406
16	0.447(37)	0.446(37)	0.444(36)	0.446	0.0416(16)	0.0422(17)	0.0421(20)	0.0406
32	0.446(18)	0.445(19)	0.445(18)	0.446	0.0428(19)	0.0415(18)	0.0412(17)	0.0406
64	0.446(9)	0.446(11)	0.445(9)	0.446	0.0426(19)	0.0427(20)	0.0420(19)	0.0406

Table 4. Comparison of observables for $L = 16$ at different couplings

coupling (β)	1 $\times$ 1 Wilson Loop				Topological Susceptibility			
	HMC	DM	Langevin	Exact	HMC	DM	Langevin	Exact
3	0.811(17)	0.811(17)	0.809(17)	0.810	0.0096(4)	0.0114(6)	0.0106(14)	0.0111
5	0.894(9)	0.894(9)	0.891(10)	0.894	0.0048(2)	0.0058(5)	0.0075(3)	0.0057
7	0.926(7)	0.926(7)	0.924(6)	0.926	0.00013(2)	0.0045(5)	0.0131(5)	0.0039
9	0.944(3)	0.942(4)	0.940(6)	0.942	0	0.0031(4)	0.0154(7)	0.0029
11	0.954(3)	0.953(4)	0.950(5)	0.953	0	0.0025(3)	0.0165(13)	0.0024

3E for U(1) Gauge Field

Q Zhu, G Aarts, W Wang, K Zhou, **LW**, JHEP(2026)

Comprehensive Comparision for 2D U(1)

Learned at $\beta = 1$ with **30k** cofigurations, $L = 16$

No Topological Freezing!

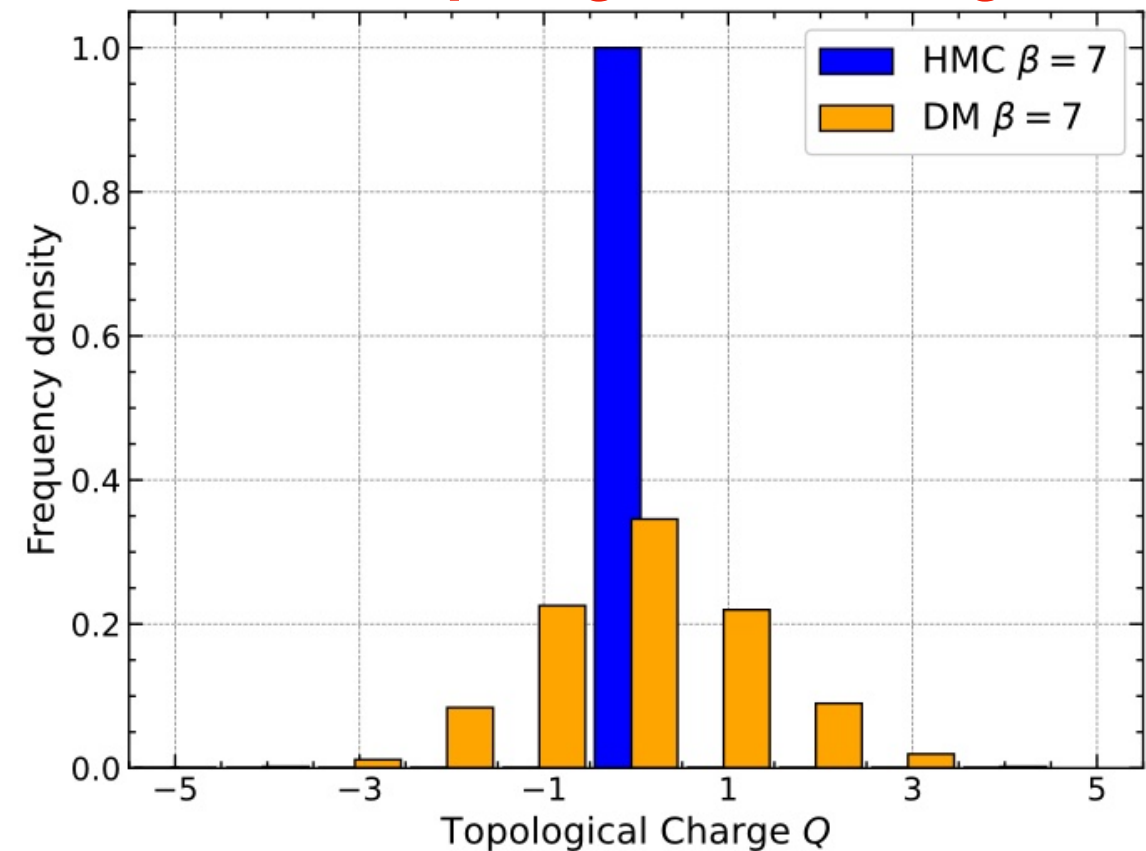
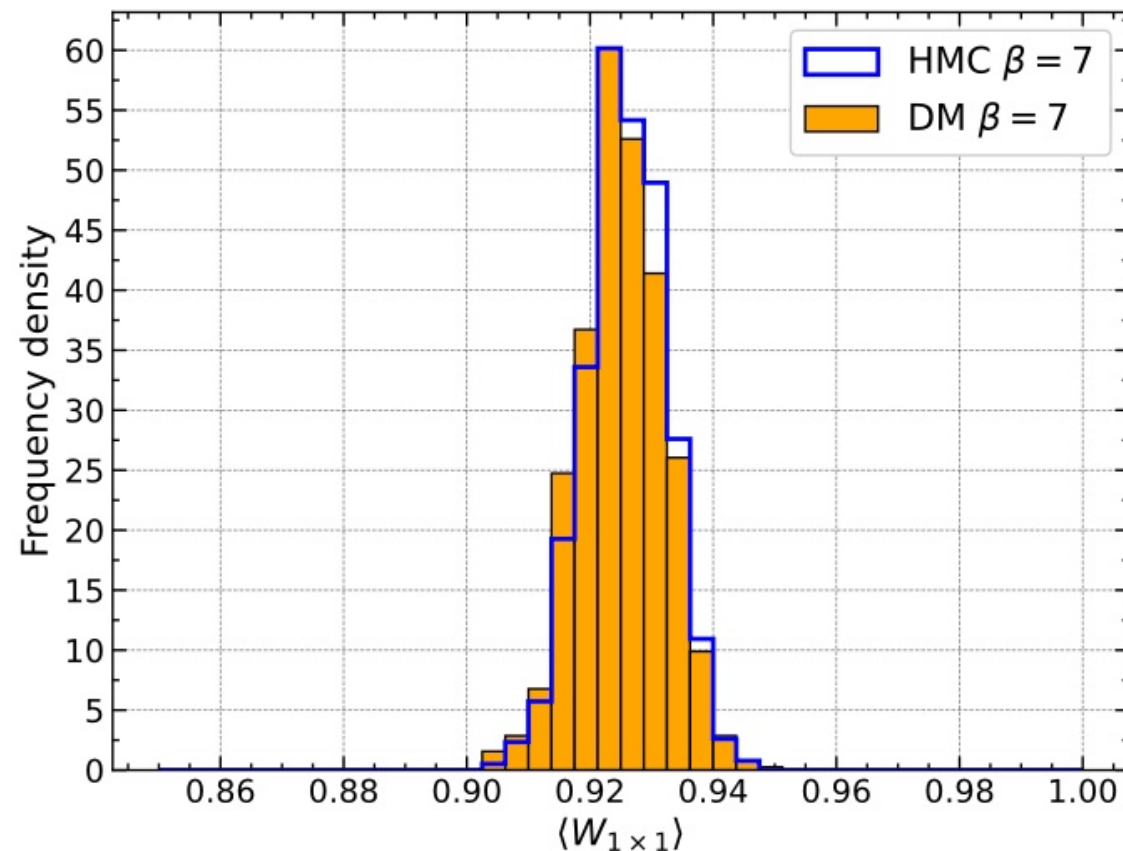


Table 2. Comparison of the $l \times l$ Wilson Loops for $L = 16, \beta = 7$

Loop size (l)	HMC	DM	Langevin	Exact
1	0.926(7)	0.926(7)	0.924(6)	0.926
2	0.737(31)	0.737(32)	0.730(34)	0.734
3	0.510(67)	0.496(72)	0.489(73)	0.498
4	0.311(97)	0.283(96)	0.283(106)	0.290

Table 3. Comparison of observables for $\beta = 7$ at different lattice sizes

Lattice Size (L)	1 $\times$ 1 Wilson Loop				Topological Susceptibility			
	HMC	DM	Langevin	Exact	HMC	DM	Langevin	Exact
8	0.927(13)	0.926(13)	0.921(13)	0.926	0.00006(3)	0.0040(12)	0.0143(5)	0.0040
16	0.926(7)	0.926(7)	0.924(6)	0.926	0.00013(2)	0.0045(5)	0.0131(5)	0.0039
32	0.926(3)	0.925(4)	0.924(4)	0.926	0.00013(2)	0.0040(4)	0.0137(6)	0.0039

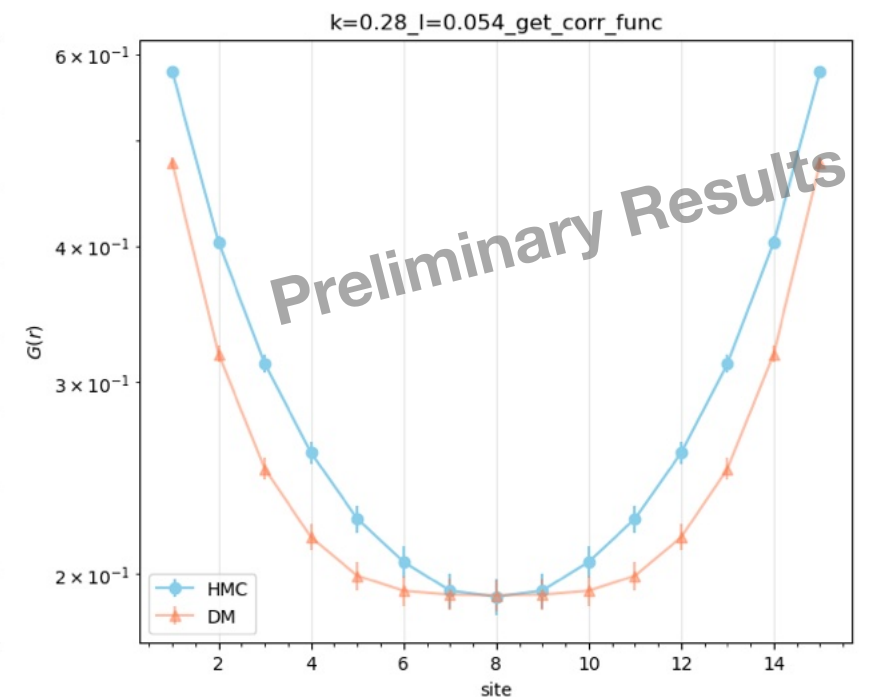
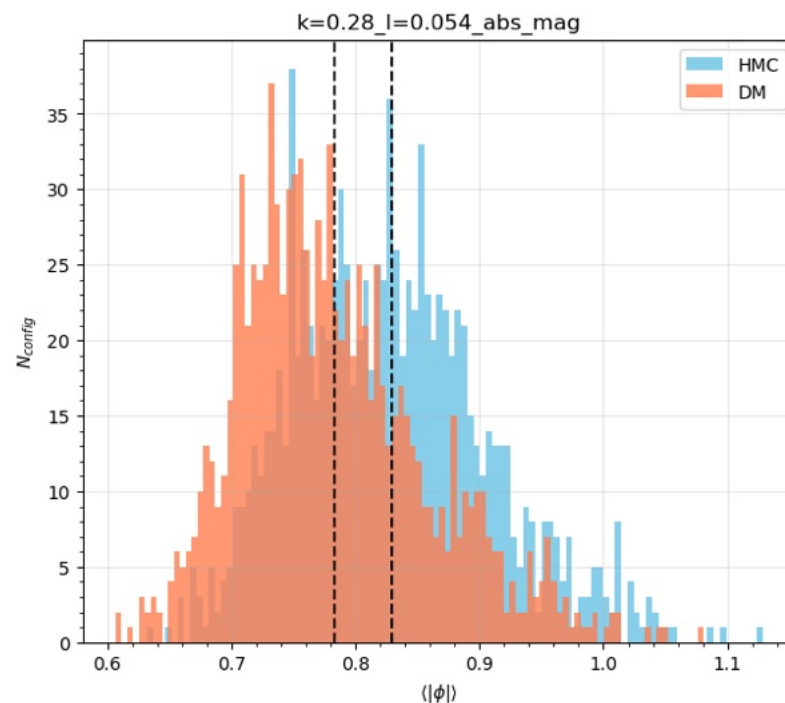
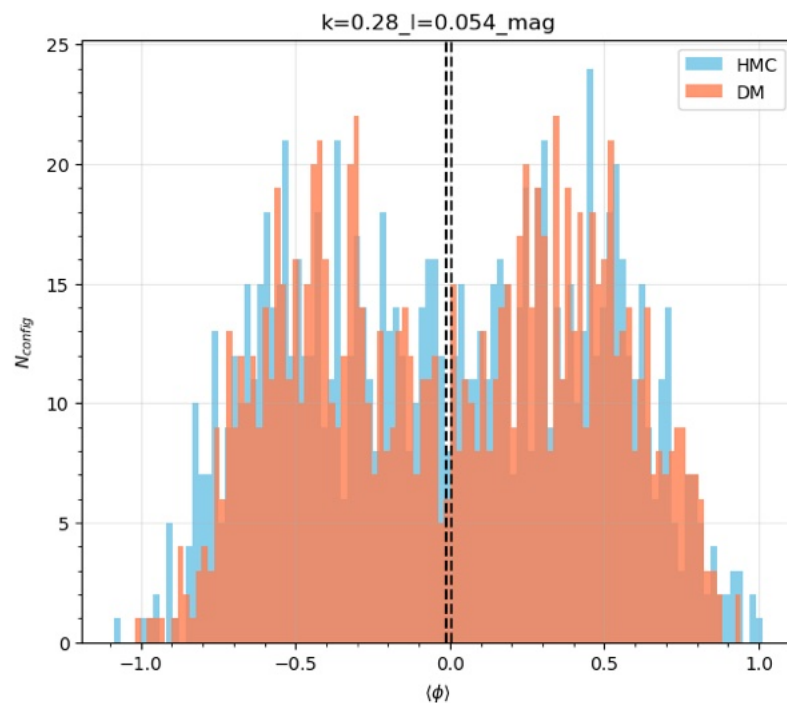
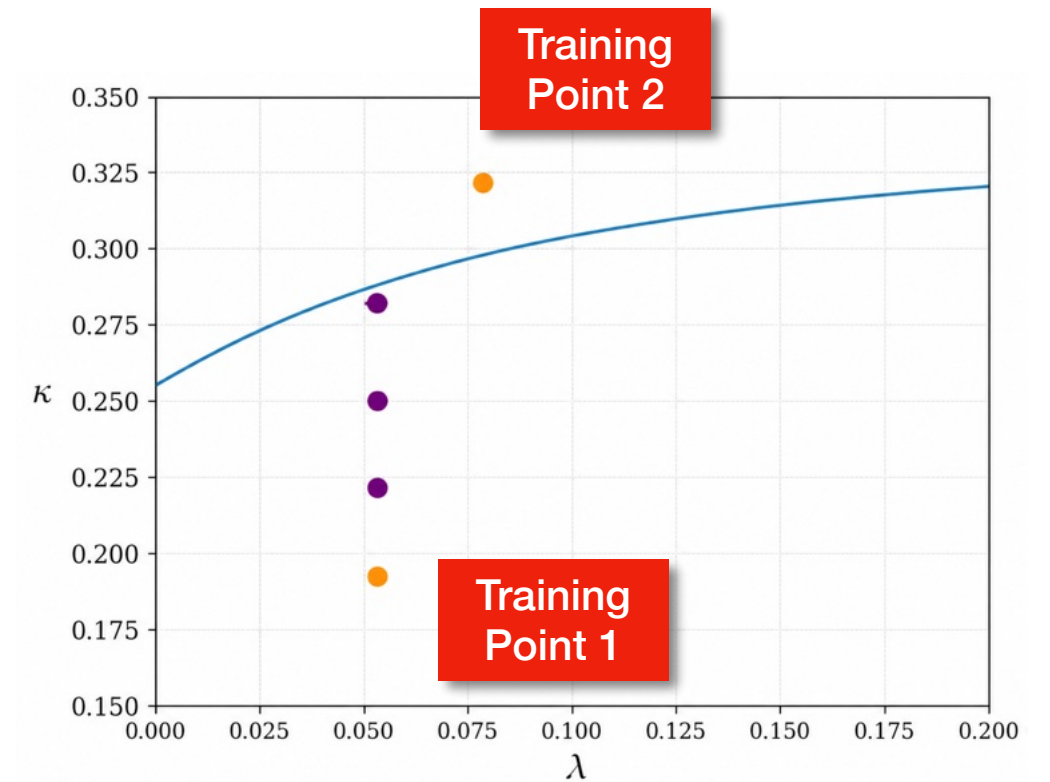
Extropolation to Criticality

$$S_E = \sum_x [-2\kappa \sum_{\mu=1}^d \phi(x)\phi(x + \hat{\mu}) + (1 - 2\lambda)\phi(x)^2 + \lambda\phi(x)^4]$$

Decomposed Score Function

$$S_\theta \equiv \kappa S_{\theta_1} + \lambda S_{\theta_2}$$

Sampling directly at {0.28, 0.054}



Back-ups: Practical Details

Diffusion Models

Score-based model

[Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, ICLR 2021](#)

[Anderson, in Stochastic Processes and their Applications, 1982](#)

- **Forward Diffusion SDE**

- **Drift term**: pulls towards mode
- **Diffusion term**: injects noise

$$\frac{d\phi}{d\xi} = \boxed{f(\phi, \xi)} + \boxed{g(\xi)\eta(\xi)}$$

- **Reverse Generative Diffusion SDE**

- Drift term is adjusted with a “**Score Function**”
- But how to get the score function ?

$$\frac{d\phi}{dt} = \boxed{\left[f(\phi, t) - g^2(t) \nabla_{\phi} \log p_t(\phi) \right]} + \boxed{g(t)\bar{\eta}(t)}$$

Model the score function with neural networks!

$$\frac{d\phi}{dt} = -\sigma^{2t} \mathbf{s}_{\hat{\theta}}(\phi, t) + \sigma^t \bar{\eta}(t)$$

$$\mathcal{L}_{\theta} = \sum_{i=1}^N \sigma_i^2 \mathbb{E}_{p_0(\phi_0)} \mathbb{E}_{p_i(\phi_i|\phi_0)} \left[\left\| \mathbf{s}_{\theta}(\phi_i, \xi) - \nabla_{\phi_i} \log p_i(\phi_i|\phi_0) \right\|_2^2 \right]$$

One example:
Variance Expanding

$$\begin{aligned} f(\phi, \xi) &= 0 \\ g(\xi) &= \sigma^{\xi} \\ \xi &\in [0, T] \end{aligned}$$

Diffusion Models

Probabilistic FlowODE

[Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, ICLR 2021](#)

[Anderson, in Stochastic Processes and their Applications, 1982](#)

- **Forward Diffusion SDE**

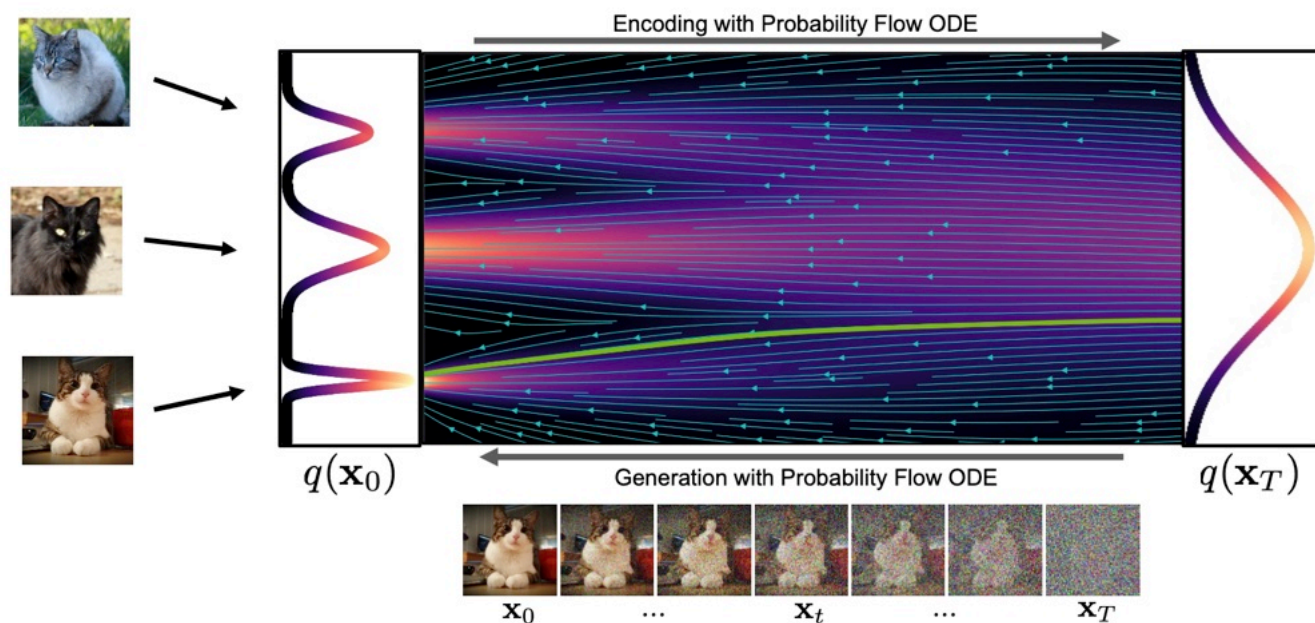
- **Drift term**: pulls towards mode
- **Diffusion term**: injects noise

$$\frac{d\phi}{d\xi} = f(\phi, \xi) + g(\xi)\eta(\xi)$$

- **Reverse Generative Diffusion SDE**

- Drift term is adjusted with a “**Score Function**”
- **Sampling from the SDE or ODE**

$$\frac{d\phi}{dt} = \left[f(\phi, t) - g^2(t) \nabla_{\phi} \log p_t(\phi) \right] + g(t)\bar{\eta}(t)$$



$$\frac{d\phi}{dt} = \left[f(\phi, t) - g^2(t) \nabla_{\phi} \log p_t(\phi) \right]$$

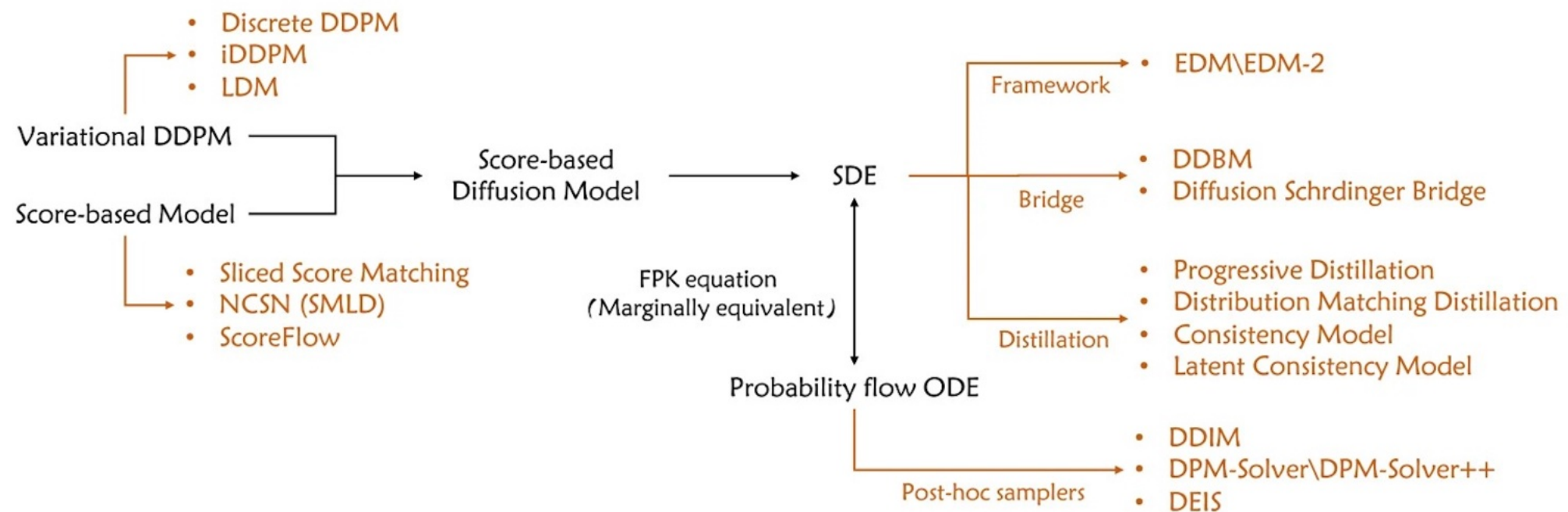
Probability FlowODE

- Enables use of advanced ODE solvers
- Exact Log-Likelihood computation
- **Recover the Flow-based models**

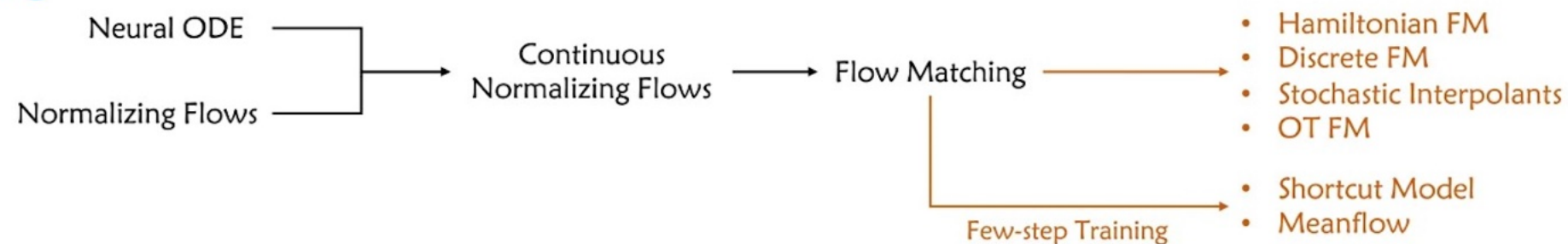
Diffusion Models

Probabilistic Flow Zoo

Diffusion



Flow matching



Rectified flow



lenny@buaa.edu.cn

Physics of Diffusion Models

Cumulants

Diffusion models: generation of correlations

- forward process $\dot{x}(t) = K(x(t), t) + g(t)\eta(t)$ $0 \leq t \leq T$
 - backward process $x'(\tau) = -K(x(\tau), T - \tau) + g^2(T - \tau)\partial_x \log P(x, T - \tau) + g(T - \tau)\eta(\tau)$ $\tau = T - t$
- noise profile $g(t) = \sigma^{t/T}$
- score

two main schemes

- variance-expanding (VE): no drift $K(x, t) = 0$
- variance-preserving (VP) or denoising diffusion probabilistic models (DDPMs):
linear drift $K(x(t), t) = -\frac{1}{2}k(t)x(t)$

Generating functionals

- moment generating

$$Z[J] = \mathbb{E}[e^{J(x,t)\phi(x,t)}] = e^{\frac{1}{2}J^2(x,t)\Xi(t)} \int D\phi_0 P_0[\phi_0] e^{J(x,t)\phi_0(x)f(t,0)}$$

- cumulant generating

$$W[J] = \log Z[J] = \frac{1}{2}J^2(x,t)\Xi(t) + \log \int D\phi_0 P_0[\phi_0] e^{J(x,t)\phi_0(x)f(t,0)}$$

- higher-order cumulants

$$\kappa_{n>2}(t) = \frac{\delta^n W[J]}{\delta J(x,t)^n} \Big|_{J=0} = \frac{\delta^n}{\delta J(x,t)^n} \log \mathbb{E}_{P_0}[e^{J(x,t)\phi_0(x)f(t,0)}] \Big|_{J=0}$$

$$\Xi(t) = \int_0^T ds f^2(t,s)g^2(s)$$

full path integral
with sources

variance
preserving

$$f(t,0) \rightarrow 0$$

variance
expanding

$$f(t,0) = 1$$

- Correlations are being destroyed and retrieved
- If score is determined exactly, full theoretical control

BEST 'PHYSICS FOR AI' PAPER AWARD 🏆

Higher-order cumulants in diffusion models

Gert Aarts, Diaa Eddin Habibi, Lingxiao Wang, Kai Zhou

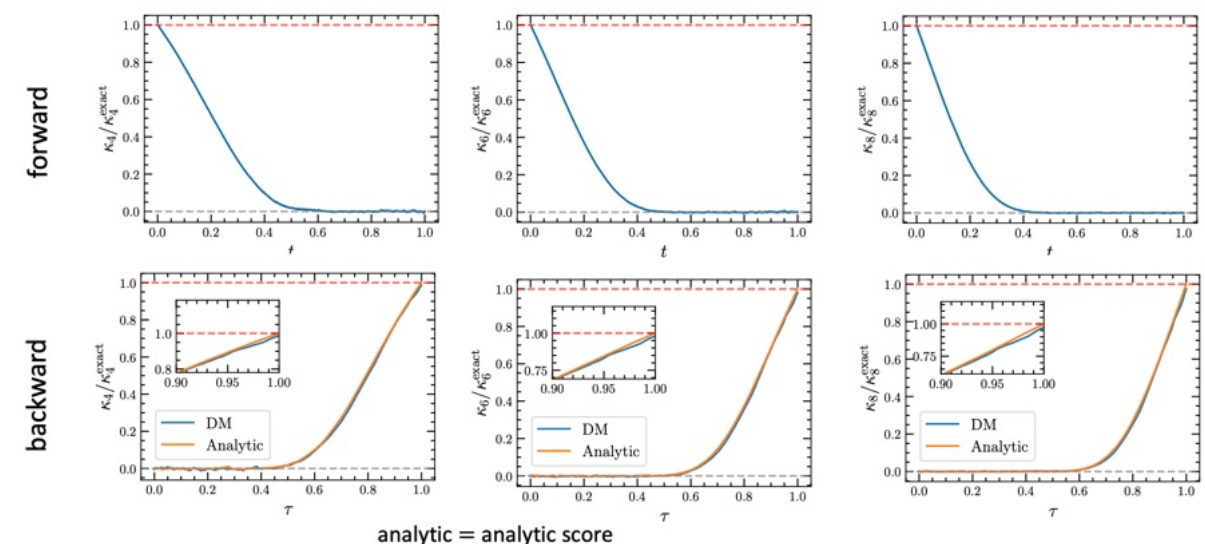
[\[paper\]](#) [\[poster\]](#)

Machine Learning and the Physical Sciences, NeurIPS 2024

4th, 6th, 8th cumulant with drift (DDPM)

$$\kappa_{n>2}(t) = \kappa_n(0)f^n(t,0)$$

$$f(t,0) \rightarrow 0$$



Marginal Note #1

Anderson's Theorem

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\frac{\partial p}{\partial t} = -\nabla \cdot (\mathbf{f}p) + \frac{1}{2}g^2\Delta p$$

$$d\mathbf{x} = \tilde{\mathbf{f}}(\mathbf{x}, t)d\bar{t} + g(t)d\bar{\mathbf{w}}$$

$$-\frac{\partial p}{\partial t} = -\nabla \cdot (\tilde{\mathbf{f}}p) + \frac{1}{2}g^2\Delta p$$

$$\frac{\partial p}{\partial t} = \nabla \cdot (\tilde{\mathbf{f}}p) - \frac{1}{2}g^2\Delta p$$



Fokker-Planck Equation

$$d\bar{t} = -t$$

Reverse Time!

Reverse Fokker-Planck Equation

Marginal Note #1

Anderson's Theorem



$$\frac{\partial p}{\partial t} = -\nabla \cdot (\mathbf{f}p) + \frac{1}{2}g^2\Delta p \quad \frac{\partial p}{\partial t} = \nabla \cdot (\tilde{\mathbf{f}}p) - \frac{1}{2}g^2\Delta p$$

$$\underbrace{-\nabla \cdot (\mathbf{f}p) + \frac{1}{2}g^2\Delta p}_{\text{Forward}} = \underbrace{\nabla \cdot (\tilde{\mathbf{f}}p) - \frac{1}{2}g^2\Delta p}_{\text{Backward}}$$

$$-\nabla \cdot (\mathbf{f}p) - \nabla \cdot (\tilde{\mathbf{f}}p) = -\frac{1}{2}g^2\Delta p - \frac{1}{2}g^2\Delta p$$

$$\nabla \cdot [(\mathbf{f} + \tilde{\mathbf{f}})p] = g^2\Delta p \quad \Delta p = \nabla \cdot (\nabla p) = \nabla \cdot (p \nabla \log p)$$

$$\nabla \cdot [(\mathbf{f} + \tilde{\mathbf{f}})p] = \nabla \cdot [g^2 p \nabla \log p]$$

$$\tilde{\mathbf{f}} = -\mathbf{f} + g^2 \nabla \log p$$

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g^2(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x})]dt + g(t)d\bar{\mathbf{w}}$$